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Pegasus: VFS 2025–2026 DBVF

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DESIGN BUILD VERTICAL FLIGHT 2026: Wildfire Response eVTOL System

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Abstract

Pegasus is an autonomous electric vertical takeoff and landing aircraft developed for the Vertical Flight Society Design-Build-Vertical Flight competition and designed around the operational needs of early-stage wildfire response. The system was developed to provide rapid aerial situational awareness, stable hover capability, modular payload deployment, and autonomous mission execution in environments where conventional reconnaissance methods may be limited by terrain, visibility, response time, or personnel risk. The final aircraft uses a hexacopter configuration with six electric propulsion units, a modular aluminum airframe, independent propulsion and avionics power systems, a Cube Orange+ flight controller, adaptive landing gear, and a bottom-mounted payload mechanism.

The aircraft was designed using a systems-engineering process that translated competition rules, stakeholder needs, and wildfire-response constraints into measurable requirements. Key design targets included a maximum takeoff weight below 25 lb, a thrust-to-weight ratio greater than 2.0, hover throttle below 60%, minimum mission endurance above 10 minutes, maximum structural deflections below 1 mm, minimum factor of safety greater than 2.0, and sufficient separation between rotor excitation frequencies and structural natural frequencies. The completed aircraft achieved an estimated takeoff weight of 18.02 lb, a thrust-to-weight ratio of 2.66, hover throttle near 47%, estimated mission endurance of 10.6 minutes, maximum structural deflection of 0.207 mm, minimum factor of safety of 2.04, and a first modal frequency of 89.985 Hz.

Validation was performed through propulsion sizing, structural finite element analysis, modal analysis, sensitivity analysis, ground testing, hover testing, manual controls testing, and autonomous mission testing. The results demonstrated that Pegasus satisfied the major DBVF competition requirements while maintaining conservative stability, structural, and propulsion margins. Although additional testing would be required before deployment in real wildfire conditions, the project demonstrated the feasibility of a modular autonomous multirotor platform for wildfire reconnaissance and lightweight payload pickup and delivery.

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Acronyms

VFS Vertical Flight Society.

DBVF Design-Build-Vertical Flight.

FAA Federal Aviation Administration.

UAV Unmanned Aerial Vehicle.

eVTOL Electric Vertical Take-off and Landing.

ESC Electronic Speed Controller.

CG Center of Gravity.

AGL Above Ground Level.

FEA Finite Element Analysis.

TO Topology Optimization.

CAD Computer Aided Design.

CNC Computer Numerical Control.

COTS Commercial Off-the-Shelf.

1 Introduction

Wildfires have become increasingly destructive across the western United States, threatening human life, infrastructure, ecosystems, and emergency response systems. Rising temperatures, prolonged drought, and expanding development near fire-prone regions have increased both the frequency and severity of wildfire events [16], [17]. Modern wildfires spread rapidly, exhibit unpredictable behavior, and require faster situational awareness than traditional response methods can consistently provide. During the early stages of ignition, responders often operate with limited information regarding fire growth, terrain accessibility, wind conditions, and nearby hazards, delaying decision-making and allowing manageable fires to escalate before adequate resources are deployed.

Conventional wildfire aviation assets such as helicopters and fixed-wing aircraft remain essential for suppression and reconnaissance but face several operational constraints. Manned aircraft require coordination, pilot availability, safe flight conditions, and suitable staging locations before deployment. Smoke, wind, terrain, and nighttime conditions can delay aerial support during the most critical stages of a wildfire. Ground crews similarly face accessibility challenges due to steep terrain, blocked roadways, active flame fronts, and rapidly changing fire boundaries.

Recent advancements in unmanned aerial vehicles (UAVs) and electric vertical takeoff and landing (eVTOL) systems have created new opportunities for emergency response [18]. Autonomous multirotor aircraft can perform vertical takeoff, precision hover, autonomous navigation, and payload delivery without runway infrastructure or exposing pilots to hazardous conditions. These systems can rapidly provide real-time aerial imagery, thermal reconnaissance, environmental sensing, and lightweight supply delivery during the earliest stages of wildfire development. As a result, UAV platforms are increasingly being explored as “Drone-as-First-Responder” systems capable of improving operational awareness while reducing response time and risk to personnel.

This thesis presents the development of Pegasus, an autonomous wildfire-response eVTOL aircraft designed for rapid deployment, modular mission adaptability, and stable operation in harsh wildfire environments. The aircraft was developed as part of the Vertical Flight Society Design-Build-Vertical Flight (DBVF) competition, which challenged student engineering teams to design and validate an autonomous electric VTOL platform capable of performing mission-oriented tasks under strict operational and safety constraints.

Pegasus was designed as a battery-powered autonomous hexacopter optimized for wildfire reconnaissance and lightweight payload deployment. The aircraft incorporates distributed electric propulsion, modular payload architecture, adaptive landing gear, autonomous flight-control integration, and a structurally optimized airframe intended to maximize reliability and flexibility. Unlike systems focused on aerodynamic efficiency or high-speed forward flight, Pegasus prioritizes stable hover performance, rapid field deployment, conservative control authority, and mission adaptability in uncertain environments.

The aircraft’s development required the integration of aerospace engineering principles across propulsion, structures, controls, flight dynamics, manufacturing, systems engineering, and autonomous operation. Extensive background research, stakeholder interviews, subsystem trade studies, analytical modeling, structural validation, and mission planning ensured the aircraft addressed realistic wildfire-response challenges while satisfying competition requirements. The final design represents both a multidisciplinary engineering effort and a practical exploration of how autonomous aerial systems can improve wildfire emergency-response effectiveness.

1.1 Motivation

The primary motivation for this project stemmed from the growing challenges of modern wildfire response [16],[17]. Wildfires have become increasingly dynamic and difficult to manage, requiring rapid situational awareness during the first minutes after ignition. Emergency personnel often must make critical decisions regarding evacuation, containment, and resource deployment with limited or delayed information. Traditional reconnaissance methods frequently cannot provide real-time visibility quickly enough to support effective response operations.

The first fifteen to thirty minutes after ignition are especially critical because early fire behavior often determines whether a wildfire can be contained [15]. However, manned aerial assets may be unavailable or unable to safely access affected areas due to smoke, visibility limitations, or terrain constraints. Ground crews also face delays caused by hazardous conditions and limited accessibility, creating a major gap in early-stage situational awareness.

Autonomous UAV systems provide a promising solution because they can be rapidly deployed, operate remotely without risking pilots, and provide persistent aerial monitoring at relatively low cost. Multirotor eVTOL platforms are particularly suited for wildfire-response missions due to their vertical takeoff capability, stable hover performance, and precise maneuverability [18]. Modular payload systems also allow UAVs to support thermal imaging, sensor deployment, communication relay, and lightweight supply delivery.

The Vertical Flight Society DBVF competition provided an ideal framework for developing these technologies in a realistic engineering environment. The project emphasized autonomous operation, reliability, payload integration, and mission adaptability while applying aerospace engineering principles across propulsion, structures, controls, manufacturing, and systems integration [27]. Ultimately, Pegasus was developed to explore how autonomous vertical flight systems can improve wildfire-response operations through faster deployment, improved situational awareness, and reduced risk to emergency personnel.

1.2 Timeline

The development of Pegasus followed a structured systems-engineering process, shown below in Figure 1.1, beginning with preliminary research and mission definition before progressing through concept selection, subsystem trade studies, analytical validation, manufacturing,

integration, and testing. The timeline aligned with both the Mechanical Engineering senior design sequence at Santa Clara University and the requirements of the Vertical Flight Society DBVF competition.

The project began with research into wildfire-response challenges and the role autonomous aerial systems could play in improving situational awareness. Existing UAV platforms, multirotor configurations, and emergency-response drone applications were studied to identify operational needs. Stakeholder interviews with firefighters, drone operators, and faculty advisors helped define major design priorities, including stability, autonomous navigation, payload flexibility, communication reliability, and operational safety [2],[3],[4].

Following mission definition, multiple aircraft configurations were evaluated through analytical trade studies comparing endurance, thrust margin, controllability, hover efficiency, and mission suitability. A hexacopter configuration was selected due to its stability, redundancy, and payload adaptability for wildfire environments.

Detailed subsystem development focused on propulsion sizing, structural optimization, avionics integration, payload systems, landing gear, and power distribution. Finite Element Analysis (FEA), topology optimization, modal analysis, and Simulink modeling were used to validate structural integrity, vibration behavior, and propulsion-system endurance.

After analytical validation, fabrication and integration were completed using machined aluminum components and additive manufacturing techniques. Avionics, propulsion electronics, telemetry systems, and payload mechanisms were integrated into the final airframe with emphasis on reliability, maintainability, and field serviceability.

The final phase involved ground testing, propulsion verification, vibration testing, hover validation, and subsystem integration testing to confirm compliance with competition requirements and design objectives. Future development goals included expanded autonomous capability, advanced sensor integration, and improved wildfire-response functionality.

1.3 Background Research

Background research for this project centered on three primary areas: wildfire-response operations, multirotor eVTOL aircraft design, and emerging autonomous UAV technologies for emergency-response applications. The purpose of this research was to identify operational capability gaps, understand environmental challenges, and establish design priorities for the Pegasus platform.

Research into wildfire-response operations revealed several recurring operational limitations associated with existing emergency-response aviation systems. Wildfire environments are characterized by high temperatures, rapidly changing wind conditions, reduced visibility from smoke and ash, difficult terrain, and limited communication reliability. These conditions create major challenges for both manned aircraft and ground crews. Additionally, emergency-response aircraft often require prepared infrastructure, extensive logistics coordination, or

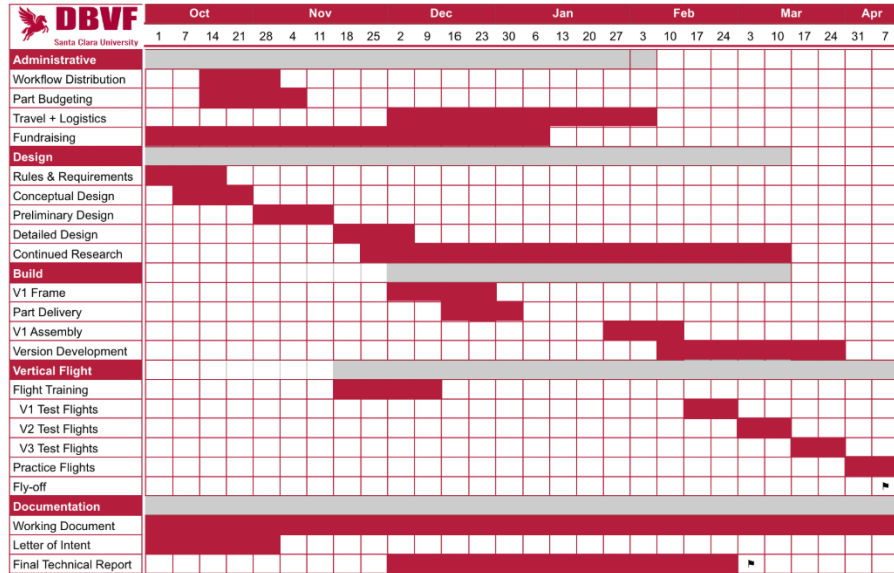


Figure 1.1: Gantt Chart

sufficient visibility conditions before deployment, limiting their usefulness during the earliest stages of a wildfire incident.

1.3.1 Literature Review and Resource Collection

D. Häusermann et al., “FireDrone: Multi-Environment Thermally Agnostic Aerial Robot,” *Advanced Intelligent Systems*, Jun. 2023.

Summary: This article presents FireDrone, a UAV designed to operate in extreme temperatures encountered in firefighting and polar environments. The drone uses a lightweight polyimide aerogel shell for thermal insulation and an internal liquid carbon dioxide cooling system to protect onboard electronics. Testing in both hot and cold conditions demonstrated the system’s ability to maintain safe operating temperatures for the drone’s components.

Kuldoshbay Ayazov et al., “Forest Fire Detection and Notification Method Based on AI and IoT Approaches,” vol. 15, no. 2, Jan. 2023.

Summary: This article proposes an early wildfire detection system combining IoT smoke sensors with the YOLOv5 deep-learning model. A Raspberry Pi with an MQ-2 smoke sensor and camera detects potential fires and sends images for AI-based verification. Once confirmed, alerts and visual data are automatically transmitted to fire authorities. Experimental results showed improved detection accuracy and reduced false alarms compared to standalone sensor systems.

Minh Long Hoang, “Smart Drone Surveillance System Based on AI and on IoT Communication in Case of Intrusion and Fire Accident,” *Drones*, vol. 7, no. 12, Dec. 2023.

Summary: This article presents a smart drone surveillance system integrating AI computer vision and IoT sensors for intrusion and fire detection. PIR and flame sensors activate a Tello drone equipped with YOLOv8 and Cascade Classifier models to identify people, suspicious objects, and fires. The drone uses PID flight control for stable tracking and streams live video to safety personnel during emergency events.

S. Weeks et al., “Additive Manufacturing Drone Design Challenge,” *2020 Intermountain Engineering, Technology and Computing (IETC)*, Oct. 2020.

Summary: This paper describes the design and testing of a 3D-printed racing drone developed for the ASME IAM3D competition. The project emphasized lightweight ABS and TPU structures, modular arm assemblies, rapid prototyping, and integrated payload capabilities. Flight and crash testing demonstrated the drone’s durability and overall performance.

T. S. Goruganthu et al., “Building an AI-Based Surveillance Drone Cloud Platform,” *IEEE International Workshop on Future Intelligent Technologies for Young Researchers (FIITYR)*, Jul. 2024.

Summary: This article discusses a cloud-based drone surveillance platform integrating AI analytics, mission planning, and real-time communication. The system supports automated alerts, threat detection, telemetry tracking, and live video streaming. Interactive mapping and simulation tools improve mission coordination, providing an intelligent and scalable solution for drone-based situational awareness.

1.3.2 Benchmarking

The team also studied commercially available UAV platforms commonly used in public safety and environmental monitoring operations shown in Table 1 below. Systems such as the DJI Matrice series and Skydio autonomous platforms demonstrated the growing capabilities of UAV systems in aerial reconnaissance, obstacle avoidance, autonomous navigation, and thermal imaging. However, these systems also revealed limitations in modularity, wildfire-specific survivability, payload adaptability, and operation in degraded environmental conditions. This benchmarking process helped identify areas where Pegasus could emphasize operational flexibility and mission-oriented design [19],[20],[21].

Additional research focused on multirotor aircraft configurations and rotorcraft flight dynamics. Multiple UAV architectures were evaluated including quadcopters, hexacopters, hybrid lift-and-cruise VTOL systems, and tail-sitter designs. The team analyzed how each configuration influenced hover efficiency, disk loading, thrust margin, controllability, integration complexity, and operational risk. While hybrid and tail-sitter configurations offered greater theoretical range, they introduced significant transition-flight complexity and integration challenges. The hexacopter configuration provided the best balance between redundancy, stable hover capability, thrust margin, and operational simplicity for the intended wildfire-response mission [28],[29].

Table 1: Examples of Other Aircraft

Parameter	DJI M350	Skydio X10	Dragonfly Commander 3XL
Footprint (prop-to-prop / unfolded)	810 × 670 × 430 mm	790 × 650 × 145 mm	64" × 63" × 12"
Max takeoff weight	9.2 kg (20.3 lb)	2.49 kg (5.49 lb)	55 lb (24.9 kg)
Max payload	2.7 kg (6.0 lb)	0.34 kg (340 g)	22 lb (10 kg)
Max flight time (no payload)	55 min	40 min	50 min
Max horizontal speed	23 m/s ≈ 83 km/h (51 mph)	45 mph (≈ 72 km/h)	45 mph (≈ 72 km/h)
Max range/link distance	20 km (O3 Enterprise link, FCC)	12 km (7.5 mi) line-of-sight with Connect SL link	24 mi (≈ 39 km)
Autonomy	GNSS/RTK waypoint missions, return-to-home, omnidirectional obstacle sensing (pilot still in loop)	High autonomy: 360° vision, AI obstacle avoidance, automated path planning & tracking, minimal pilot input	GNSS waypoint missions, basic automation; pilot-in-the-loop for obstacle avoidance

Propulsion-system research included detailed analysis of brushless motor performance, propeller efficiency, ESC compatibility, battery discharge characteristics, and hover power requirements. Manufacturer test data and analytical performance models were used to estimate hover thrust, endurance, throttle margins, and total system power consumption. This research directly informed propulsion sizing methodology and allowed the team to establish conservative operational margins necessary for turbulent wildfire environments [30],[31].

Structural and manufacturing research also played a major role in the development process. The team investigated lightweight structural materials, topology optimization techniques, vibration behavior in multirotor airframes, and additive manufacturing methods suitable for rapid prototyping and modular design iteration. Studies on rotor excitation frequencies and structural resonance informed the modal-analysis process used later during validation[32].

Another important research area involved modular payload systems and adaptive mission architecture. The team examined how emergency-response UAVs could transition between surveillance, sensor deployment, and lightweight supply delivery roles without requiring complete aircraft redesign. This research ultimately led to the development of Pegasus’s integrated bottom-plate payload architecture and magnetic linear-actuator release system, which emphasized simplicity, maintainability, and rapid mission reconfiguration [33],[34].

Overall, the background research phase established the technical and operational foundation

for the aircraft. Rather than pursuing maximum aerodynamic efficiency or aggressive performance envelopes, the research consistently demonstrated that wildfire-response operations demand reliability, stability, maintainability, rapid deployment capability, and operational adaptability under uncertain environmental conditions.

1.4 Customer Needs Assessment

A comprehensive customer needs assessment was conducted to ensure the aircraft addressed realistic operational challenges encountered during wildfire-response missions. Stakeholder input was gathered below in Table 2 through discussions with firefighters, drone operators, public safety personnel, and faculty advisors with expertise in controls systems, aerospace engineering, and autonomous flight technologies. The purpose of these interviews was to identify the most critical operational capabilities required for a practical wildfire-response UAV platform which were derived in Table 3.

Table 2: List of Stakeholders and Customers

Stakeholder/ Customer	Description of Stakeholder/Customer	Affected Design Considerations
Firefighter Lucas Learner	Firefighter at Santa Clara Fire station 9 who has seen drones used in emergency operations.	Heat resistance is an important factor to include as these drones will have close hand encounters with fires. Stability is crucial when carrying payloads like cameras and sensors.
Professor Hiremath	Professor at Santa Clara University who has in-depth knowledge on sensors and controls.	Obstacle detection and navigation systems are important factors to ensure they are working efficiently and accurately.
Officer Clark	Officer at the Maui Police Department who has first hand experience flying and operating drones during missions.	Helped guide our design to autonomy, as this will decrease response time and reduce staffing burden.

One of the most consistent themes identified during stakeholder interviews was the importance of rapid deployment capability. Firefighters emphasized that the value of an aerial reconnaissance platform decreases significantly if lengthy setup procedures delay launch during the initial stages of a wildfire. As a result, the aircraft needed to support rapid assembly, simple operation, minimal field infrastructure requirements, and autonomous mission execution immediately after arrival on-site.

Stakeholders also emphasized the importance of stable hover performance and reliable controllability in turbulent environments. Wildfire conditions often produce unpredictable updrafts, wind shear, and rapidly changing thermal conditions that can destabilize lightweight aircraft. Because many mission tasks involve sustained hover for observation, payload deployment,

or environmental monitoring, the aircraft required significant thrust margin, conservative control tuning, and high resistance to disturbances. This directly influenced the decision to adopt a hexacopter configuration with substantial hover authority and redundancy.

Another major customer need involved modularity and mission adaptability. Emergency responders indicated that mission requirements may change rapidly throughout the lifecycle of a wildfire incident. During early detection phases, thermal cameras and mapping sensors may be prioritized, while later stages may require communication relay systems, marker deployment, or lightweight supply transport. Therefore, stakeholders strongly preferred a platform capable of rapidly switching payload systems without requiring extensive reconfiguration or structural redesign.

Communication reliability and autonomous operation were also identified as critical requirements. Public safety operators emphasized that maintaining telemetry, navigation, and video-streaming reliability is difficult in mountainous terrain, smoke-obstructed environments, and infrastructure-damaged regions. Therefore, the aircraft required reliable telemetry systems, robust flight-control integration, redundant sensing capability, and fail-safe operating modes to ensure safe operation even in degraded communication conditions.

Ease of maintenance and field-serviceability also emerged as important operational considerations. Firefighters and operators preferred systems that could be repaired, inspected, or reconfigured quickly in the field using simple tools. This directly influenced the aircraft's modular airframe architecture, externally accessible payload systems, replaceable landing gear configurations, and simplified mechanical integration strategy.

Finally, safety and operational robustness were consistently emphasized throughout the customer needs assessment process. Stakeholders prioritized predictable behavior, conservative operating margins, redundant systems, and stable autonomous flight over aggressive maneuverability or maximum speed. This reinforced the project's overarching design philosophy that reliability and mission availability are more important in emergency-response systems than achieving peak theoretical performance.

1.5 Identifying Needs to Address

The customer needs assessment, operational research, and subsystem trade studies identified several key engineering challenges that guided the Pegasus design process.

The first major requirement was stable autonomous operation in turbulent wildfire environments. Wildfires create strong thermal updrafts, erratic wind conditions, and poor visibility that can reduce aircraft controllability. To address this, Pegasus required high thrust margin, stable hover capability, conservative flight-control tuning, and propulsion redundancy. This directly influenced the selection of the hexacopter configuration, which offered improved stability, controllability, and redundancy compared to alternative concepts.

Table 3: Customer Needs Derived from Interviews

Customer Need	Description
Drone Stability	Drone stability is crucial because wildfire environments are turbulent and unpredictable. The drone must remain stable when encountering wind, rain, and other harsh conditions to ensure that any payloads being carried remain stable throughout the mission.
Accuracy	Accuracy is an important factor because the drone will be autonomous, so it must be able to guide itself to the exact coordinates given.
Payload Capacity	Missions vary widely and situationally, so payload versatility is very important. The drone has to be able to carry a wide variety of payloads depending on mission requirements.
Efficiency	Efficiency is essential to ensure the drone can complete long-duration wildfire missions without excessive battery consumption. Because wildfire response often involves extended search, monitoring, or supply-delivery tasks, the system must maximize flight time through efficient propulsion, smart energy management, and lightweight design. Higher efficiency directly increases mission capability and reduces operational downtime between battery swaps.
Communications	Reliable communications are critical for transmitting live video, sensor data, and telemetry between the drone and the ground control station. In wildfire zones where smoke, heat, terrain, and infrastructure damage can interfere with signals, the drone must maintain a stable, long-range communication link. This ensures operators have real-time situational awareness and can issue commands or receive alerts without interruption.
Safety	Safety is the primary concern when flying a drone, and extensive measures will be taken to ensure the drone operates safely with no risk to surrounding areas.

Rapid mission adaptability was another critical need. Because wildfire-response missions can change quickly, the aircraft needed to support multiple operational roles without extensive redesign. Pegasus addressed this through a modular payload architecture capable of supporting sensors, cameras, communication relays, and payload-delivery systems. A magnetic linear-actuator mechanism was developed to provide reliable payload pickup and release while minimizing mechanical complexity.

Terrain adaptability also played an important role in the design process. Wildfire environments often lack ideal launch and landing conditions, requiring the aircraft to operate on uneven or soft terrain. To improve ground stability, Pegasus incorporated adaptive landing gear capable of converting into a ski-style configuration.

Structural efficiency and vibration resistance were also key requirements. Since endurance and hover efficiency are highly sensitive to weight, the airframe needed to maintain adequate stiffness and strength while minimizing mass. At the same time, the structure needed to

avoid resonance with rotor excitation frequencies. These requirements led to the use of Finite Element Analysis (FEA), topology optimization, modal analysis, and lightweight aluminum structures.

Endurance and propulsion efficiency represented another major challenge. Wildfire-response missions require sustained hover operation, payload transport, and autonomous navigation while maintaining safe battery reserves. Propulsion sizing therefore prioritized efficient motor-propeller combinations, conservative hover throttle margins, and sufficient thrust authority for maneuverability and gust rejection.

Finally, Pegasus needed to balance DBVF competition requirements with realistic wildfire-response functionality. Rather than optimizing solely for competition scoring, the aircraft was designed around operational reliability, maintainability, modularity, and deployment practicality. This systems-level approach shaped Pegasus into a platform intended not only for competition success, but also for demonstrating the potential of autonomous eVTOL in real-world emergency-response operations.

2 Conceptual Design

The conceptual design phase of the Pegasus wildfire-response eVTOL aircraft established the foundational engineering framework for the entire project. During this phase, mission objectives, operational constraints, customer needs, and competition requirements were translated into quantifiable engineering requirements and subsystem-level design decisions. The primary goal of the conceptual design process was to develop an aircraft architecture capable of performing reliable autonomous wildfire-response operations while maintaining simplicity, stability, modularity, and manufacturability.

The conceptual design process followed a systems-engineering methodology beginning with mission definition and requirement identification before progressing into subsystem decomposition, aircraft configuration trade studies, propulsion sizing, and architecture integration. Multiple aircraft concepts and subsystem configurations were evaluated using analytical modeling, customer needs assessments, and engineering tradeoff analyses. The resulting design prioritized operational reliability and mission adaptability over maximum aerodynamic efficiency or aggressive flight performance.

Pegasus was ultimately developed as a battery-powered autonomous hexacopter optimized for stable hover, modular payload integration, autonomous waypoint navigation, and lightweight payload delivery in wildfire-response environments. The aircraft architecture integrates distributed electric propulsion, centralized flight-control systems, modular structural interfaces, and field-serviceable subsystem integration to support rapid deployment and adaptable mission execution.

2.1 System Requirements

The system requirements for Pegasus were derived from three primary sources: the Vertical Flight Society Design-Build-Vertical Flight (DBVF) competition requirements, stakeholder/-customer needs assessments, and operational constraints associated with wildfire-response missions. Together, these requirements established the performance targets and engineering constraints that guided all major conceptual design decisions.

The DBVF competition imposed several critical top-level requirements that directly influenced aircraft configuration and subsystem selection. The aircraft was required to remain below a maximum takeoff weight (MTOW) of 25 lb while utilizing commercially available electric propulsion systems powered by LiPo battery packs limited to 6S operation. Additionally, the aircraft was required to use separate power systems for propulsion and avionics, include external propulsion disconnect methods, and incorporate communications-based kill-switch functionality for safety compliance.

Beyond competition requirements, wildfire-response operations imposed additional mission-specific design constraints. The aircraft needed to operate reliably in turbulent and thermally unstable environments while maintaining stable hover performance for observation and payload deployment tasks. Wildfire environments also required operation from unprepared terrain

without runway infrastructure, emphasizing the importance of vertical takeoff capability, adaptable landing systems, and compact deployment logistics. Additionally, because wildfire response missions are highly time-sensitive, the aircraft required rapid deployment capability and minimal setup complexity.

Customer needs assessments further refined the system requirements by identifying operational priorities from firefighters, drone operators, and technical advisors. Stakeholders consistently emphasized the importance of the following characteristics:

- Stable autonomous hover capability
- Reliable communication systems
- Payload modularity
- Rapid field deployment
- Flight stability in gusting wind conditions
- Ease of maintenance and repair
- Operational reliability over aggressive maneuverability
- Adaptability to multiple mission profiles

These stakeholder-driven priorities strongly influenced the selection of the aircraft configuration and subsystem architecture.

To translate these operational needs into measurable engineering objectives, several derived performance requirements were established during the conceptual design phase. One of the most important derived requirements was maintaining a thrust-to-weight ratio greater than 2.0 at the design operating weight. This requirement ensured that the aircraft retained sufficient control authority for gust rejection, stable hover operation, and payload maneuvering without approaching throttle saturation.

A hover endurance requirement was also established based on DBVF mission demands and projected wildfire-response operational needs. Simulations and propulsion analyses targeted a minimum mission endurance exceeding ten minutes while maintaining a conservative battery reserve margin. Hover throttle percentage was similarly constrained to remain below approximately 85% throttle during nominal operation to preserve maneuverability and improve system reliability under dynamic environmental conditions.

Structural requirements were additionally established to ensure sufficient airframe durability and vibration resistance. These included:

- Structural factor of safety greater than or equal to 2.0
- Maximum arm deflection under load below 1 mm

- Adequate frequency separation between hover excitation frequencies and structural natural frequencies

The final set of conceptual system requirements therefore represented a balance between:

1. Competition compliance,
2. Realistic wildfire-response operational needs,
3. Stakeholder expectations,
4. Manufacturability constraints,
5. Safety and reliability considerations.

The resulting design philosophy emphasized operational robustness, conservative control margins, modularity, and mission adaptability as the primary drivers of system-level design decisions.

2.2 Subsystem Breakdown

Following the establishment of system-level requirements, the aircraft was decomposed into several major subsystems to simplify design development, analytical modeling, and integration planning. Breaking the aircraft into modular engineering subsystems allowed the team to independently analyze performance tradeoffs while ensuring compatibility with overall system objectives.

Figure 2.1 below shows the breakdown of the Pegasus aircraft subsystems.

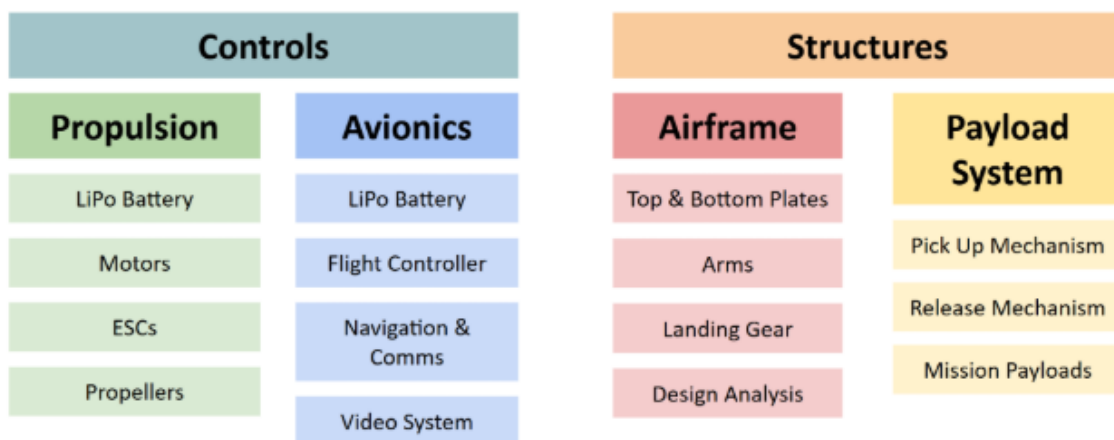


Figure 2.1: Subsystem breakdown

Each subsystem was designed around the operational needs of wildfire-response missions while supporting integration into the overall aircraft architecture.

2.2.1 Airframe and Structural System

The airframe subsystem provides the primary structural backbone of the aircraft and supports all propulsion, avionics, payload, and landing components. Structural design focused heavily on stiffness-to-weight optimization, modularity, and manufacturability.

The conceptual airframe, shown in Figures 2.2 and 2.3 below, adopted a hexacopter configuration with six aluminum arms connected to a reinforced central plate assembly. Aluminum was selected due to its favorable balance between stiffness, manufacturability, weight, and thermal durability. The central plate architecture was specifically designed to support modular payload integration while maintaining efficient structural load paths between suspended payloads and the primary frame.

The bottom plate became one of the most important structural features in the aircraft because it simultaneously supported payload attachment, landing gear integration, and actuator mounting systems. This integrated mounting strategy simplified maintenance while lowering the aircraft center of gravity to improve hover stability in turbulent conditions.

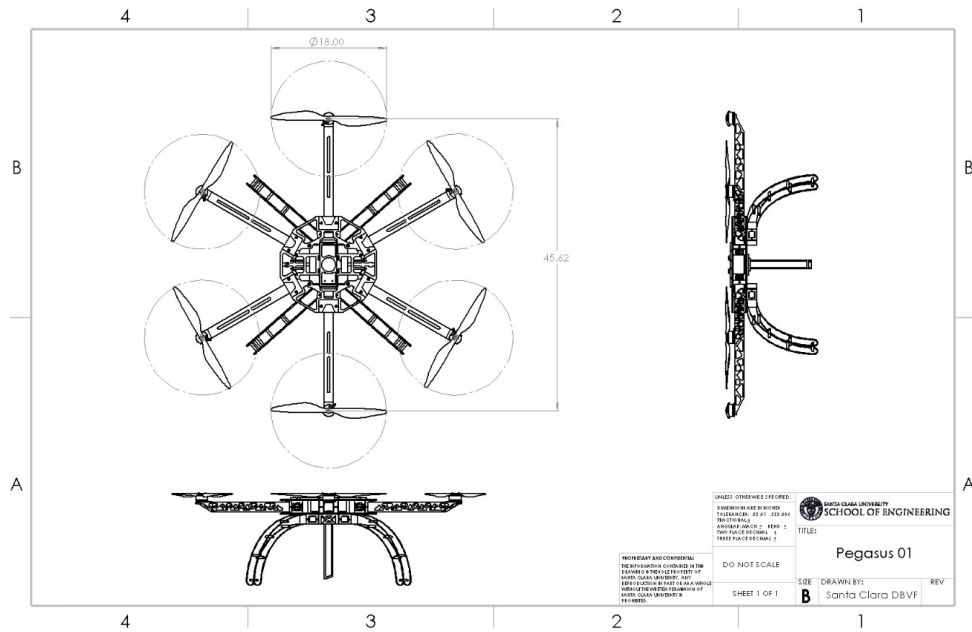


Figure 2.2: Orthogonal View

2.2.2 Propulsion System

The propulsion subsystem was responsible for generating sufficient thrust, hover authority, and endurance capability for wildfire-response operations. During conceptual design, multiple motor and aircraft configurations were evaluated in Tables 4 and 5 below, using thrust, efficiency, mass, reliability, and integration complexity as tradeoff criteria.

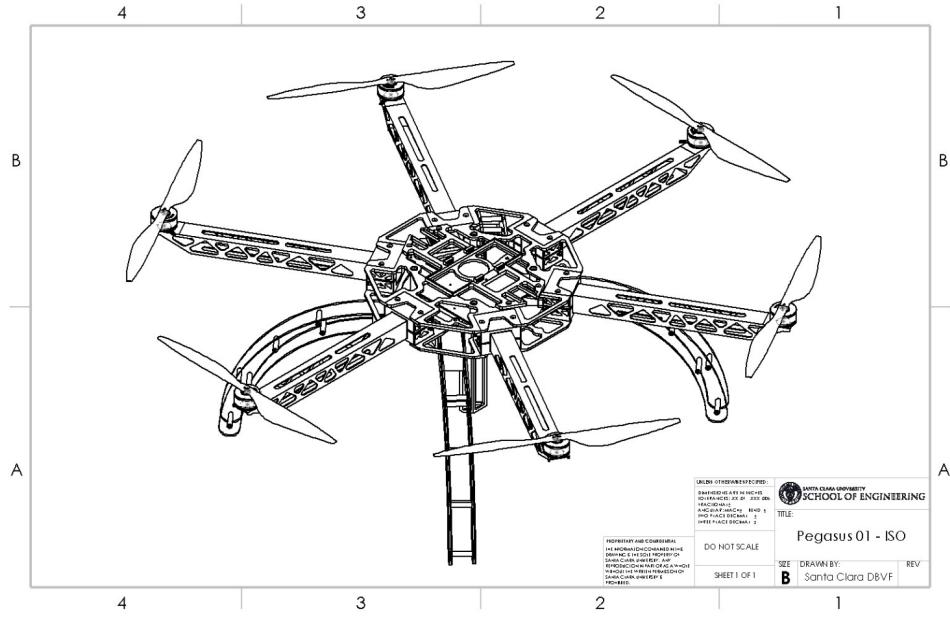


Figure 2.3: Isometric view

The final propulsion configuration selected six T-Motor MN4110 KV340 brushless motors paired with 18×6.1 carbon-fiber propellers operating on a 6S LiPo battery system. This configuration provided sufficient thrust margin while maintaining reasonable hover efficiency and controllability.

The hexacopter architecture offered several major conceptual advantages:

- Improved stability during hover
- Greater control authority in wind
- Lower disk loading
- Increased redundancy during motor failure
- Better payload handling characteristics

These advantages outweighed the slightly improved theoretical endurance offered by some quadcopter concepts.

The propulsion subsystem was also conceptually designed around conservative operating margins. Hover throttle operation near 50% ensured sufficient reserve thrust for gust rejection, maneuvering, and emergency stabilization. The propulsion performance breakdown is shown below in Table 6.

Table 4: Concept Selection Matrix

	Multiplier	Fixed Wing Hybrid	Quadcopter	Hexacopter	Tail Sitter
Maneuverability	4	2 = 8	4 = 16	4 = 16	3 = 12
Simplicity	2	2 = 4	5 = 10	4 = 8	2 = 4
Reliability/Stability	4	2 = 8	3 = 12	4 = 16	2 = 8
Payload Capacity	3	3 = 9	3 = 9	4 = 12	3 = 9
Totals		= 29	= 47	= 52	= 33

Table 5: Concept Compare Output Summary

Concept	Endurance Prop (min)	Disk Loading N/m ²	Thrust Margin Ratio	Range (km)
Hexacopter	11.5	87.1	2.52	6.69
Hybrid VTOL (lift+cruise)	10.8	138.9	2	20.99
Quadcopter	13.1	104.5	2.1	7.66
Tail-sitter (twin-prop)	10.8	188.4	1.17	32.65

Table 6: Propulsion Performance Summary

Parameter	Symbol	Value	Units	Source (Test/Manufacturer/CAD)
All-up weight	W	18.08	lb	Calculated
Number of rotors	n	6		CAD
Prop diameter	D	18	in	Manufacturer
Prop disk area (per rotor)	$A = \pi(D/2)^2$	0.164	m ²	Calculated
Total disk area	$A_{\text{total}} = nA$	0.985	m ²	Calculated
Hover thrust per motor	$T_{\text{hover}} = W/n$	3.01	lbf	Calculated
Max thrust per motor	T_{max}	8.01	lbf	Manufacturer
Total max thrust	$T_{\text{total}} = nT_{\text{max}}$	48.1	lbf	Calculated
Thrust-to-weight ratio	$T/W = T_{\text{total}}/W$	2.66		Calculated
Hover RPM	$\text{RPM}_{\text{hover}}$	3450	rpm	Manufacturer
Hover excitation frequency	$f_{\text{hover}} = \text{RPM}/60$	57.5	Hz	Calculated
Hover power per motor	$P_{\text{hover motor}}$	137.3	W	Manufacturer
Total hover power	$P_{\text{hover total}} = nP_{\text{hover motor}}$	823.8	W	Calculated
Mission power factor	k	1.15		
Mission power	$P_{\text{mission}} = kP_{\text{hover total}}$	947.37	W	Calculated

2.2.3 Avionics and Flight-Control System

The avionics subsystem served as the centralized control and navigation architecture for the aircraft. The conceptual design prioritized autonomous flight capability, sensor redundancy, communication reliability, and safe manual override operation.

The system architecture centered around the Cube Orange+ flight controller integrated with GPS navigation systems, telemetry radios, video transmission hardware, and RC communication links. The flight controller was responsible for:

- Stabilization control
- Autonomous waypoint navigation
- Sensor fusion
- Failsafe management
- Telemetry integration
- Mission execution logic

The conceptual design also separated propulsion and avionics power systems to improve operational safety and reduce electrical noise interference.

Communication reliability was treated as a major conceptual design priority because wildfire environments often degrade telemetry quality through terrain obstruction and smoke interference. As a result, the avionics subsystem emphasized redundant sensing, robust telemetry integration, and conservative autonomous control behavior.

2.2.4 Payload and Mission System

The payload subsystem was designed to maximize mission adaptability and support rapid reconfiguration between surveillance, sensor deployment, and lightweight delivery missions.

A key innovation during conceptual development was the integrated bottom-mounted payload architecture which can be seen in Figures 2.4 and 2.5 below. This subsystem combined:

- Modular payload mounting
- Linear actuator deployment
- Magnetic payload pickup
- Passive payload release geometry

The system allowed payloads to be attached and released without complex gripping mechanisms or high-power active release systems. Instead, payloads utilized embedded magnets that interfaced with the actuator-mounted magnetic pickup system. An angled release cover mechanically separated the payload during actuator retraction to ensure repeatable and reliable deployment.

This subsystem architecture provided several operational advantages:

- Reduced mechanical complexity

- Improved reliability
- Rapid payload swapping
- Simplified maintenance
- Lower structural loading
- Shorter force transmission paths

Most importantly, it allowed the aircraft to adapt quickly between wildfire mission phases without requiring redesign of the primary airframe structure.

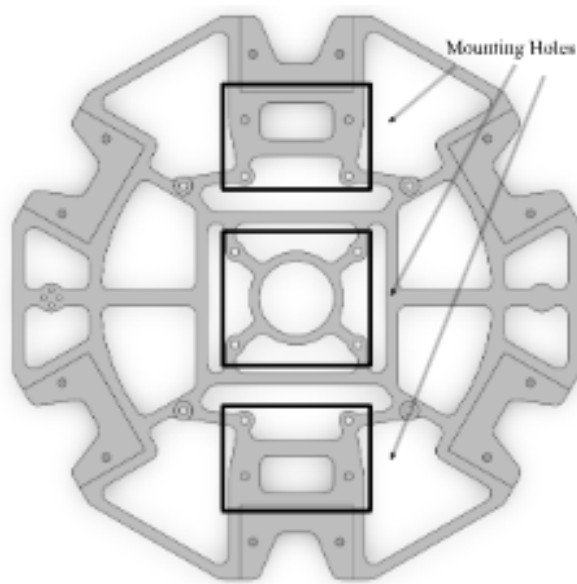


Figure 2.4: Bottom Plate Architecture

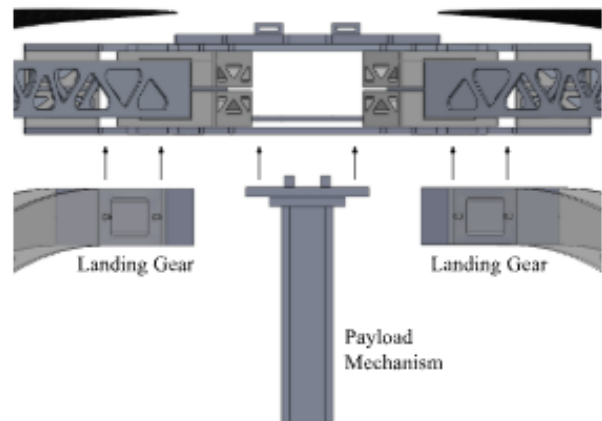


Figure 2.5: Modular Payload Mechanism & Landing Gear

2.2.5 Landing Gear System

The landing gear subsystem was conceptually designed around terrain adaptability and operational survivability rather than minimum aerodynamic drag.

Wildfire operations rarely occur on prepared surfaces, requiring the aircraft to operate from ash-covered terrain, uneven clearings, rocky surfaces, and soft soil conditions. To address these challenges, Pegasus incorporated modular adaptive landing gear capable of conversion into ski-style configurations for improved ground pressure distribution and stability.

The landing gear was intentionally mounted directly to the reinforced bottom plate rather than propulsion arms to minimize bending loads transmitted into sensitive propulsion structures during hard landings. Additionally, lightweight replaceable components improved

field-serviceability and reduced operational downtime.

The conceptual design therefore treated landing gear not as a secondary support structure, but as a critical operational subsystem directly influencing deployment capability and mission survivability.

2.3 System Architecture

The final conceptual system architecture integrated all major subsystems into a centralized multirotor platform optimized for wildfire-response operations. The architecture was designed to balance stability, modularity, controllability, and maintainability while minimizing integration complexity and operational risk.

At the highest level, Pegasus utilized a distributed electric propulsion architecture centered around a reinforced modular airframe. Six independently controlled brushless motors generated thrust through large-diameter propellers optimized for hover efficiency and stable low-speed flight. The distributed propulsion layout improved controllability and provided redundancy in the event of partial propulsion failure.

The avionics architecture was centralized around the flight controller, which interfaced with:

- GPS navigation systems
- Telemetry radios
- RC communication hardware
- Video transmission systems
- Power distribution modules
- Payload-control systems

The flight controller acted as the central processing node responsible for autonomous navigation, stabilization, mission execution, and failsafe behavior.

The aircraft architecture also separated avionics and propulsion power systems to improve reliability and safety. This separation reduced the likelihood of avionics brownout conditions caused by propulsion power fluctuations and minimized electromagnetic interference affecting sensitive navigation hardware.

Structurally, the architecture emphasized centralized mass distribution and low center-of-gravity placement. Batteries, avionics hardware, and payload systems were intentionally positioned near the geometric center of the aircraft to reduce trim penalties and improve hover controllability.

The integrated bottom-plate architecture served as the primary interface between the payload subsystem, landing gear subsystem, and central airframe. This design simplified maintenance while improving payload adaptability and structural efficiency.

The resulting conceptual system architecture reflected the overall design philosophy of the project:

- Conservative flight stability
- Operational robustness
- Modular mission adaptability
- Ease of maintenance
- Reliability in degraded environments
- Practical wildfire-response capability

Rather than maximizing speed or aerodynamic efficiency, the Pegasus architecture prioritized dependable mission execution in unpredictable operating conditions, making it well-suited for wildfire reconnaissance and emergency-response operations.

3 Controls

The controls subsystem of Pegasus was developed to provide stable autonomous flight, reliable hover performance, mission execution capability, and safe operation in highly dynamic wildfire-response environments. Unlike conventional fixed-wing aircraft, multirotor eVTOL systems require continuous closed-loop stabilization because all six propulsion units simultaneously contribute to lift generation, attitude control, and translational maneuvering. As a result, the controls architecture became one of the most critical subsystems in the overall aircraft design.

Wildfire-response missions impose particularly demanding control-system requirements due to turbulent atmospheric conditions, thermal updrafts, wind shear, uneven terrain, and degraded visibility environments. The aircraft therefore required a flight-control system capable of maintaining stable hover performance, precise attitude regulation, autonomous waypoint navigation, and reliable failsafe operation while minimizing pilot workload and maintaining mission reliability.

The controls subsystem was conceptually designed around four major operational objectives:

1. Stable multirotor attitude control
2. Autonomous navigation capability
3. Reliable communication and telemetry integration
4. Safe fault-tolerant operation

To satisfy these objectives, Pegasus adopted a layered control architecture centered around a dedicated autopilot flight controller integrated with GPS navigation systems, telemetry communication hardware, and electronic speed controllers (ESCs) which can be seen in the wiring diagram below in Figure 3.1. This architecture allowed the aircraft to transition between manual control, assisted flight modes, and fully autonomous mission execution while maintaining stable closed-loop flight behavior.

Because wildfire environments are highly unpredictable, the design philosophy for the controls subsystem emphasized operational reliability over aggressive maneuverability or high-speed flight performance. The aircraft was intentionally optimized for controlled hover, low-speed translation, payload stability, and autonomous mission consistency rather than aerobatic performance or rapid maneuvering capability.

Additionally, the controls subsystem had to integrate seamlessly with the aircraft's propulsion, payload, and avionics systems. The flight controller therefore served as the central coordination node responsible for:

- Attitude stabilization
- Sensor fusion

- Autonomous mission logic
- Navigation processing
- Failsafe management
- Payload actuation commands
- Telemetry and communication handling

The controls architecture ultimately enabled Pegasus to operate as a stable, modular, and autonomous wildfire-response platform capable of supporting reconnaissance, sensor deployment, and lightweight payload delivery missions under uncertain environmental conditions.

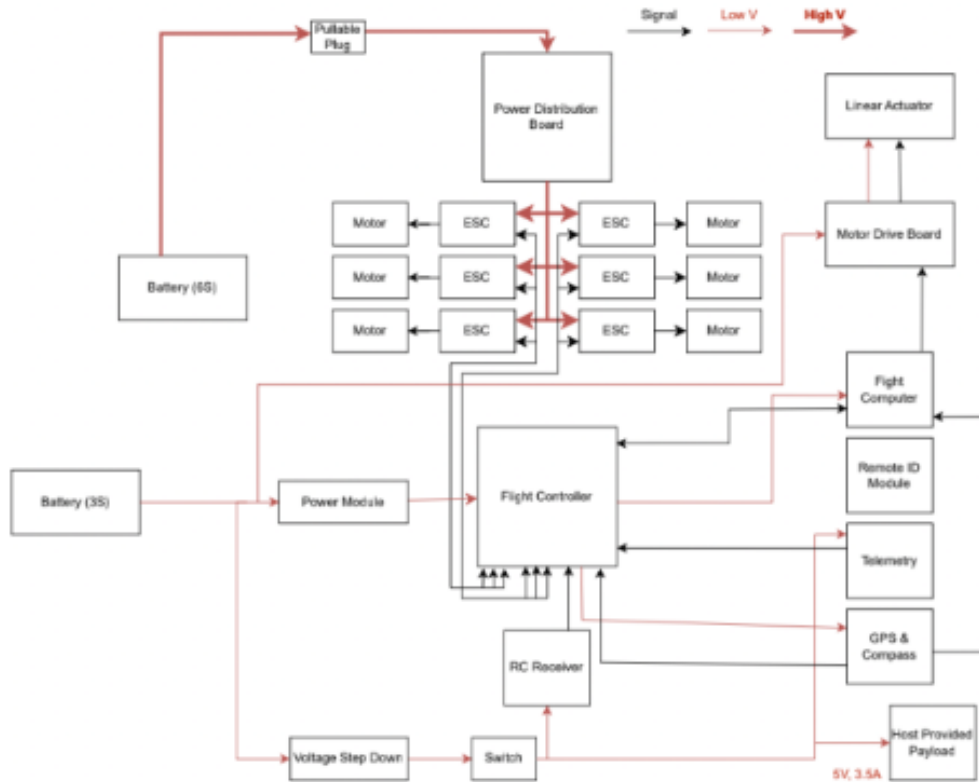


Figure 3.1: Wiring Diagram

3.1 Flight Controller Selection

The selection of the flight controller represented one of the most important design decisions within the controls subsystem because it directly influenced aircraft stability, autonomous capability, sensor integration, telemetry reliability, and overall mission safety. The flight controller functions as the “brain” of the aircraft, continuously processing sensor inputs and generating real-time motor control outputs to stabilize the multirotor system during all

phases of operation.

For Pegasus, the flight controller needed to satisfy several major operational and technical requirements:

- Reliable autonomous flight capability
- Stable multirotor hover control
- Compatibility with hexacopter configurations
- GPS waypoint navigation support
- Robust telemetry integration
- Expandability for future sensor integration
- Redundant sensing capability
- Compatibility with established autopilot ecosystems
- Strong community documentation and support
- High reliability under outdoor operating conditions

Because wildfire-response operations demand reliable autonomous behavior and robust failsafe handling, the team prioritized mature and widely validated flight-control hardware rather than experimental or proprietary systems. Several candidate flight controllers were initially considered during subsystem trade studies, including:

- Pixhawk-series controllers
- CubePilot systems
- DJI integrated flight systems
- Matek flight controllers
- Custom embedded solutions

3.1.1 Requirements and Resource Limits (hard constraints):

- Thrust-to-weight ratio $\geq 1.5:1$ at maximum gross weight.
- Maximum takeoff weight ≤ 8 kg, excluding payload.
- Power architecture: 22.2 V nominal battery system, corresponding to a 6S LiPo configuration.
- Endurance target: ≥ 60 minutes at the system level; propulsion must be efficient enough to support this requirement.

- Wind stability: maintain hover and control in gusts up to approximately 25 mph.
- Budget for motors and propellers: $\leq$ \$800 total.
- Motor, propeller, and ESC mass budget: $\leq$ 2.5 kg combined, if possible.

Identify Alternatives:

Table 7: Alternative Motors

Option	Description/Configuration	Key Specifications	Approximate Cost (Total)
A – 6× T-Motor MN4110 KV340	6× brushless MN3510-15 360 KV with 13–15 in carbon propellers; efficient, low-KV motor suitable for endurance and payload lift.	Max thrust per motor = 2.2 kg with 84.36 W power; 100% efficiency; mass = 118 g each.	\$640
B – 6× T-Motor MN3110 470 KV Hexacopter Setup	High-efficiency low-KV motors for redundancy; greater overall thrust but heavier.	Max thrust per motor = 1.2 kg with 118.40 W power; 100% efficiency; mass = 98 g each.	\$720
C – 6× iFlight 2806.5 FPV Cinelifter Motor	Designed for a 7–8 in quad frame. High thrust and responsive performance.	Mass = 53.3 g each; for a smaller configuration, not compatible with our design.	\$750
D – 8× SunnySky X2814 900 KV Motors	Offers a balance of power and efficiency. Guarantees durability and reliability. Ideal for aircraft weights between 1200 g and 1400 g.	Max continuous power = 750 W; weight = 106 g each.	\$790
E – Hybrid Gas-Electric System	Gas generator powering small electric motors for extended range.	>15 kg	\$900

Decision Making Drivers:

Table 8: Decision Making Drivers

Driver	Description / How Evaluated	Weight (%)
Thrust / Performance	Meets 1.5:1 thrust-to-weight ratio with control margin. Static thrust and RPM vs. current were considered.	30%
Efficiency	Current draw and propeller efficiency were evaluated based on flight-time impact.	20%
Mass	Total motor assembly mass was evaluated relative to the design limit.	15%
Reliability	Mechanical robustness and thermal durability were considered.	15%
Cost	Must fit within the \$800 budget.	10%
Complexity	Integration and maintenance difficulty were considered.	10%
Total		100%

Scoring scale: integer 0–5, where 5 is best.

Scoring guidance:

- **Thrust:** 5 = exceeds thrust requirement by large margin; 3 = meets requirement; 1 = below requirement.

- **Efficiency:** 5 = clearly highest expected flight efficiency, meaning lower current at cruise; 3 = typical; 1 = poor.
- **Mass:** 5 = light; 3 = moderate; 1 = too heavy compared with the mass budget.
- **Reliability:** 5 = strong redundancy and robustness; 3 = typical; 1 = fragile or complex failure modes.
- **Cost:** 5 = well under budget; 3 = near budget; 1 = over budget.
- **Complexity:** 5 = simple off-the-shelf integration; 3 = moderate; 1 = very complex to tune or maintain.

Tradeoff scoring weighted:

Each option was scored from 0–5 for each decision driver using team consensus. The weighted contribution for each cell was calculated as:

$$\text{Weighted contribution} = \frac{\text{score}}{5} \times \text{driver weight} \quad (3.1)$$

The sum of all weighted contributions gives the total weighted percent, with a maximum possible score of 100.

Table 9: Tradeoff Scoring

Option	Thrust (30%)	Efficiency (20%)	Mass (15%)	Reliability (15%)	Cost (10%)	Development Complexity (10%)	Total
A	$(4/5) \times 30$ = 24	$(4/5) \times 20$ = 16	$(4/5) \times 15$ = 12	$(4/5) \times 15$ = 12	$(4/5) \times 10$ = 8	$(4/5) \times 10$ = 8	80
B	$(4/5) \times 30$ = 24	$(4/5) \times 20$ = 16	$(2/5) \times 15$ = 6	$(4/5) \times 15$ = 12	$(3/5) \times 10$ = 6	$(3/5) \times 10$ = 6	70
C	$(5/5) \times 30$ = 30	$(5/5) \times 20$ = 20	$(3/5) \times 15$ = 9	$(3/5) \times 15$ = 9	$(3/5) \times 10$ = 6	$(3/5) \times 10$ = 6	83
D	$(3/5) \times 30$ = 18	$(2/5) \times 20$ = 8	$(2/5) \times 15$ = 6	$(5/5) \times 15$ = 15	$(2/5) \times 10$ = 4	$(2/5) \times 10$ = 4	55
E	$(5/5) \times 30$ = 30	$(4/5) \times 20$ = 16	$(1/5) \times 15$ = 3	$(2/5) \times 15$ = 6	$(1/5) \times 10$ = 2	$(1/5) \times 10$ = 2	59

3.2 Conclusions from Analysis

Ranking Summary:

1. Option C (280 KV Large-Prop): 83
2. Option A (T-Motor MN4110 KV340): 80
3. Option B (MN3110 470 KV Hex): 70
4. Option E (Hybrid): 59
5. Option D (8× Coaxial): 55

Interpretation:

- The T-Motor MN4110 KV340 performs near the top overall. It provides an excellent balance of thrust, mass, efficiency, and reliability, while staying within cost and integration budgets.
- The 280 KV large-prop system offers slightly higher efficiency and thrust margin, but adds larger propellers and additional airframe stress considerations.
- The hexacopter option, Option B, adds redundancy but penalizes mass and endurance.
- The hybrid and coaxial options are either over budget or too complex for the DBVF timeline.

Conclusion:

- **Option A – T-Motor MN4110 KV340:** Selected because it offers a strong balance of thrust, efficiency, mass, cost, and ease of integration. It meets the system’s performance targets without excessive complexity.
- **Option C – 280 KV Long-Prop Setup:** Slightly better performance, including higher efficiency and higher thrust margin, but introduces larger propellers and airframe design or clearance issues. This option is worth keeping as a backup or future optimization.

Non-Contenders:

- **Option B – MN3110 470 KV Hex:** Although it provides redundancy and stability, it sacrifices endurance and increases mass and cost, which are critical factors for the 60-minute endurance target.
- **Option D – 8× 900 KV Coaxial:** Complexity and integration risk are high, while efficiency and endurance suffer.
- **Option E – Hybrid Gas-Electric:** Exceeds the budget, adds fuel, thermal, and maintenance complexity, and is beyond the time budget for this prototype stage.

Next Steps:

1. Procure the T-Motor MN4110 KV340 motors and ESCs for bench testing.
2. Perform static thrust versus current tests with 14–15 in propellers to validate efficiency.
3. Update system mass and power budgets after empirical data are collected.
4. If endurance falls short of the target, re-evaluate the larger propeller option, Option C.

Summary:

This analysis confirms that the T-Motor MN4110 KV340 is an optimal propulsion choice for the wildfire-response eVTOL prototype. It balances thrust, efficiency, and cost while remaining straightforward to implement and service, making it ideal for a competition-grade, field-deployable design.

3.3 Detailed Subsystem Description – Propulsion (T-Motor MN3510-15 KV360)

3.3.1 Summary of Motor Selection and Test Data

We evaluated the T-Motor MN3510 Navigator series, KV = 360 configuration (MN3510-KV360). T-Motor provides a set of bench test results for various propellers and voltages. The recommended paired propeller family includes 14×4.8 , 15×5 , and 16×5.4 carbon folding propellers. The MN3510 test table lists thrust, current, power, and efficiency for these propellers at 14.8 V and 22.2 V test conditions.

Key published propeller and motor data include:

- Motor physical dimensions: diameter = 40.1 mm $\times$ 28.5 mm; shaft diameter = 4 mm; motor mass > 97 g without cables and > 117 g with cables.
- Idle current at 10 V: 0.4 A for the KV360 configuration.
- Maximum continuous current specified for KV360: 15 A.
- Representative bench data for MN3510 KV360 at 22.2 V using a 15×5 carbon-fiber propeller:
 - At 65% throttle: current = 3.0 A, electrical power = 44.4 W, static thrust = 580 g.
 - At 100% throttle: current = 6.8 A, electrical power = 100.6 W, static thrust = 1000 g, or approximately 1.0 kg.

These values were read from the T-Motor test table and used as the basis for the propulsion analysis.

3.3.2 Propeller Selection

Based on the T-Motor matching guide and test table, the 15×5 carbon-fiber propeller is a recommended match for the MN3510-360KV motor when operating on a 6S, 22.2 V power system.

The 15×5 propeller was selected for the following reasons:

- It produces one of the highest thrust values per electrical power among the practical propeller sizes listed for 22.2 V operation.
- It provides a good thrust-to-weight tradeoff for a mid-size multirotor aircraft.
- Test data show that the 15×5 propeller at 22.2 V produces approximately 580 g of thrust at 44 W under 65% throttle and approximately 1000 g of thrust at 100 W under 100% throttle.
- It is practical for multirotor designs where propeller clearance and torque are constrained.

Therefore, the baseline propeller choice for the MN3510-360KV evaluation is the 15×5 carbon-fiber folding propeller.

3.3.3 Thrust Margin and Flight Time Estimates

To evaluate the feasibility and performance of the propulsion subsystem, a full thrust and power analysis was performed for a hexacopter configuration using six T-Motor MN4110 KV340 motors paired with the selected propeller type. A hexacopter architecture offers improved stability, redundancy in failure scenarios, and higher total available thrust compared to quadrotor systems, making it suitable for an eVTOL designed for wildfire-response operations.

Total Thrust Capacity

Based on T-Motor performance data for the MN3510-15 360 KV motor, the measured thrust per motor depends on the propeller chosen and the operating voltage. Assuming a representative 6S LiPo power system and a carbon-fiber propeller consistent with T-Motor testing, each motor can produce approximately:

- Maximum static thrust per motor: approximately 2.2 kgf.
- Safe continuous thrust per motor: approximately 0.660 kgf.

Using six motors, the vehicle's total thrust capacity becomes:

$$T_{\text{max,total}} = 6 \times 2.2 \text{ kgf} = 13.2 \text{ kgf} \quad (3.2)$$

$$T_{\text{continuous,total}} = 6 \times 0.660 \text{ kgf} = 3.96 \text{ kgf} \quad (3.3)$$

Thrust-to-Weight Ratio

A conservative maximum takeoff weight used in the consideration phase was 11.34 kg, including payload. This ensures stable takeoff, maneuvering capability, and adequate control authority in turbulent wildfire conditions.

$$TWR = \frac{T_{\text{max,total}}}{m_{\text{total}}} = \frac{13.2 \text{ kgf}}{11.34 \text{ kg}} = 1.16 \quad (3.4)$$

Further refinements, including realistic frame-weight reduction, battery sizing, and higher-efficiency propeller selection, will continue to increase the operational thrust-to-weight ratio toward the target range of 1.8–2.0.

Ongoing optimization will ensure sufficient thrust margin by adjusting:

- Propeller diameter and pitch.
- Motor KV and efficiency.
- Total system mass.
- Battery configuration.

Flight Time Estimate

Flight time is approximated using average current draw per motor during hover. MN3510-15 data indicate:

- Hover power is approximately 95–120 W per motor.
- Hover current on a 6S battery is approximately 4.0–5.0 A per motor.

Thus, the total system hover current is:

$$I_{\text{hover,total}} = 6 \times 4.5 \text{ A} = 27 \text{ A} \quad (3.5)$$

Using a 6S 22 Ah LiPo battery, typical for 15-minute endurance goals:

$$\text{Flight Time} = \frac{0.8 \times 22 \text{ Ah}}{27 \text{ A}} = 0.65 \text{ hr} = 39 \text{ min} \quad (3.6)$$

Factoring in wind, turbulence, maneuvering, and nonlinear battery discharge effects:

$$\text{Expected Actual Endurance} = 18\text{--}25 \text{ min} \quad (3.7)$$

This meets the preliminary requirement of 15 minutes minimum flight endurance with nominal payload, with margin for additional load and operational uncertainty.

Commercial integrated systems such as DJI flight-control ecosystems offered strong stabilization performance and simplified integration; however, they limited flexibility for future payload expansion, autonomy development, and custom mission integration. Custom embedded solutions provided maximum flexibility but introduced significant development complexity, increased integration risk, and reduced reliability for a time-constrained senior design project.

Ultimately, the Cube Orange+ autopilot system was selected as the primary flight controller for Pegasus due to its strong balance between reliability, modularity, autonomous capability, and ecosystem support.

The Cube Orange+ provided several advantages that aligned directly with the mission requirements of the aircraft:

High-Performance Processing Capability

The Cube Orange+ incorporates a high-speed STM32H7 processor capable of supporting advanced multirotor control algorithms, sensor fusion operations, autonomous navigation tasks, and telemetry management simultaneously. This processing capability was particularly important because Pegasus required stable flight behavior while also supporting autonomous waypoint execution, payload integration, and future mission expansion capability.

Redundant Sensor Architecture

One of the most important advantages of the Cube Orange+ was its redundant inertial sensing architecture. The controller includes multiple IMUs and vibration-isolated sensor integration, improving flight reliability and robustness in noisy vibration environments common to multirotor aircraft. Because rotor vibration can negatively affect attitude estimation and autonomous stability, sensor redundancy significantly improves confidence in reliable operation.

PX4 and ArduPilot Compatibility

The Cube Orange+ supports both PX4 and ArduPilot autopilot ecosystems, providing extensive flexibility for autonomous mission development and flight tuning. This compatibility allowed the team to leverage:

- Existing autonomous flight modes
- GPS waypoint navigation
- Return-to-launch functionality
- Failsafe behaviors
- Telemetry integration
- Mission scripting capability

Additionally, the mature software ecosystems surrounding PX4 and ArduPilot provided extensive documentation, simulation support, and troubleshooting resources, significantly reducing development risk during integration and testing.

Hexacopter Configuration Support

Because Pegasus utilized a six-rotor propulsion architecture, the flight controller needed to support advanced multirotor motor mixing and stable six-motor control allocation. The Cube Orange+ fully supports hexacopter configurations and provides robust ESC communication compatibility, making integration with the distributed propulsion system straightforward.

Telemetry and Communication Integration

The selected controller also supported multiple telemetry and communication interfaces including:

- GPS modules
- Telemetry radios
- RC receiver integration
- MAVLink communication
- Data logging systems
- External sensor interfaces

This connectivity was essential because wildfire-response operations require reliable telemetry transmission, mission-state awareness, and autonomous mission management.

Expandability for Future Development

Another major factor influencing controller selection was future expandability. The project envisioned future development involving:

- Thermal imaging systems
- AI-assisted fire detection
- Autonomous docking systems
- Obstacle avoidance sensors
- Advanced payload automation
- Swarm coordination concepts

The Cube Orange+ provided sufficient processing capability and interface flexibility to support these future expansions without requiring major avionics redesign.

Safety and Reliability Considerations

Operational safety was also a primary consideration during flight-controller selection. The Cube Orange+ supports numerous built-in safety features including:

- Geofencing
- Return-to-home functionality
- GPS failsafe behavior
- Battery failsafes
- RC signal loss handling
- Flight-mode redundancy

These features were particularly important because wildfire environments may involve communication degradation, visibility limitations, and rapidly changing operational conditions. Conservative failsafe behavior was therefore prioritized to minimize operational risk and improve mission reliability.

Following controller selection, the avionics architecture was developed around the Cube Orange+ as the central control node. The controller interfaced directly with the ESCs, GPS systems, telemetry radios, RC communication hardware, and payload-control systems through a modular wiring architecture. The final configuration separated propulsion and avionics power systems to reduce electrical interference and improve reliability during transient propulsion loading conditions.

The flight controller selection process ultimately reinforced the broader design philosophy of Pegasus: prioritize reliability, modularity, and operational robustness over unnecessary complexity or aggressive performance optimization. The Cube Orange+ provided a stable and expandable platform capable of supporting both current competition objectives and future wildfire-response mission development.

4 Propulsion System

The propulsion system was one of the primary design drivers for Pegasus because the aircraft mission required stable vertical takeoff, sustained hover, payload carriage, and reliable operation within the DBVF weight and battery constraints. Unlike a fixed-wing aircraft, the hexacopter must generate all lift through its rotors during the entire mission, so propulsion sizing directly determines endurance, control authority, disturbance rejection, and payload capability. The final propulsion architecture was therefore selected to provide conservative thrust margin while remaining compatible with the 6S battery limit, the 25 lb maximum takeoff weight requirement, and the team’s goal of maintaining low hover throttle during wildfire-response operations.

The completed propulsion system uses six T-Motor MN4110 KV340 brushless motors, six 18 x 6.1 carbon-fiber propellers, six T-Motor FLAME 60A 12S V2.0 electronic speed controllers (ESCs), and a 6S 10000 mAh lithium-polymer propulsion battery. This combination was chosen because it provided a thrust-to-weight ratio well above the project target of 2.0, maintained hover throttle near 47 percent, and supported a mission-endurance estimate above the 10 minute DBVF requirement.

4.1 Background and Propulsion Requirements

Propulsion requirements were derived from both competition constraints and mission-specific wildfire-response needs. The aircraft was required to remain below the DBVF 25 lb maximum takeoff weight limit, operate from a commercially available 6S LiPo propulsion battery, and complete a mission duration of at least 10 minutes. In addition to these top-level requirements, Pegasus needed enough thrust margin to reject wind gusts, recover from disturbances, maneuver while carrying payloads, and avoid operating near maximum throttle during normal hover.

The team established a minimum thrust-to-weight ratio target of 2.0 at the design operating weight. A value above 2.0 means that the propulsion system can theoretically produce more than twice the aircraft weight in static thrust, leaving significant throttle authority for attitude control and vertical acceleration. This margin was especially important because wildfire environments can include turbulent airflow, thermal updrafts, and rapidly changing wind conditions that require continuous controller correction.

A second propulsion target was to keep nominal hover throttle below approximately 60 percent. Lower hover throttle reduces the risk of throttle saturation, improves control authority, and gives the flight controller additional margin during payload handling, wind disturbances, and emergency maneuvers. The final aircraft achieved hover operation near 47 percent throttle, which satisfied this design intent and confirmed that the selected motor-propeller combination was not operating near its performance limit during normal flight.

4.2 Rotor Configuration

The six-rotor hexacopter configuration was selected because it provided the best balance between stability, redundancy, payload capability, and integration simplicity for the DBVF mission. A quadcopter configuration offered slightly higher theoretical endurance in early modeling, but it required each rotor to carry a larger fraction of the vehicle weight. This increased per-motor loading reduced thrust margin and made the vehicle less attractive for payload operations in gusting conditions.

The hexacopter layout distributes the lift requirement across six propulsion units, reducing the hover thrust required from each motor and improving control authority. The larger total rotor disk area also lowers disk loading relative to more compact alternatives, which improves hover efficiency and reduces the induced power required to generate lift. These characteristics were particularly valuable for Pegasus because the mission emphasized vertical takeoff, hover, autonomous waypoint flight, and payload deployment rather than high-speed cruise.

The configuration also improved operational robustness. With six independently controlled motors, the flight controller has more control allocation authority for roll, pitch, yaw, and vertical thrust than a four-rotor layout. Although the prototype was not designed to intentionally continue mission operations after a propulsion failure, the hexacopter architecture provides greater inherent fault tolerance and more graceful degradation than a quadcopter, which aligns with the project emphasis on safe and conservative wildfire-response operation.

4.3 Motor, Propeller, and ESC Selection

The selected propulsion system consisted of the T-Motor MN4110 KV340 brushless motor, shown in Figure 4.1, paired with an 18 x 6.1 carbon-fiber propeller and T-Motor Flame 60A 12S V2.0 electronic speed controllers. This motor-propeller-ESC combination was selected because it provided the required thrust margin, electrical compatibility, and control reliability for a hover-dominant hexacopter operating from a 6S propulsion battery.

The T-Motor MN4110 KV340 motor was appropriate for Pegasus because its 340 KV rating, 6S voltage compatibility, and manufacturer-recommended 17 to 18 inch propeller range matched the low-speed, high-thrust requirements of a multirotor aircraft. The motor also provided sufficient maximum power and current capability while keeping individual motor mass low enough to support the 25 lb competition weight constraint. Since Pegasus was designed primarily for stable vertical flight, payload transport, and autonomous hover operation, the propulsion system prioritized thrust authority and efficiency over high forward-speed performance.

The 18 x 6.1 carbon-fiber propeller was selected to increase rotor disk area and improve hover efficiency. Larger propellers accelerate a larger column of air at a lower induced velocity, which reduces the power required for a given amount of thrust compared with smaller, higher-speed propellers. This selection was consistent with the overall aircraft design philosophy: Pegasus prioritized stable hover, endurance, and payload controllability rather than high-speed flight.

The T-Motor Flame 60A 12S V2.0 ESC was selected to provide reliable current handling and compatibility with the chosen motor and battery system. Each ESC was rated above the expected operating current of the MN4110 motor during hover and mission flight, giving the propulsion system electrical margin for transient throttle increases, maneuvering, and disturbance rejection. The ESCs also supported standard PWM control, which simplified integration with the Cube Orange+ flight controller and allowed the team to maintain a straightforward, serviceable propulsion architecture.

At the design all-up weight of approximately 18.08 lb, each motor must provide approximately 3.01 lbf of hover thrust. Manufacturer data for the selected motor-propeller combination indicated that this thrust level occurs near the mid-throttle range and requires approximately 137 W per motor. With six rotors, the predicted total hover power is approximately 823.8 W. The maximum thrust per motor is approximately 8.01 lbf, giving a total available thrust of 48.1 lbf and a resulting thrust-to-weight ratio of 2.66. These results confirmed that the selected motor, propeller, and ESC combination provided sufficient thrust, power, and electrical margin for the Pegasus mission profile. Table 10 shows each component, the selected hardware, their specifications, and the role they each play in the design.

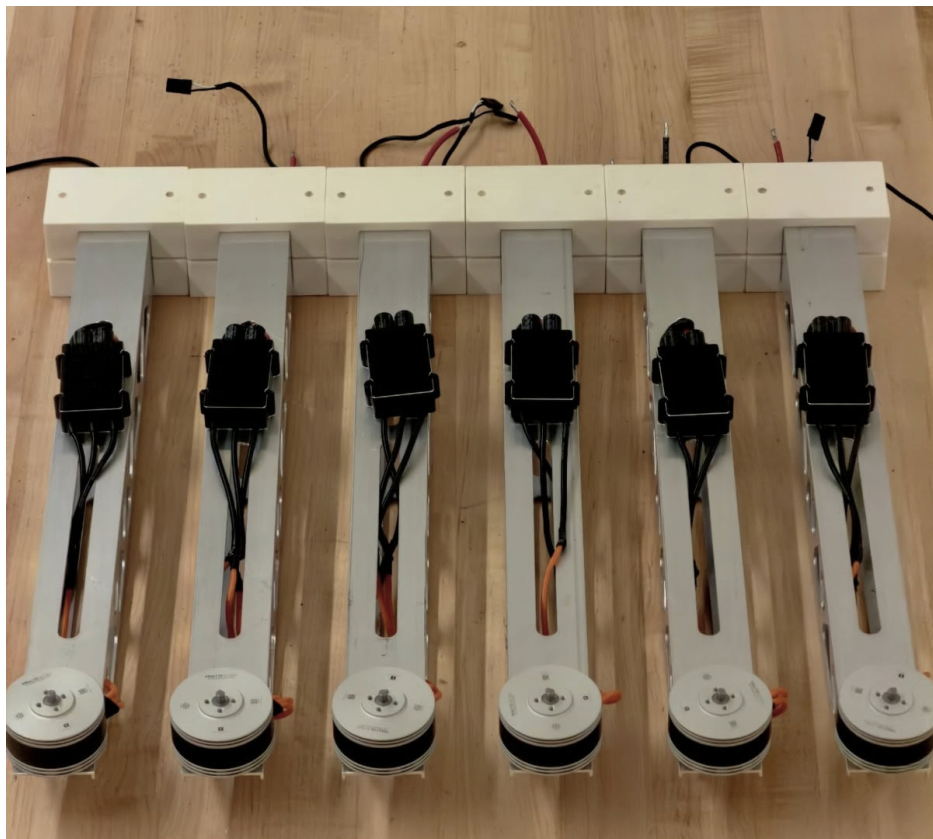


Figure 4.1: Selected MN4110 KV340 motor and Flame 60A 12S V2.0 ESC Installed

Table 10: Propulsion Hardware Selected for Pegasus

Component	Selected Hardware	Key Specification	Design Role
Motor	T-Motor MN4110 KV340	340 KV, 6S LiPo rating, 152 g motor mass, 30 A peak current for 180 s	Provides hover thrust and control authority for each rotor station.
Propeller	18 × 6.1 carbon-fiber propeller	18 in diameter; compatible with MN4110 recommended propeller range	Increases disk area and improves hover efficiency.
ESC	T-Motor FLAME 60A 12S V2.0	60 A continuous current, 6–12S LiPo compatibility	Regulates motor speed and provides current margin relative to motor demand.
Propulsion battery	6S 10000 mAh LiPo	22.2 V nominal voltage, 10000 mAh capacity	Primary energy source for all six propulsion motors.

4.4 ESC and Power Distribution Integration

Each motor is driven by an individual T-Motor FLAME 60A 12S V2.0 ESC. The ESC rating provided a large electrical margin relative to the expected operating point. The motor peak current rating is approximately 30 A for 180 seconds, while the selected ESC is rated for 60 A continuous current and higher short-duration peak current. This means that the ESCs are not operating near their continuous-current limit during expected hover or mission-average conditions, reducing thermal risk and improving reliability.

A dedicated power distribution board routes the 6S propulsion-battery power to the six ESCs. This architecture keeps high-current propulsion power separate from the avionics power system and supports the DBVF requirement for independent propulsion and flight-control power sources. The propulsion system is powered from the main 6S pack, while the avionics system is powered separately from a 3S battery through regulated power electronics. This separation reduces the chance that propulsion voltage sag or current transients will affect the flight controller, telemetry, GPS, or receiver.

The physical routing of the propulsion wiring was also treated as a design concern. High-current wires were routed through or alongside the aluminum arms where practical, while signal wiring was kept organized to reduce snag hazards and limit electrical-noise coupling. This integration approach improved field maintainability and supported the project goal of a clean, serviceable aircraft architecture.

4.5 Battery and Endurance Modeling

The propulsion battery was modeled as a 6S 10000 mAh LiPo pack with a nominal voltage of 22.2 V. The nominal stored energy is therefore estimated as:

$$E_{\text{battery}} = \frac{C_{\text{battery}} V_{\text{battery}}}{1000} \quad (4.1)$$

$$E_{\text{battery}} = \frac{(10000 \text{ mAh})(22.2 \text{ V})}{1000} = 222 \text{ Wh} \quad (4.2)$$

Because it is not practical or safe to use the entire nominal battery energy during flight, the endurance analysis was based on a reserve-limited usable-energy estimate. The team used battery reserve assumptions in both the Simulink model, shown in Figures 4.2 and 4.3, and the later Monte Carlo sensitivity analysis to avoid overstating available flight time. The deterministic Simulink model treated the propulsion system as an equivalent constant-power load applied to a table-based 6S LiPo battery model. The model tracked battery voltage, current, and state of charge over time and stopped the simulation when the battery reached a reserve threshold.

This Simulink battery model was useful because it provided a direct time-domain prediction of battery discharge behavior under representative propulsion loads. Under hover-equivalent loading, the model predicted approximately 12.35 to 13.56 minutes to reach the reserve threshold. To account for climb, descent, horizontal translation, controller activity, and other non-ideal mission effects, the team applied a conservative mission power factor of 1.15 to the hover power estimate:

$$P_{\text{mission}} = kP_{\text{hover,total}} \quad (4.3)$$

$$P_{\text{mission}} = 1.15(823.8 \text{ W}) = 947.37 \text{ W} \quad (4.4)$$

Using this mission-adjusted power level, the predicted endurance was approximately 10.6 to 11.7 minutes to reserve. This result provided analytical support that the propulsion system could satisfy the 10-minute DBVF mission requirement while maintaining practical battery reserve.

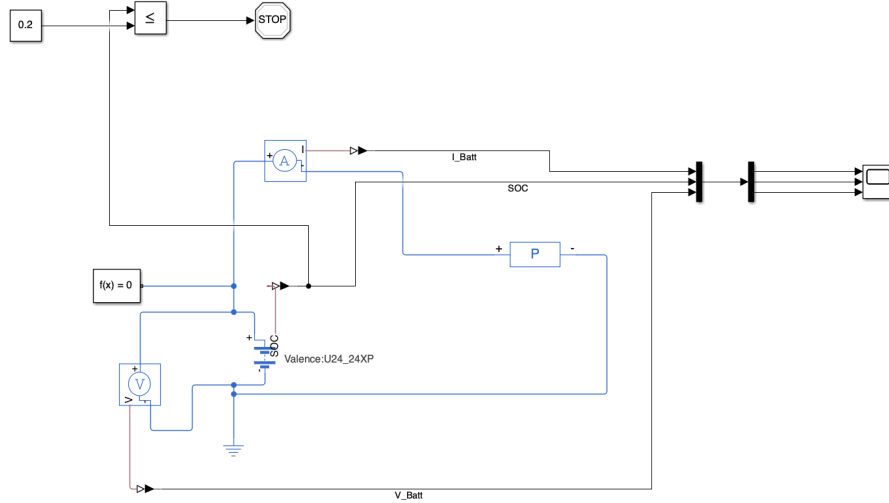


Figure 4.2: Simulink battery endurance model used for deterministic propulsion endurance prediction

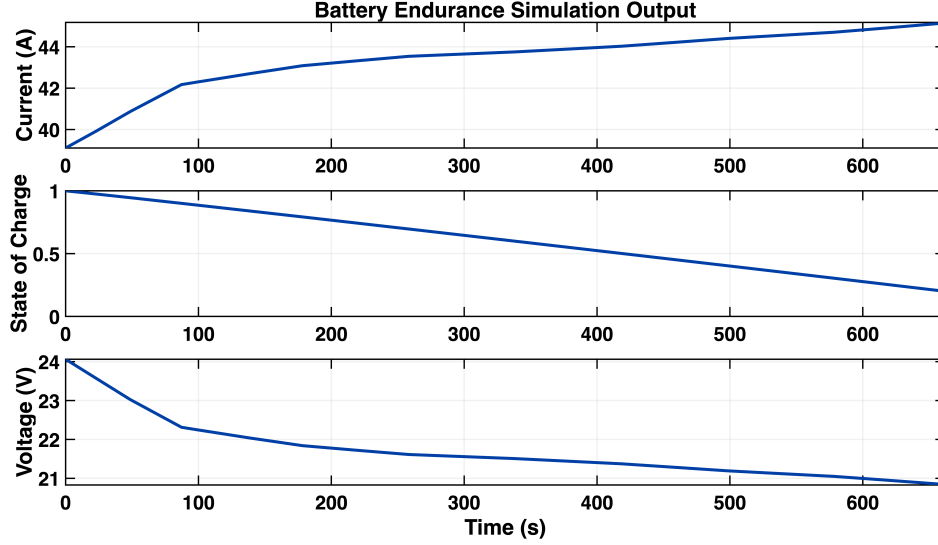


Figure 4.3: Representative battery endurance simulation output for mission-equivalent power demand

4.6 Propulsion Sizing and Performance Calculations

The major propulsion sizing calculations were based on aircraft weight, number of rotors, propeller disk area, manufacturer thrust data, and battery-energy estimates. The required hover thrust per motor was calculated by dividing the aircraft weight by the number of rotors:

$$T_{\text{hover,motor}} = \frac{W}{n} \quad (4.5)$$

$$T_{\text{hover,motor}} = \frac{18.08 \text{ lb}}{6} = 3.01 \text{ lbf} \quad (4.6)$$

The total maximum thrust was calculated from the manufacturer maximum thrust per motor:

$$T_{\text{total}} = nT_{\text{max,motor}} \quad (4.7)$$

$$T_{\text{total}} = 6(8.01 \text{ lbf}) = 48.1 \text{ lbf} \quad (4.8)$$

The resulting thrust-to-weight ratio was:

$$\frac{T}{W} = \frac{T_{\text{total}}}{W} = \frac{48.1}{18.08} = 2.66 \quad (4.9)$$

This value exceeded the project target of 2.0 and confirmed that the aircraft retained substantial thrust margin. The propeller disk area was also calculated to evaluate hover efficiency:

$$A_{\text{rotor}} = \pi \left(\frac{D}{2} \right)^2 \quad (4.10)$$

$$A_{\text{rotor}} = \pi \left(\frac{0.4572 \text{ m}}{2} \right)^2 = 0.164 \text{ m}^2 \quad (4.11)$$

$$A_{\text{total}} = nA_{\text{rotor}} = 6(0.164 \text{ m}^2) = 0.985 \text{ m}^2 \quad (4.12)$$

The resulting total disk area helped keep disk loading low and supported efficient hover performance. Together, the thrust-to-weight ratio, hover throttle estimate, and endurance model showed that the selected propulsion system provided a conservative operating envelope for the DBVF mission.

4.7 Propulsion Validation

Propulsion performance was validated through a combination of analytical modeling, ground testing, hover testing, and flight observations. During ground testing, the motors were run incrementally while the team monitored throttle response, vibration behavior, motor direction, electrical continuity, and ESC operation. This confirmed that the propulsion system was correctly wired, that each motor produced thrust in the intended direction, and that the power-distribution system could support simultaneous motor operation.

Hover testing provided the most important full-system validation of the propulsion sizing process. The aircraft achieved stable hover near 47 percent throttle, matching the expected mid-throttle operating point from the manufacturer data and confirming that sufficient thrust margin remained available for maneuvering and disturbance rejection. Flight testing also confirmed that the aircraft could sustain mission durations above the 10 minute DBVF endurance requirement while maintaining acceptable battery reserve.

Overall, the propulsion system satisfied the main design requirements established for Pegasus. It remained compatible with the 6S battery constraint, provided a thrust-to-weight ratio of 2.66, operated at a low hover throttle, and supported mission-adjusted endurance near the required 10 minute target. These results validated the motor, propeller, ESC, and battery selections and confirmed that the propulsion architecture was appropriate for a stable wildfire-response hexacopter.

5 Airframe Design

The airframe of the Pegasus aircraft, displayed in Figure 5.1, serves as the primary structural foundation, engineered to facilitate stable flight while carrying mission specific equipment in hazardous wildfire environments. The design prioritizes a high strength to weight ratio, modular repairability, and operational durability. To compete effectively in the VFS DBVF competition as well as improving early wildfire response, the airframe architecture was developed through a data driven approach. This involved utilizing topology optimization to minimize parasitic mass without compromising structural safety.

The airframe was designed around a hexacopter configuration to provide the motor redundancy necessary for operating in turbulent fire zones. By selecting 6061-T6 aluminum as the primary material for the central frame and propulsion system, the team was able to balance cost effectiveness with the high yield strength required for heavy lift operations. This section details the design of the four primary sub assemblies that comprise the airframe:

- **Central Frame (Top and Bottom Plates):** The structural core of the aircraft, which houses the avionics and serves as the junction for all primary load paths.
- **Propulsion Arms:** Six 18 inch cantilevered structures that transfer motor thrust to the central frame while providing protected routing for electronic cabling.
- **Landing Gear:** A modular, hybrid material system designed with a wide, rounded footprint to ensure stability on uneven or soft terrain.
- **Payload Mechanism:** An integrated release system located at the center of gravity, designed to deploy emergency supplies.

5.1 Central Frame: Top and Bottom Plates

The central frame serves as the primary structural junction for the Pegasus aircraft. It provides the mechanical interface between the six propulsion arms and the mission critical avionics stack. The architecture utilizes a sandwich plate design, consisting of a top and bottom plate shown in Figure 5.2, that prioritizes a high area moment of inertia to resist the significant bending moments generated by the 18 inch propulsion arms during high thrust maneuvers.

5.1.1 Material Selection and Geometry

Both the top and bottom plates are designed as regular dodecagons fabricated from 0.19 inch thick 6061-T6 aluminum. The geometry was defined by a 15 inch width measured between opposite parallel edges. This flat to flat distance ensures that the internal bay provides sufficient surface area for mounting the square propulsion arms while maintaining a compact overall profile.

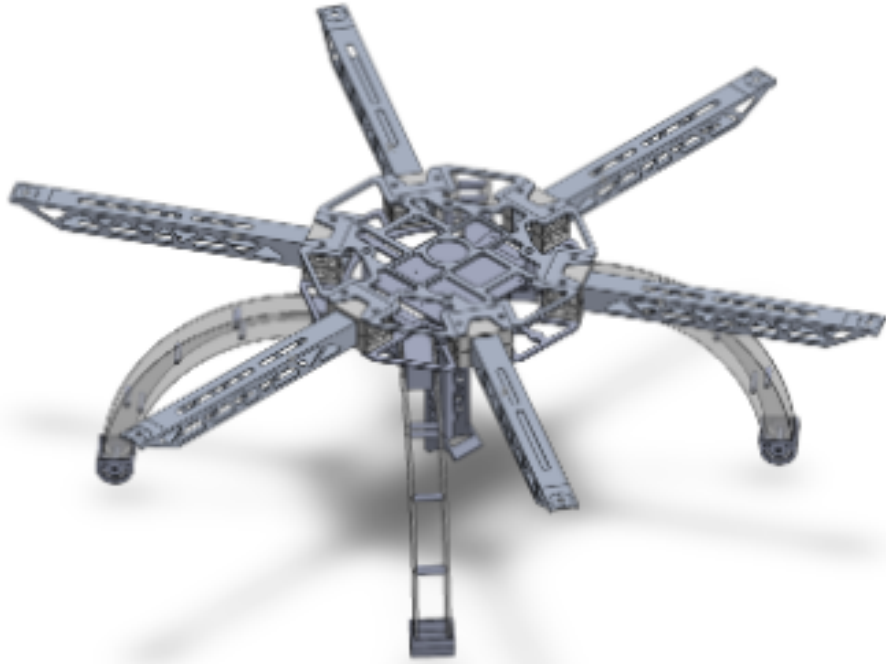


Figure 5.1: Full Structural Design

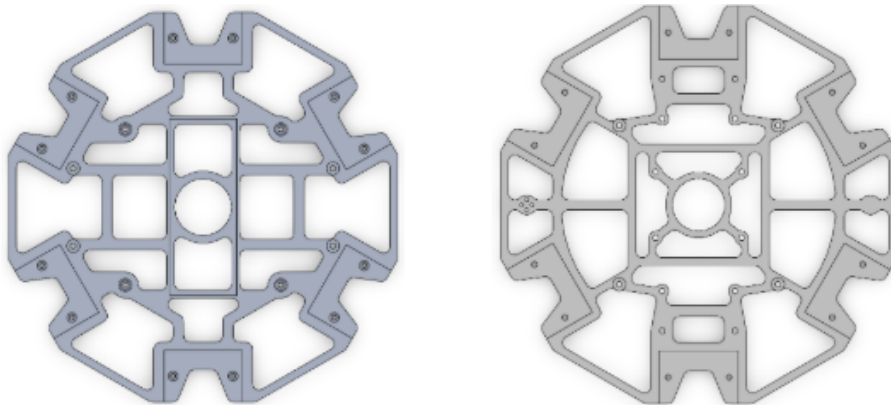


Figure 5.2: Top (left) and Bottom (right) Plates

The selection of 6061-T6 aluminum was driven by several technical and logistical factors. While carbon fiber composites offer high strength-to-weight ratios, aluminum was chosen primarily due to the limited project budget and onsite manufacturing restrictions. Aluminum provided a cost-effective alternative that was compatible with the precision machining equipment available in the university facilities. This allowed the team to maintain high tolerances for the mounting faces without the long lead times or high costs associated with external composite fabrication.

Additionally, aluminum offers a yield strength of approximately 275 MPa and acts as a passive heat sink for internal power electronics in high-temperature wildfire zones. The 12-sided geometry provides six flat, 1.5 inch wide faces that perfectly match the profile of the square propulsion arms. This ensures precise 120 degree motor alignment, which is critical for maintaining stable flight control authority.

5.1.2 Generative Design and Topology Optimization

To maximize the thrust-to-weight ratio and the payload capacity of the aircraft, the central plates underwent a rigorous topology optimization process during the design phase. This allowed the team to refine the geometry based on calculated flight loads rather than relying on solid, over engineered plates that would add unnecessary parasitic mass.

The optimization followed a five stage iterative methodology utilizing SOLIDWORKS Simulation to achieve a balance between weight reduction and structural stiffness. Each iteration began by running a topology optimization simulation to identify low stress regions where the material contributed little to the load path. In Figures 13 and 14 below, each iteration of TO can be seen starting with the basic geometry to the final result. Based on these computational results, specific manual cuts were modeled into the CAD geometry to remove unnecessary material.

After each set of cuts was performed, the team validated the new design iteration using a static structural simulation. This secondary analysis ensured that the material removal did not create unexpected stress concentrations or drop the factor of safety below the target value of 2.0. This cycle of simulation, manual design adjustment, and static validation was repeated for five iterations until the design converged on an optimal geometry.

This process resulted in a 54.6 percent mass reduction for the top plate, dropping from its original calculated mass to 1.42 lbs. The bottom plate saw a 57.2 percent reduction, ending at a final mass of 1.34 lbs. These savings were critical in achieving the final 6.91 lb mass margin and ensuring the aircraft remained well under the 25 lb competition limit.

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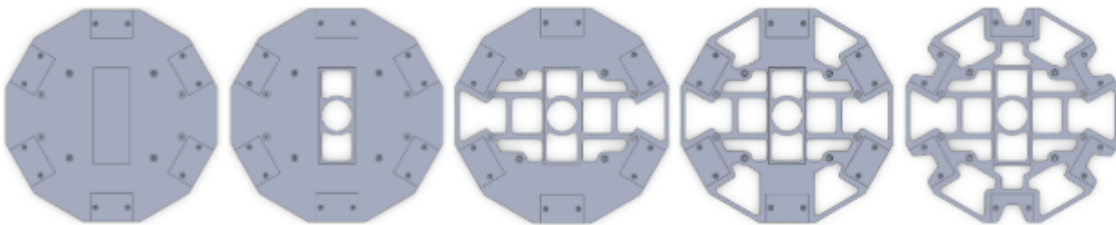


Figure 5.3: Topology Optimization (Top Plate)

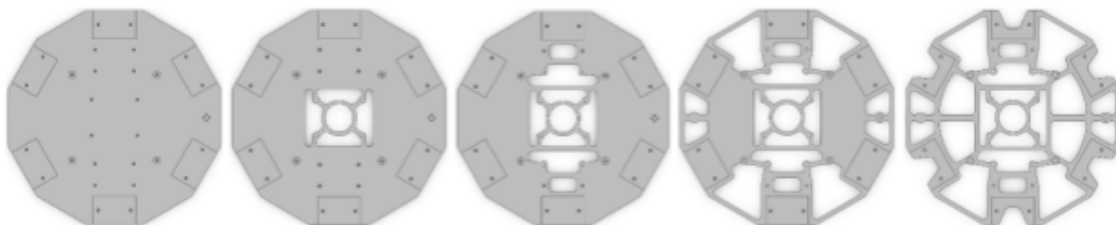


Figure 5.4: Topology Optimization (Bottom Plate)

After each set of cuts was performed, the team validated the new design iteration using a static structural simulation. This secondary analysis ensured that the material removal did

not create unexpected stress concentrations or drop the factor of safety below the target value of 2.0. This cycle of simulation, manual design adjustment, and static validation was repeated for five iterations until the design converged on an optimal geometry.

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5.2 Propulsion Arms

The propulsion arms are the primary structural extensions of the aircraft. They are responsible for transferring the thrust generated by the motors to the central frame. To meet the performance requirements of a high capacity hexacopter, the arms were designed to provide maximum bending stiffness while maintaining a low mass profile.

5.2.1 Geometry and Manufacturability

The arms are fabricated from 1.5 inch square 6061-T6 aluminum tubing with a 1/16 inch wall thickness. Each arm has a total length of 18 inches. This length provides the necessary lever arm to ensure flight stability and sufficient clearance for the 18 inch propellers.

Aluminum was selected as the primary material over carbon fiber composites because of manufacturing accessibility and facility constraints. While carbon fiber is common in high performance aerospace applications, the team could not work with the material due to the lack of specialized equipment and safety protocols required for composite fabrication on campus. In contrast, 6061-T6 aluminum could be processed using the CNC machinery available in the university machine shop. This allowed for an entirely in house fabrication process and faster design iterations.

The choice of square tubing over round alternatives was a deliberate design decision to simplify the mechanical interfaces. The flat surfaces of the square profile offer significantly easier alignment of the motors during assembly. This geometry also ensures a stable, flush connection to the dodecagon central plates without the need for complex anti-rotation hardware or specialized mounting brackets.

5.2.2 Mass Reduction

To ensure the aluminum arms remained competitive with the weight of composite alternatives, the team applied the same iterative topology optimization process used for the central plates. The optimization targeted material removal along the neutral axis of the square tubing where the internal stresses were calculated to be lowest, and the iterations are shown in Figure 5.5 below.

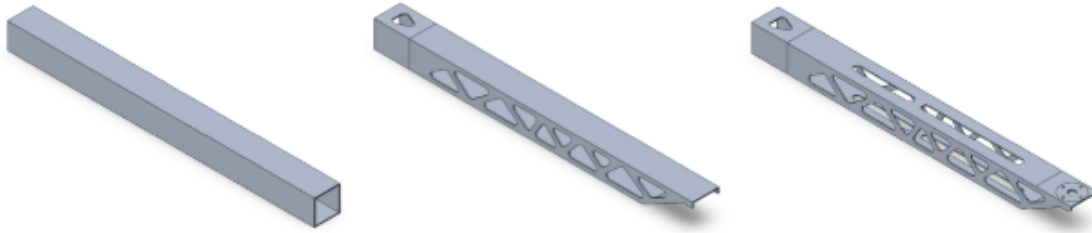


Figure 5.5: Topology Optimization (Arm)

While the central plates required five stages of refinement, the propulsion arms reached an optimized state after three iterations of simulation and manual design adjustments. By following this cycle of static validation and material removal, the team successfully reduced the mass of each arm from an initial 0.63 lbs to a final weight of 0.42 lbs. This 33.3 percent reduction in mass was achieved while maintaining a high factor of safety of 33.4. This high level of structural conservatism ensures that the arms remain rigid and provide stable motor mounting points even when subjected to the erratic gusts and thermal updrafts of a wildfire environment.

5.2.3 Onsite CNC Manufacturing and Alignment

The propulsion arms were manufactured onsite using three-axis CNC milling. This onsite fabrication strategy allowed the team to maintain direct oversight of the tolerances for the motor mounting patterns and the frame interface. Precision was especially critical for the alignment of the bolt holes at the tip of each 18-inch arm. Any angular deviation at the mounting point would be magnified over the length of the arm, which would lead to significant thrust vectoring errors and reduced flight efficiency.

To ensure perfectly repeatable alignment across all six arms, the team designed a custom alignment feature. Each propulsion arm was CNC machined with specific alignment holes at the base. These holes interface directly with a physical key that was printed into the internal geometry of the 3D printed ASA clamps. During assembly, this keyed alignment mechanism locks into the arm, providing a mechanical hard stop that secures the arm in place and ensures that the motor orientation is perfectly perpendicular to the central frame. This method eliminated the risk of human error during the assembly process and guaranteed that each arm was positioned at the correct offset.

5.2.4 Structural Logic and Architecture

From a structural standpoint, the propulsion arms function as cantilever beams fixed at the central hub. The 1.5 inch square cross section was selected to provide a high area moment of inertia. This is essential for resisting the vertical bending moments produced during maximum thrust maneuvers.

The use of hollow tubing rather than solid bar stock allows for a significantly higher strength

to weight ratio. This architecture also provides a protected internal conduit for the power wires and signal cables. Routing the electronics through the center of the 6061-T6 tubing shields the wiring from environmental heat and physical damage while keeping the external profile of the aircraft aerodynamic and free of snag hazards. This integrated approach ensures the robust operation of the system in the challenging conditions of a wildfire environment.

5.3 Landing Gear

The landing gear system provides the necessary ground clearance for the propulsion system and the payload mechanism while ensuring stability during takeoff and landing cycles. For the Pegasus aircraft, the landing gear, Figure 5.6, was designed specifically to handle the unpredictable terrain of wildfire zones, which often includes loose ash, fallen debris, and uneven soil.

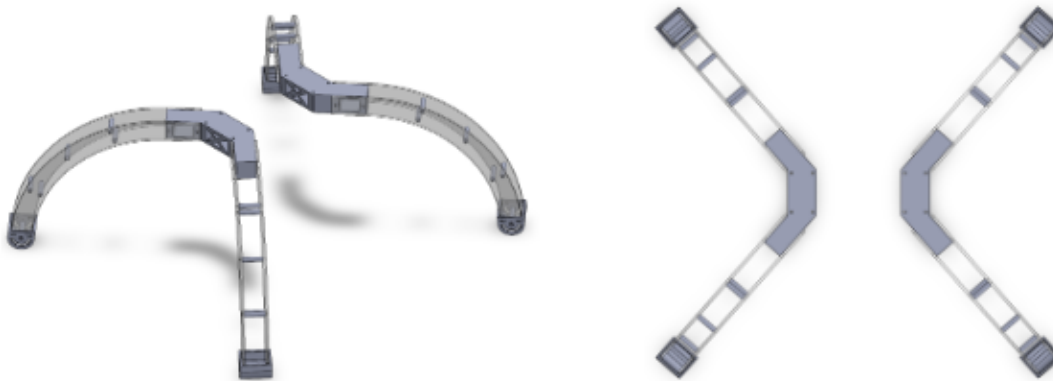


Figure 5.6: Landing Gear Design

5.3.1 Hybrid Material Architecture

The landing gear assembly utilizes a hybrid material approach to balance structural rigidity with ease of manufacturing. The primary structural legs are fabricated from cast acrylic. This material was selected for its cost effectiveness and its compatibility with onsite laser cutting equipment. The use of laser cutting allowed for the creation of complex, curved geometries that would be difficult to achieve with manual machining.

To join these legs to the airframe, the design incorporates 3D printed connecting pieces. These components serve as the mechanical interface between the acrylic legs and the bottom central plate. By using 3D printed polymers for these complex junctions, the team was able to create a secure, form fitting attachment point that maintains the necessary leg angle for a wide, stable stance.

5.3.2 Structural Reinforcement

While the cast acrylic provides the necessary stiffness to maintain the height of the aircraft, its brittleness under shear loads required additional reinforcement. To address this, the design

includes multiple 3D printed pegs integrated throughout the length of the acrylic legs.

These pegs are strategically placed to act as structural spacers and load distributors. By physically linking the parallel layers of the leg assembly, the pegs increase the overall structural integrity and prevent the acrylic components from buckling or shifting under the weight of the aircraft. This reinforcement strategy ensures that the landing gear can withstand the vertical impact forces of a landing without compromising the brittle material.

5.3.3 Modular Integration and Terrain Adaptability

A key feature of the landing gear design is its modularity. By mounting each leg assembly directly to the bottom central plate, the system remains independent of the propulsion arms and the internal electronics. This modular approach allows for rapid field maintenance. If a leg is damaged during a hard landing, it can be removed and replaced as a standalone unit without requiring a full disassembly of the airframe.

At the base of each leg, the system features specialized shoes constructed from a combination of 3D printed housing and high density foam. These shoes utilize a rounded profile to improve the versatility of the ground interface. The rounded geometry ensures that the landing gear does not snag on fallen branches or debris during a landing that involves lateral movement. Additionally, the rounded shape allows for consistent contact with the ground even if the aircraft is tilted at an angle due to uneven terrain.

The shoes serve two primary functions:

1. **Ground Interface:** The wide profile of the shoes increases the surface area of the landing gear. This prevents the aircraft from sinking into soft ash or loose soil during deployment.
2. **Shock Absorption:** The integrated foam pads act as a passive damping system. This layer dissipates kinetic energy during the moment of touchdown, which protects the central frame and sensitive avionics from high frequency vibrations and mechanical shock.

The combination of the curved leg geometry, the keyed structural reinforcement, and the modular mounting interface provides the Pegasus with a robust foundation. This ensures the aircraft remains upright and operational across a variety of environmental conditions.

5.4 Payload Mechanism

The payload mechanism is the primary mission system for the Pegasus aircraft. It is designed to facilitate the secure transport and precise release of sensor packages or emergency supplies. To meet the requirements of the VFS DBVF competition and early wildfire response, the mechanism was engineered to be repeatable, lightweight, and electronically controlled without the need for complex mechanical latches or pyrotechnic release systems.

5.4.1 Architecture

The system is built around a high torque linear actuator mounted vertically to the underside of the bottom central plate. A high strength permanent magnet is fixed to the end of the actuator shaft. This magnet serves as the primary attachment point for the payload, which is also equipped with a permanent magnet.

The mechanism also includes a stationary outer casing that surrounds the actuator shaft. This casing is fixed to the bottom plate and remains stationary while the actuator moves. The internal area of the casing is sized to allow the magnet to pass through freely, while the external rim provides a flat, rigid surface that acts as a mechanical stop for the payload.

5.4.2 Operation Sequence and Slip-Off

The release of the payload is achieved through a mechanical slip-off process. During the transport phase, the linear actuator is in the extended position, which holds the magnet and the payload against the stationary outer casing. This ensures that the payload does not swing or shift during flight, maintaining the stability of the center of gravity.

When the release command is triggered, the linear actuator retracts the magnet upward through the casing. As the magnet moves beyond the stationary rim of the casing, the payload interface plate is physically blocked by the edges of the casing. Because the casing is rigid, it prevents the payload from following the upward movement of the magnet. As the actuator continues to retract, the physical gap between the magnets increases until the magnetic flux is no longer strong enough to overcome the force of gravity. This results in a clean, vertical release of the payload.

5.4.3 Integration and Reusability

By utilizing a magnetic decoupling method, the mechanism eliminates the risk of mechanical binding that can occur with traditional hook or pin release systems. This design is also fully reusable and can be reset instantly by retracting the actuator, allowing the team to perform multiple deployment cycles during a single mission or across several competition heats.

The entire payload assembly is mounted directly to the center of the bottom dodecagon plate. This positioning ensures that the payload weight is concentrated at the center of gravity of the hexacopter, which minimizes the trim adjustments required by the flight controller during transport. The modular mounting interface allows the mechanism to be removed or adjusted easily, supporting the overall goal of airframe serviceability and mission flexibility in a wildfire environment.

6 System Analysis

6.1 Finite Element Analysis and Structural Validation

The structural integrity of the airframe was validated using a static structural analysis FEA within SOLIDWORKS Simulation. The primary objective of this analysis was to ensure that the optimized geometries for the central plates and propulsion arms could withstand the maximum operational loads expected during flight missions while maintaining a minimum factor of safety of 2.0.

6.1.1 Analysis Setup and Material Properties

All structural components were modeled using 6061-T6 aluminum. The material properties defined for the simulation included a yield strength, σ_y , of 275 MPa and a Young's modulus, E , of 68.9 GPa. The simulations used high-quality parabolic tetrahedral solid elements. To ensure accurate results, a refined mesh density was applied to the internal radii and webbed structures created during the topology optimization process. This refinement allowed the software to accurately capture stress gradients in the areas where material was removed.

6.1.2 Central Frame Plate Analysis

The top and bottom plates were analyzed under a worst case loading scenario to simulate the maximum stresses encountered during a high thrust climb. The simulation modeled the plates as being fixed at the central avionics mounting points while being subjected to the cantilevered moments transferred from the propulsion system. Figures 6.1 and 6.2 present the von Mises stress distributions obtained from the static structural FEA simulations.

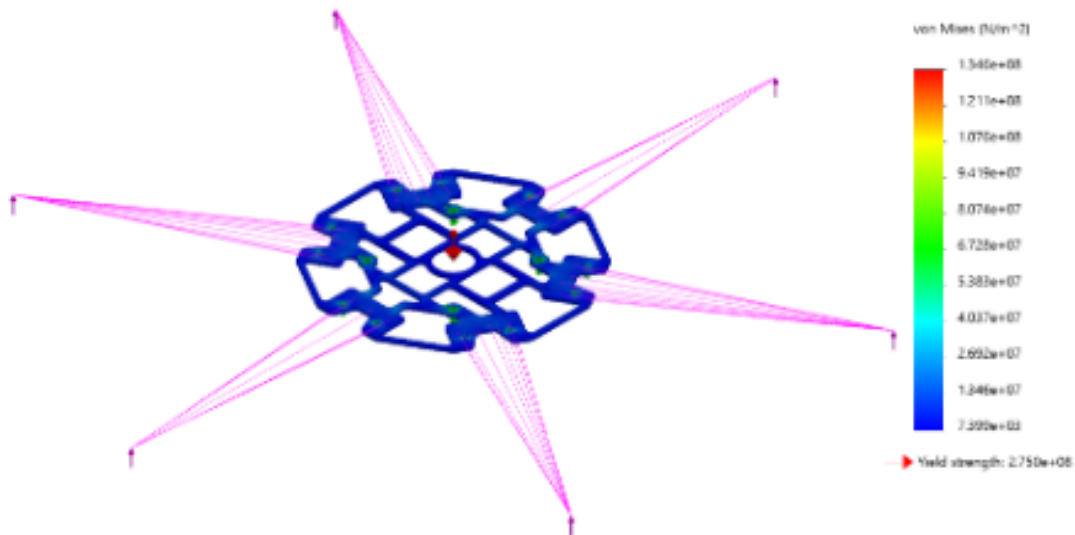


Figure 6.1: Top Plate Static Structural Analysis (von Mises)

- **Boundary Conditions:** Each of the six arm interfaces was subjected to a vertical force of 5.5 lbf modeled as remote loads, representing the maximum design thrust per

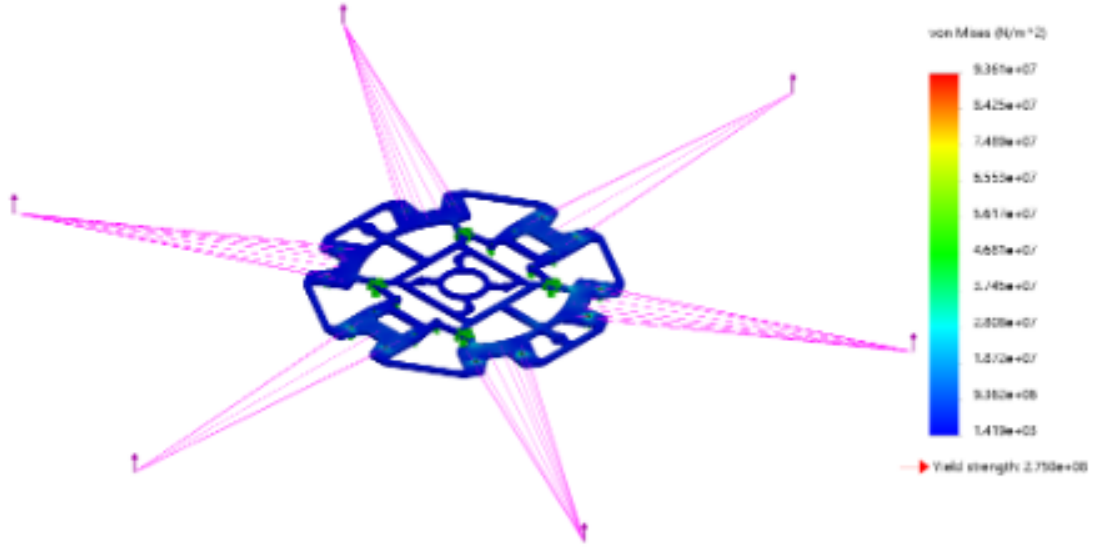


Figure 6.2: Bottom Plate Static Structural Analysis (von Mises)

motor. The central mounting holes were set as fixed conditions, while the propulsion arm mount holes were modeled as slider conditions. Additionally, the battery weight was modeled as a distributed load on the top plate.

- **Stress Results:** The analysis revealed a maximum von Mises stress of 135 MPa in the top plate and 93.6 MPa in the bottom plate. As expected, the peak stress concentrations were located at the fillets of the internal webbed structures created during the five-stage optimization process.
- **Safety Factor:** Using the yield strength of the aluminum alloy, the calculated factor of safety is:

$$FoS = \frac{\sigma_y}{\sigma_{\max}} = \frac{275 \text{ MPa}}{135 \text{ MPa}} \approx 2.04 \quad (6.1)$$

- **Displacement Analysis:** The maximum resultant displacement was recorded as 0.118 mm for the top plate and 0.0767 mm for the bottom plate at the outer edges. This minimal deflection confirms that the plates provide a rigid foundation for the propulsion arms, which is necessary to prevent vibration coupling with the flight controller's inertial measurement units.

6.1.3 Propulsion Arm Analysis

The propulsion arms were analyzed as cantilever beams fixed at the interface with the central frame. This setup simulated the upward force exerted by the motors at the tip of the 18 inch span. Figure 19 below present the von Mises stress distributions obtained from the static structural FEA simulation.

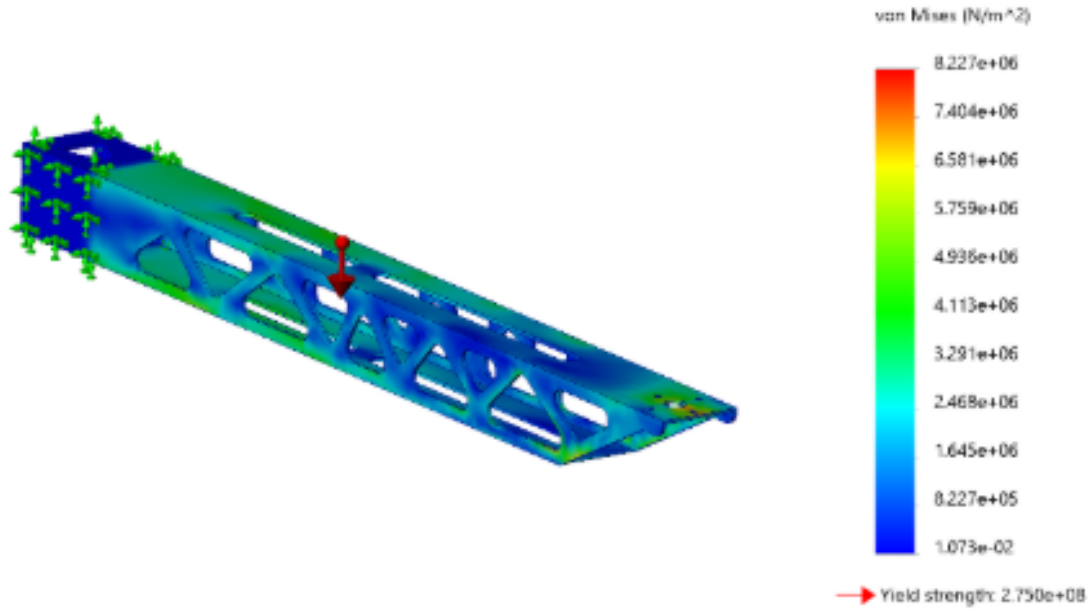


Figure 6.3: Propulsion Arm Static Structural Analysis (von Mises)

- **Boundary Conditions:** A point load of 5.5 lbf was applied to the motor mounting holes at the end of the 1.5 inch square tube. On the opposite end, the section mounted to the central frame was set as fixed.
- **Stress and Safety Factor:** Due to the high area moment of inertia provided by the square cross section, the peak von Mises stress remained well below the material limits. The analysis yielded a factor of safety of 33.4. While this value significantly exceeds the minimum requirement of 2.0, this level of structural conservatism was maintained to ensure extreme rigidity.
- **Vibration and Deflection:** High rigidity is required to prevent aeroelastic flutter and to ensure the propellers remain parallel to the central frame during high thrust maneuvers. The minimal deflection observed in the simulation validates that the aluminum arms provide the necessary stiffness to isolate the flight controller from high frequency motor noise.

6.2 Modal Analysis

Modal analysis was performed to evaluate the dynamic behavior of the Pegasus propulsion arms and determine whether rotor-induced excitation could create resonance during flight. This analysis was necessary because multirotor aircraft generate periodic vibration from motor rotation, propeller imbalance, aerodynamic loading, and ESC-driven torque fluctuations. If one of these excitation frequencies aligns with a structural natural frequency of the airframe, vibration amplitudes can increase significantly, potentially affecting flight-controller sensor measurements, structural fatigue life, motor alignment, and overall flight stability.

The propulsion arms were selected as the primary components for modal evaluation because they behave approximately as cantilever beams extending from the central frame. Each arm supports a motor and propeller assembly near its free end, where added mass has the greatest effect on the first bending mode. In the SolidWorks modal model, the inboard 2 in of the arm was assigned a fixed boundary condition to approximate the clamped region where the arm is secured inside the central frame assembly. This fixed region represented the structural constraint provided by the arm clamps, fasteners, and top/bottom plate interface.

A remote mass was applied at the motor mounting location to approximate the mass of the assembled propulsion hardware. The motor mass was modeled as 152 g, and the propeller mass was modeled as approximately 31.5 g, giving a total remote mass of:

$$m_{\text{remote}} = 152 \text{ g} + 31.5 \text{ g} = 183.5 \text{ g} \quad (6.2)$$

This remote mass was used instead of modeling the full motor and propeller geometry because the goal of the modal analysis was to capture the effect of the tip-mounted propulsion mass on the arm's bending behavior. Since mass located near the free end of a cantilever beam strongly lowers the first natural frequency, including the remote mass produced a more realistic estimate than analyzing the aluminum arm alone.

The expected hover excitation frequency was estimated from the motor operating speed required to generate hover thrust. Based on propulsion data and motor operating estimates, Pegasus was expected to hover at approximately 3578 rpm. The corresponding excitation frequency is:

$$f_{\text{hover}} = \frac{\text{RPM}}{60} \quad (6.3)$$

$$f_{\text{hover}} = \frac{3578}{60} = 59.63 \text{ Hz} \quad (6.4)$$

The modal analysis predicted a first natural frequency of 87.891 Hz for the propulsion arm. The first mode shape was dominated by vertical bending of the arm, which is consistent with cantilever-beam behavior. This result is important because the first bending mode is the lowest-frequency structural mode and is therefore the most likely mode to be excited during normal motor operation.

The first bending mode of the Pegasus propulsion arm from modal analysis is shown in Figure 6.4. The inboard 2 in of the arm was fixed to approximate the clamped base connection to the central frame, and a 183.5 g remote mass was applied at the motor mounting location to represent the combined motor and propeller mass. The first natural frequency was predicted to be 87.891 Hz and was compared against the estimated motor operating frequency to evaluate resonance risk during flight.

The separation ratio between the first natural frequency and the expected hover excitation frequency was calculated as:

Mode Shape: 1 Value = 89.959 Hz
Deformation scale: 0.0208473

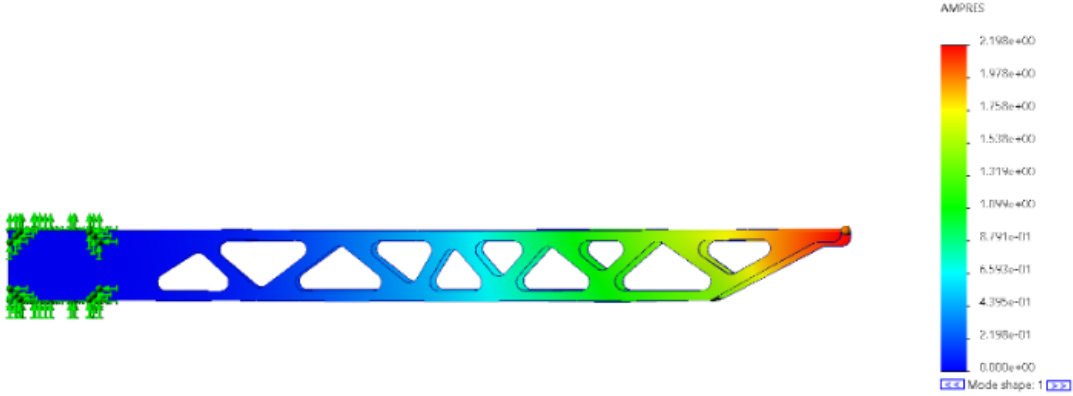


Figure 6.4: Arm Frequency Analysis (Mode Shape 1)

$$\text{Separation Ratio} = \frac{f_1}{f_{\text{hover}}} \quad (6.5)$$

$$\text{Separation Ratio} = \frac{87.891}{59.63} = 1.47 \quad (6.6)$$

This result indicates that the first natural frequency is above the expected hover excitation frequency, so resonance is not expected during nominal hover operation. However, the margin is not large enough to ignore high-throttle operation. As throttle increases, motor speed and rotor excitation frequency also increase. At full throttle, the motor speed was estimated as approximately 5500 rpm, giving:

$$f_{\text{full}} = \frac{5500}{60} = 91.67 \text{ Hz} \quad (6.7)$$

Because 91.67 Hz is above the first natural frequency of 87.891 Hz, the aircraft would pass through the first bending-mode frequency before reaching full throttle. The motor speed corresponding to the first natural frequency is:

$$RPM_{\text{resonance}} = f_1(60) \quad (6.8)$$

$$RPM_{\text{resonance}} = 87.891(60) = 5273.46 \text{ rpm} \quad (6.9)$$

This means that operation near approximately 5273 rpm could excite the first bending mode if the aircraft were held near that speed. To reduce this risk, an 85% throttle limit was selected during testing. At this limit, the maximum motor speed was estimated as approximately 5000 rpm, corresponding to:

$$f_{85\%} = \frac{5000}{60} = 83.33 \text{ Hz} \quad (6.10)$$

The 85% throttle limit keeps the maximum expected motor excitation frequency below the first natural frequency of the arm. Although the margin between 83.33 Hz and 87.891 Hz is limited, this operating restriction gives approximately 5% separation between the first bending mode and the maximum expected excitation frequency during testing.

Table 11: Modal Analysis Results for Pegasus Propulsion Arm

Mode No.	Freq. (rad/sec)	Freq. (Hz)	Period (s)
1	552.23	87.891	0.011378
2	711.95	113.31	0.0088253
3	3995.2	635.86	0.0015727
4	5348.8	851.29	0.0011747
5	5388.7	857.64	0.001166

The modal analysis results should be interpreted as an engineering approximation rather than an exact prediction of the physical aircraft response. The simulation simplified several real-world factors, including clamp compliance, fastener preload, damping, motor-mount stiffness, propeller imbalance, and interaction between the arms and central frame. Physical validation using onboard IMU logs, external accelerometers, or impact-hammer testing would provide stronger confirmation of the simulated natural frequencies. Despite these limitations, the modal analysis provided useful evidence that the Pegasus arm design maintains acceptable dynamic separation during hover, while also showing the need for a conservative throttle limit to avoid sustained operation near the first bending mode.

6.3 Sensitivity Analysis

To complement deterministic propulsion sizing and physical flight testing, a Monte Carlo sensitivity analysis was performed to evaluate how uncertainty in aircraft loading, battery condition, temperature, and wind could affect mission endurance. The revised analysis was intentionally calibrated to the final Pegasus hexacopter configuration and to the mission profile observed during competition testing, rather than to a conservative hover-only worst-case model. The goal was to preserve the same analytical flow used in the standalone sensitivity report while presenting it in the thesis format.

The analysis estimated the probability of satisfying the DBVF 10-minute mission requirement, identified which active uncertainties most strongly affected flight time, and quantified design margins in weight, hover throttle, and endurance. A simulated case was considered successful only when it satisfied both the competition weight limit and the minimum flight-time requirement:

$$t_{\text{flight}} \geq 10 \text{ min} \quad (6.11)$$

$$W_{\text{total}} \leq 25 \text{ lb} \quad (6.12)$$

6.3.1 Objective and Modeling Approach

The updated sensitivity model was calibrated to the final Pegasus hexacopter configuration rather than using a generic worst-case hover model. Earlier versions of the analysis stacked several conservative penalties together, which made the model useful for preliminary sizing but less representative of the aircraft flown during final DBVF testing. The final formulation instead used the team’s selected propulsion data, expected operating weight range, and observed mission profile to estimate realistic mission robustness.

The primary success condition was defined as completing the mission while satisfying both the DBVF maximum takeoff weight constraint and the minimum flight-time requirement. In mathematical form, a simulated case was considered successful when the total all-up weight was less than or equal to 25 lb and the predicted flight time was greater than or equal to 10 minutes. This allowed the model to directly evaluate the aircraft against the same high-level endurance and weight constraints used during design validation.

6.3.2 Weight, Power, and Battery Modeling

Instead of varying individual payload masses and then summing them into total takeoff weight, the simulation treated total all-up weight as the uncertain quantity of interest. The modeled all-up weight followed a triangular distribution with a lower bound of 18 lb, a most likely value of 20 lb, and an upper bound of 22 lb:

$$W_{\text{total}} \sim \text{Triangular}(a, c, b) \quad (6.13)$$

$$a = 18 \text{ lb}, \quad c = 20 \text{ lb}, \quad b = 22 \text{ lb} \quad (6.14)$$

The margin relative to the DBVF maximum takeoff weight was calculated for every simulation case as:

$$\Delta W = W_{\text{max}} - W_{\text{total}} = 25 - W_{\text{total}} \quad (6.15)$$

Total maximum thrust was computed from the six selected motors:

$$T_{\text{max}} = nT_{\text{motor}} \quad (6.16)$$

$$T_{\text{max}} = 6(3634 \text{ g}) = 21804 \text{ g} \quad (6.17)$$

For each sampled aircraft weight, hover throttle was estimated using the ratio of required vehicle weight to total available thrust:

$$\theta_{\text{hover}} = \left(\frac{W_{\text{total}}^{(g)}}{T_{\text{max}}} \right) \times 100\% \quad (6.18)$$

The corresponding thrust-to-weight ratio was then calculated as:

$$TWR = \frac{T_{\max}}{W_{\text{total}}^{(g)}} = \frac{1}{\theta_{\text{hover}}/100\%} \quad (6.19)$$

Hover power was obtained from manufacturer thrust-to-power data for the T-Motor MN4110 KV340 motor with the 18×6.1 propeller. Figure 6.5 shows the electrical power per motor plotted against thrust per motor.

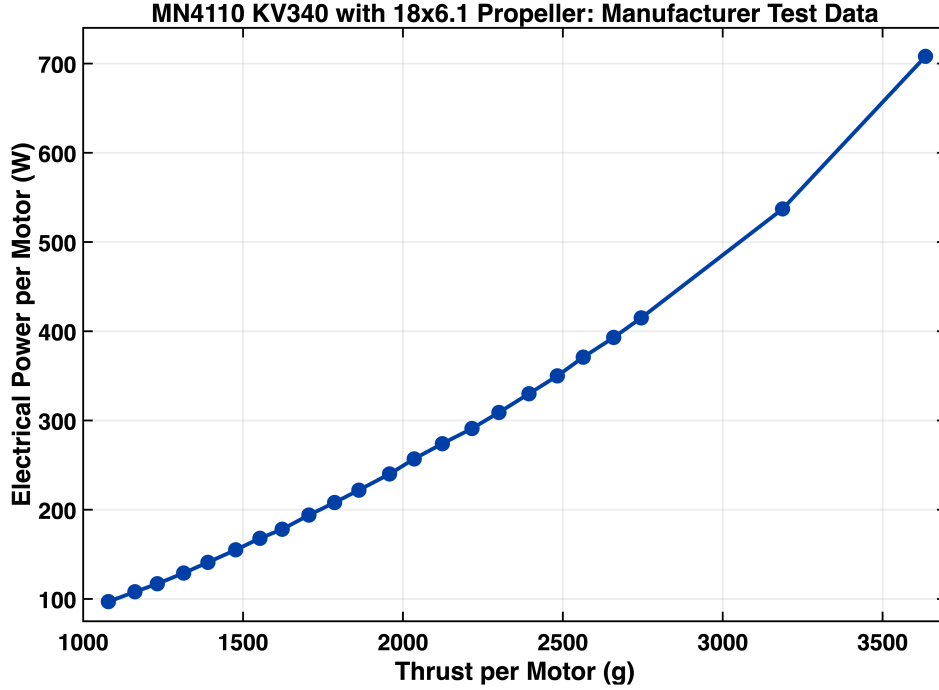


Figure 6.5: Manufacturer thrust-to-power data used for the Monte Carlo hover-power interpolation

For each Monte Carlo trial, the required thrust per motor was first calculated by dividing the sampled vehicle weight by the six rotors:

$$T_m = \frac{W_{\text{total}}^{(g)}}{n} \quad (6.20)$$

The model then interpolated the manufacturer test data to determine electrical hover power per motor. The MATLAB implementation used piecewise cubic Hermite interpolation (PCHIP) with a low-thrust anchor point so that the hover-power curve behaved realistically below the first manufacturer data point:

$$P_{\text{hover,per motor}} = I(T_m; \{(T_{m,j}, P_{m,j})\}) \quad (6.21)$$

The total hover power was then calculated by multiplying the interpolated power per motor by the number of rotors:

$$P_{\text{hover,total}} = nP_{\text{hover,per motor}} \quad (6.22)$$

This empirical method was preferred over a simplified momentum-theory estimate because it directly reflected the selected motor-propeller combination and included practical electrical losses from the hardware data.

Because the DBVF mission was not a continuous steady-hover mission, the hover-power estimate was converted into a mission-average power estimate. The mission included landing segments, ground-contact periods, low-power transitions, and other intervals where the average electrical power was lower than steady hover power. This behavior was captured using a calibrated mission-average factor:

$$k_{\text{mission}} = 0.90 \quad (6.23)$$

$$P_{\text{mission,base}} = k_{\text{mission}} P_{\text{hover,total}} \quad (6.24)$$

Wind was included as a modest power penalty over a 0 to 12 mph range. The wind factor increased mission-average power by up to 12 percent at the maximum modeled wind speed:

$$k_{\text{wind}}(v_w) = 1 + \left(\frac{v_w}{12}\right) (0.12), \quad v_w \in [0, 12] \text{ mph} \quad (6.25)$$

The pilot factor from the earlier version of the analysis was intentionally neutralized and removed from the stochastic input set:

$$k_{\text{pilot}} = 1.0 \quad (6.26)$$

The total mission-average electrical power used for each simulation case was therefore:

$$P_{\text{total}} = P_{\text{hover,total}} k_{\text{mission}} k_{\text{wind}} \quad (6.27)$$

Battery energy was modeled using the 6S, 10000 mAh propulsion battery. The nominal battery energy was calculated as:

$$E_{\text{battery}} = \frac{C_{\text{battery}} V_{\text{battery}}}{1000} \quad (6.28)$$

$$C_{\text{battery}} = 10000 \text{ mAh}, \quad V_{\text{battery}} = 22.2 \text{ V} \quad (6.29)$$

Battery capacity and temperature effects were retained as active stochastic factors:

$$\beta_C \sim U(0.90, 0.98) \quad (6.30)$$

$$\beta_T \sim U(0.95, 1.00) \quad (6.31)$$

An 85 percent usable-energy fraction was applied to preserve practical landing reserve while avoiding the overly conservative assumptions used in early sizing models:

$$\alpha_{\text{usable}} = 0.85 \quad (6.32)$$

$$E_{\text{usable}} = E_{\text{battery}} \beta_C \beta_T \alpha_{\text{usable}} \quad (6.33)$$

Finally, predicted flight time was calculated from usable energy and mission-average power:

$$t_{\text{flight}} = \frac{E_{\text{usable}}}{P_{\text{total}}} \times 60 \text{ min} \quad (6.34)$$

Equivalently, substituting the battery-energy expression gives:

$$t_{\text{flight}} = \frac{\left(\frac{C_{\text{battery}} V_{\text{battery}}}{1000} \right) \beta_C \beta_T \alpha_{\text{usable}}}{P_{\text{total}}} \times 60 \quad (6.35)$$

In summary, because the DBVF mission was not a continuous steady-hover mission, the total power model included a calibrated mission-average power factor. The final model used a factor of 0.90 to account for lower-power portions of the competition mission, including landing, ground-contact periods, and low-power transitions. Wind was included as a modest average power penalty over a 0 to 12 mph range, with the maximum modeled wind condition increasing mission-average power by 12 percent. Pilot-efficiency effects were intentionally removed from the final uncertainty set so that the model focused on vehicle and environmental parameters rather than subjective operator variation.

Battery energy was modeled using the nominal 6S, 10000 mAh propulsion battery capacity and voltage. The usable battery energy was then adjusted by a battery-capacity factor, a temperature factor, and an 85 percent usable-energy fraction. This retained a practical landing reserve while avoiding the overly conservative assumptions used in earlier versions of the model. The resulting flight-time estimate for each simulation case was computed from the usable energy divided by the predicted mission-average electrical power.

6.3.3 Active Uncertain Parameters

Only parameters that remained active in the cleaned MATLAB implementation were varied in the final analysis. This helped avoid double-counting losses that were already captured by the manufacturer thrust-to-power data and prevented outdated assumptions from influencing the final interpretation. The active uncertain parameters are summarized in Table 12.

Parameters such as pilot efficiency, individual sandbag-weight variation, and separate motor or ESC efficiency sweeps were removed from the stochastic input set. Payload uncertainty was represented more cleanly by the total all-up weight distribution, and motor/ESC losses were already reflected in the manufacturer propulsion data. This made the final model more representative of the actual aircraft and reduced the risk of compounding several conservative assumptions.

6.3.4 Monte Carlo Simulation Framework

The final numerical implementation used 3000 Monte Carlo trials. For uniformly distributed variables, each sample was generated using:

Table 12: Active Uncertain Parameters Used in the Final Monte Carlo Sensitivity Analysis

Parameter	Distribution / Range	Role in Analysis
All-Up Weight	Triangular distribution from 18–22 lb, mode at 20 lb	Captures expected flight-ready loading variation.
Battery Capacity Factor	Uniform distribution from 0.90–0.98	Represents battery health and usable capacity variation.
Temperature Factor	Uniform distribution from 0.95–1.00	Represents mild temperature-related battery performance variation.
Wind Speed	Uniform distribution from 0–12 mph	Applies a modest mission-average power penalty for wind.

$$x^{(i)} = a + r^{(i)}(b - a), \quad r^{(i)} \sim U(0, 1) \quad (6.36)$$

For the triangular all-up-weight distribution, the MATLAB code used inverse-transform sampling. With lower bound a , mode c , upper bound b , and u sampled from a uniform distribution, the sampled vehicle weight was computed as:

$$W_{\text{total}} = a + \sqrt{u(b - a)(c - a)}, \quad u < \frac{c - a}{b - a} \quad (6.37)$$

$$W_{\text{total}} = b - \sqrt{(1 - u)(b - a)(b - c)}, \quad u \geq \frac{c - a}{b - a} \quad (6.38)$$

The simulation workflow followed the report sequence:

1. Select the number of simulations, baseline hardware data, and mission constants.
2. Sample W_{total} , β_C , β_T , v_w , and τ .
3. Compute weight margin using $\Delta W = 25 - W_{\text{total}}$.
4. If W_{total} exceeds 25 lb, mark the case unsuccessful.
5. Compute hover throttle and per-motor hover thrust.
6. Interpolate manufacturer thrust-to-power data to obtain hover power per motor.
7. Compute total hover power, mission-average power, and wind-adjusted power.
8. Compute usable battery energy.
9. Calculate predicted flight time.
10. Mark the trial successful only if $W_{\text{total}} \leq 25$ lb and $t_{\text{flight}} \geq 10$ min.

11. Store weight, weight margin, hover throttle, flight time, success status, and simplified mission points.

After all trials were completed, the model calculated summary statistics and Pearson correlation coefficients between each active input and predicted flight time. These correlations were used to determine which uncertain parameters had the strongest influence on endurance performance:

$$\rho_{X_k, t_{\text{flight}}} = \frac{\text{Cov}(X_k, t_{\text{flight}})}{\sigma_{X_k} \sigma_{t_{\text{flight}}}} \quad (6.39)$$

6.3.5 Results and Design Interpretation

The calibrated Monte Carlo analysis placed Pegasus near the upper end of the acceptable mission-robustness range. Across repeated 3000-sample runs, the predicted probability of satisfying the 10 minute endurance requirement generally fell in the approximate range of 94.0 percent to 95.5 percent. The standalone report used a 95 percent threshold for an excellent rating, so results in the mid-90 percent range were interpreted as near-excellent and robust, especially because Monte Carlo sampling variation can shift the estimated probability by several tenths of a percent.

The sensitivity trends matched the physical intuition developed during propulsion sizing. Increasing all-up weight reduced flight time because each motor had to generate more thrust, which increased electrical power draw through the manufacturer thrust-to-power map. Lower battery capacity factor and lower temperature factor reduced usable stored energy. Higher wind speed increased mission-average power through the wind penalty factor. As a result, weight management, battery health, and wind conditions were confirmed as the most important operational factors for maintaining endurance margin. The final design also maintained useful margin in the two most important feasibility constraints. The entire modeled weight range remained below the 25 lb DBVF limit, giving a positive weight margin for all simulated cases. The hover-throttle calculation remained within a usable control range, supporting the conclusion that Pegasus retained sufficient thrust authority for disturbance rejection and maneuvering during the mission.

The Monte Carlo results were first evaluated using the overall aircraft weight distribution. As shown in Figure 6.6, the simulated all-up weight remained well below the DBVF maximum takeoff weight limit of 25 lb across all trials. The distribution is centered near the expected operating weight of approximately 20 lb, with the upper end of the modeled range remaining below 22 lb. This result confirms that the final Pegasus configuration retained a meaningful weight margin even when loading uncertainty was included.

The flight-time distribution is shown in Figure 6.7. Most simulated trials exceeded the 10-minute mission requirement, with the mean flight time falling above the required threshold. A small portion of the distribution falls below 10 minutes, which represents cases where unfavorable combinations of higher weight, reduced battery capacity, lower temperature

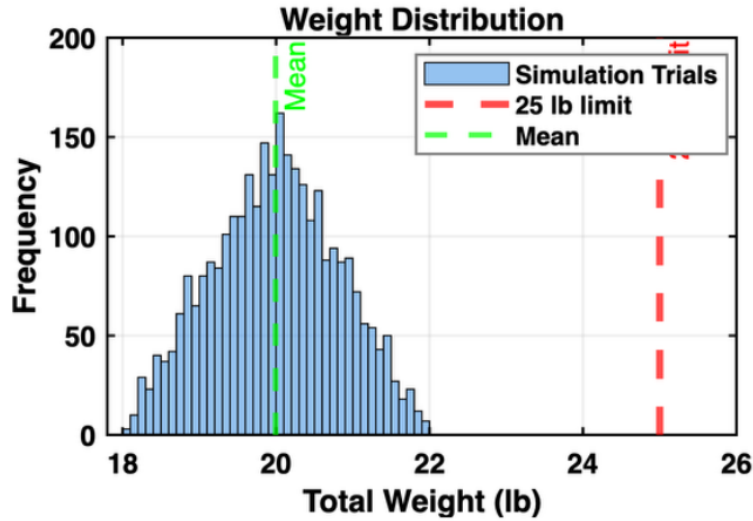


Figure 6.6: Monte Carlo distribution of simulated all-up weight relative to the 25 lb DBVF maximum takeoff weight limit.

factor, and increased wind reduced the available endurance. Overall, the distribution shows that Pegasus has a strong probability of completing the required mission duration while preserving practical battery reserve.

Figure 6.8 shows the simulated hover throttle distribution. The hover throttle remained well below the 50 percent reference line for most trials, indicating that the propulsion system was not operating near its maximum capability during nominal hover. This result supports the propulsion sizing conclusion that the selected motor-propeller combination provides adequate thrust margin for disturbance rejection, maneuvering, and payload operations.

To better understand which environmental and vehicle parameters most strongly affected endurance, individual parameter trends were evaluated against flight time. Figure 6.9 shows the effect of wind speed on predicted flight time. The negative correlation indicates that increasing wind speed reduces endurance because the aircraft must use more power to maintain stable flight and complete the mission. Although the effect is moderate, wind remains an important operational consideration because it directly increases mission-average power demand.

Battery condition was also found to influence predicted endurance, as shown in Figure 6.10. The positive correlation indicates that higher battery capacity factors resulted in longer flight times. This trend is expected because healthier batteries provide more usable energy before reaching the reserve threshold. The result emphasizes the importance of battery condition monitoring, proper charging practices, and avoiding degraded packs during competition or field operation.

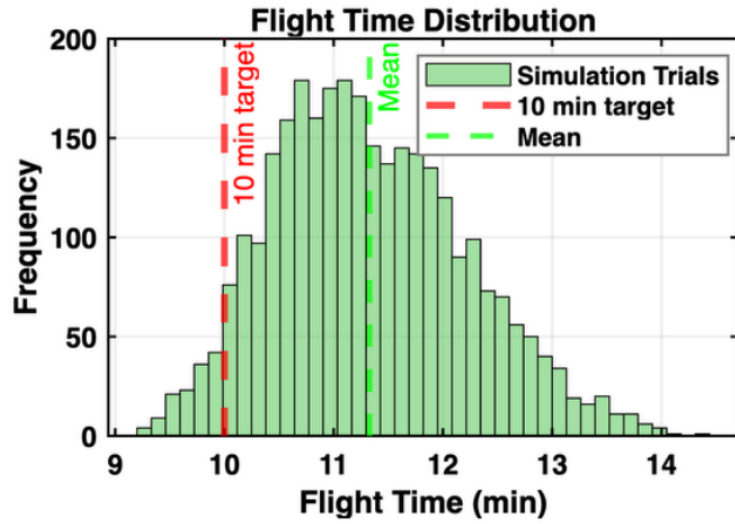


Figure 6.7: Monte Carlo distribution of simulated all-up weight relative to the 25 lb DBVF maximum takeoff weight limit.

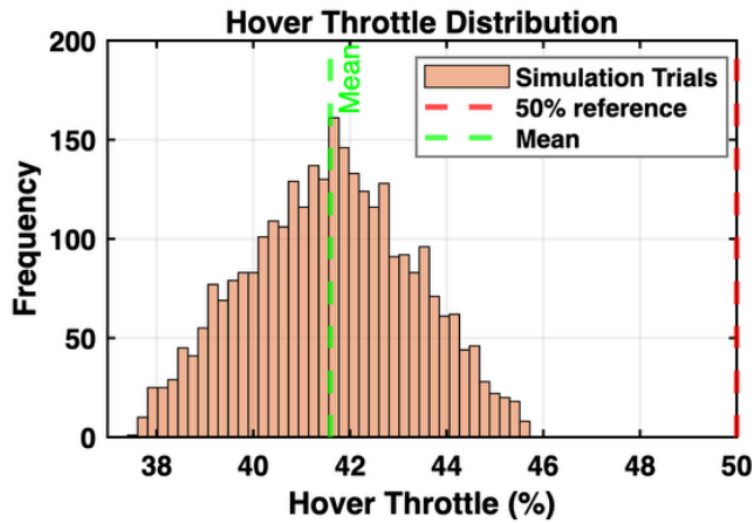


Figure 6.8: Monte Carlo distribution of predicted hover throttle relative to the 50% reference value

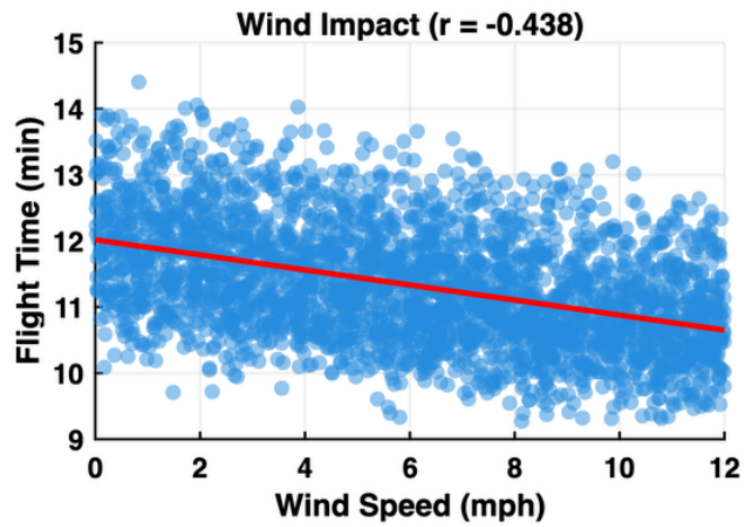


Figure 6.9: Effect of wind speed on predicted flight time, showing the negative correlation between wind and endurance.

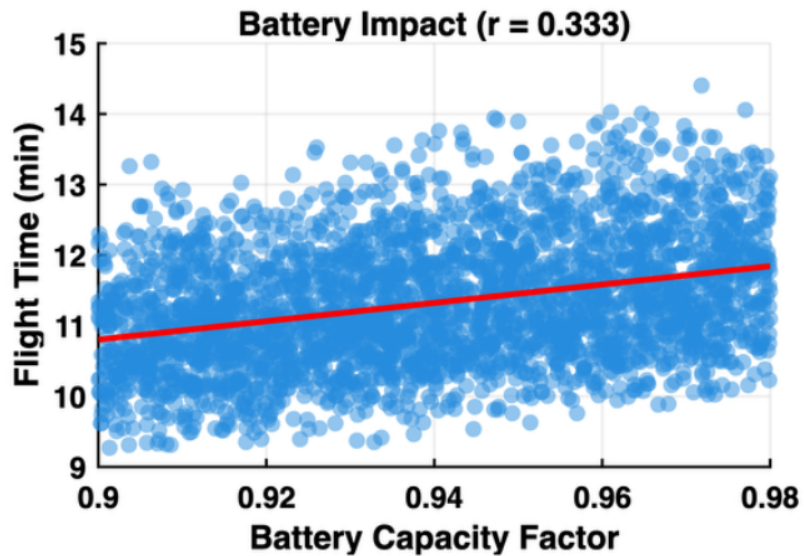


Figure 6.10: Effect of battery capacity factor on predicted flight time, showing increased endurance for healthier battery conditions.

The temperature factor trend is shown in Figure 6.11. The positive correlation indicates that improved temperature-related battery performance slightly increases flight time. Compared with weight and wind effects, temperature had a weaker influence in this analysis because only a mild operating range was modeled. However, the trend still supports the broader conclusion that battery performance can affect endurance, especially in more extreme environmental conditions.

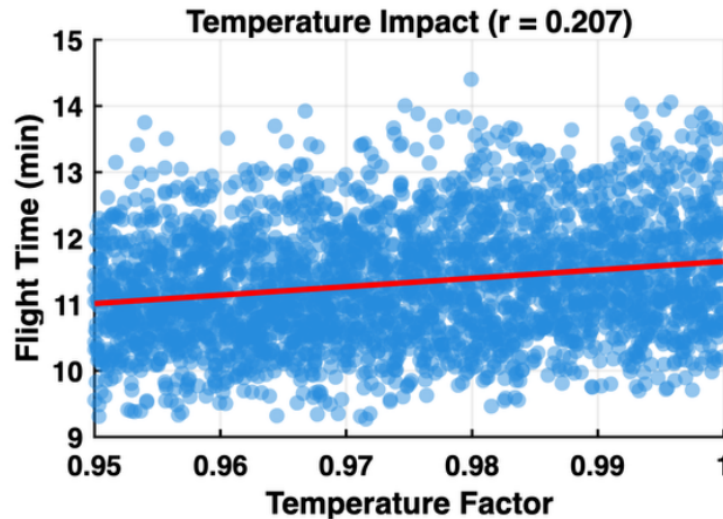


Figure 6.11: Effect of temperature factor on predicted flight time, showing a smaller positive correlation with endurance.

Figure 6.12 compares all-up weight against predicted flight time. This relationship produced the strongest trend in the analysis, with increasing aircraft weight clearly reducing endurance. Heavier cases require more thrust from each motor, which increases hover power and reduces available flight time. The strong negative correlation confirms that weight management is the most important design and operational factor for maintaining endurance margin.

The overall mission success rate is summarized in Figure 6.13. A trial was classified as successful only if the aircraft remained below the 25 lb DBVF weight limit and achieved at least 10 minutes of predicted flight time. The resulting success rate of approximately 95.3 percent indicates that Pegasus is robust across the modeled uncertainty range. This result supports the conclusion that the aircraft has sufficient endurance margin for the DBVF mission profile, though the small failure portion still highlights the need for conservative battery reserve and favorable operating conditions.

Finally, Figure 6.14 summarizes the relative sensitivity of the active uncertain parameters using correlation coefficients. All-up weight had the strongest influence on flight time, followed by wind speed, battery capacity factor, and temperature factor. This ranking confirms that

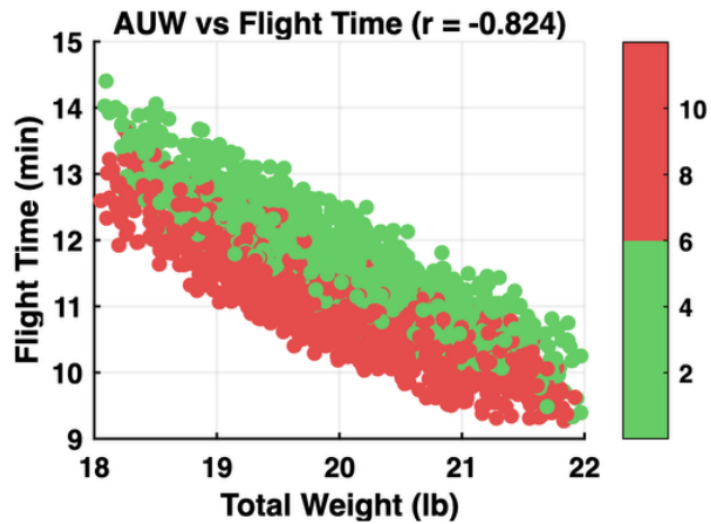


Figure 6.12: Relationship between total all-up weight and predicted flight time, with successful and unsuccessful trials shown separately.

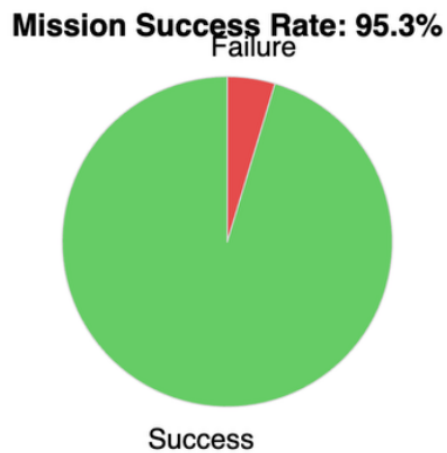


Figure 6.13: Monte Carlo mission success rate showing the percentage of trials satisfying both the 25 lb weight limit and 10 minute flight-time requirement.

reducing aircraft mass and avoiding high-wind operation are the most effective ways to improve endurance reliability. Battery health also remains important, while temperature had a smaller effect under the mild conditions modeled.

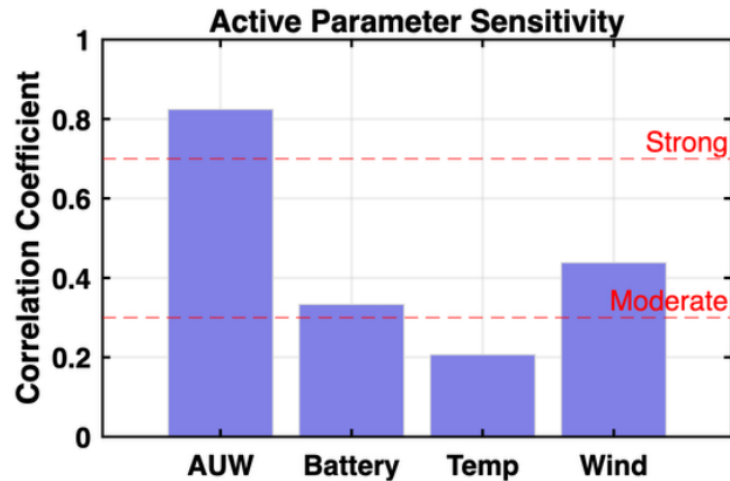


Figure 6.14: Active parameter sensitivity ranking based on correlation coefficient magnitude for all-up weight, battery condition, temperature, and wind.

Taken together, the sensitivity analysis demonstrates that Pegasus maintains positive margins in weight, hover throttle, and endurance under realistic uncertainty. The results show that the aircraft is most sensitive to all-up weight, confirming the importance of lightweight structural design and careful payload management. The analysis also shows that battery condition and wind can meaningfully affect flight time, which should be considered during preflight planning and mission execution. Overall, the Monte Carlo framework supports the conclusion that Pegasus is capable of satisfying the DBVF 10-minute endurance requirement with a high probability of success.

7 Manufacturing

7.1 Bill of Materials

Table 13 summarizes the major components used to manufacture and assemble the Pegasus aircraft. The bill of materials includes the primary airframe, propulsion, avionics, controls, and payload-system components. Each item is organized by subsystem, quantity, part name, description, unit cost, and actual cost. This table was used to track procurement, estimate total project cost, and verify that the selected hardware remained within the project budget.

Table 13: Bill of Materials

Part No.	Qty	Subsystem	Part	Part Description	Unit Cost	Actual Cost
1	2	Controls	DC-DC Step-Down	Buck converter module for proper power distribution	5.00	9.99
2	1	Avionics	Receiver	Radiomaster ELRS Receiver 2.4 GHz	23.99	23.99
3	1	Avionics	Flight Controller	Cube Orange+	264	264
4	1	Avionics	GPS	Here4 GPS	264	264
5	1	Avionics	Telemetry Set	RFD900x Telemetry Set	264	264
6	2	Propulsion	6S Battery	10000 mAh 22.2 V Battery	150.6	301.21
7	2	Avionics	3S Battery	33000 mAh 11.1 V	22.95	45.89
8	1	Controls	Remote ID Module	Holy Stone FAA Compliant Remote ID Module	34.99	34.99
9	1	Controls	Charger	LiPo Battery Charger 1S–6S	38.99	38.99
10	1	Airframe	Bottom Plate	Aluminum 6061-T6 bottom plate with machining	76.89	76.89
11	1	Airframe	Top Plate	Aluminum 6061-T6 top plate with machining	76.89	76.89
12	1	Payload System	Linear Actuator	Progressive Automations 12 V, 39 lbf force	78.08	78.08
13	1	Airframe	M3 Hardware	M3 screws assortment kit	9.99	9.99
14	1	Airframe	M4 Hardware	M4 passivated 18-8 stainless steel Phillips flat head screw	53.19	53.19
15	1	Airframe	Heat Set Inserts	M2, M3, M4, M5 threaded inserts assortment kit	15.99	15.99
16	2	Airframe	Acrylic Plates	TAP Plastics acrylic plates	100	100
17	1	Airframe	Steel Nuts (M4)	Zinc plated steel press-fit nut, sheet metal	7.56	7.56
18	1	Airframe	Rectangular Tubing	Aluminum rectangular tubing	80.82	80.82
19	6	Propulsion	ESC	FLAME 60A 12S V2.0 ESC	99.99	599.94
20	6	Propulsion	Motors	MN4110 T-Motor motors	86.99	521.94
21	4	Propulsion	Propellers	18 × 6.1 carbon-fiber propeller, 1 pair per package	82.90	331.60
22	1	Controls	PDB	MATEKSYS FCHUB 12S V2 PDB	27.11	27.11
23	1	Payload System	Linear Actuator ESC	60A 2–3S electronic speed controller	29.52	29.52
24	1	Controls	Wires/Misc.	Different wire types and connectors	40	40

7.2 Machining

The primary structural components of the drone were manufactured from 6061-T6 aluminum due to its high strength-to-weight ratio, good machinability, and overall durability. A 3-axis

CNC milling machine was used to machine the main aluminum components, including the arms and the central frame assembly. The central frame consists of both the top and bottom plates, which were CNC machined to ensure accurate hole placement, consistent geometry, and proper alignment during assembly.

The arms were also CNC machined from 6061-T6 aluminum tubes. Since the arms have a more complex geometry and needed to be securely held during machining, a custom fixture was designed and manufactured to properly position and support each arm during the CNC process.

7.3 3D Printing

Several drone components were manufactured using 3D printing, including the arm clamps, payload mechanism casing, landing gear connection pieces, and other nonstructural parts. 3D printing was selected because it allows complex geometries to be produced more easily than traditional machining, while also enabling faster design iteration and lower manufacturing cost.

Different printing materials were used depending on the function and load requirements of each part. ASA was used for the arm clamps because these components connect the arms to the central frame and must withstand structural loads during flight. ASA was also used for the landing gear connection pieces because they experience impact loads during landing and must support approximately two to three times the weight of the drone. In contrast, PLA was used for the payload mechanism because it only needs to support the maximum payload weight of 10 lb. Other nonstructural components, such as the landing gear shoes and the connection pieces for the acrylic landing gear, were also printed in PLA because they do not carry significant loads.

This material selection allowed the team to balance strength, weight, cost, and manufacturability while ensuring that each printed component met its functional requirements.

7.4 Laser Cutting

Laser cutting was used to manufacture the reinforced acrylic components of the landing gear. This method was selected because it provides precise cuts, clean edges, and consistent part dimensions, which are important for maintaining proper alignment in the landing gear assembly. Since acrylic sheets are difficult to shape accurately using manual cutting methods, laser cutting allowed the team to produce the landing gear parts efficiently while reducing manufacturing errors.

The reinforced acrylic was chosen because it provides a balance of strength, stiffness, and low weight. These properties are important because the landing gear must support the drone during takeoff and landing while also absorbing impact loads. By using laser cutting, the team was able to create accurate and repeatable landing gear components that fit properly with the 3D-printed connection pieces. Another benefit is that the landing gear can be used as a buffer if the aircraft crashes and it can break on impact.

8 Testing & Validation

The testing and validation phase of the Pegasus project was conducted to verify that the aircraft satisfied all competition requirements, subsystem performance objectives, and operational safety criteria established during the conceptual and detailed design phases. These requirements are summarized in Table 14. Because Pegasus was designed for wildfire-response applications, testing emphasized not only aerodynamic and structural performance, but also reliability, stability, vibration behavior, subsystem integration, and operational robustness under realistic mission conditions.

The overall validation philosophy of the project followed an incremental systems-engineering approach in which subsystem functionality was verified individually before progressing toward full-system integration and flight testing. This methodology minimized operational risk by identifying structural, electrical, control-system, and vibration-related issues early in development before exposing the aircraft to higher-risk flight conditions.

Testing procedures were divided into several progressive stages:

1. Ground testing and structural validation
2. Hover and stability testing
3. Basic controls testing
4. Autonomous mission testing

Each stage was designed to validate increasingly complex aspects of aircraft operation while maintaining conservative safety margins and controlled operating conditions. Validation methods included analytical modeling, finite element analysis, ground analysis, propulsion calculations, subsystem inspections, telemetry logging, and physical flight testing.

In addition to validating aircraft performance, the testing phase also served as a critical learning process for refining subsystem integration, improving control tuning, and verifying maintainability and operational readiness. Because multirotor aircraft are highly sensitive to vibration, center-of-gravity placement, propulsion mismatch, and control-loop tuning, iterative testing became essential for ensuring stable and repeatable flight behavior.

The final testing strategy therefore focused on three major objectives:

- Ensuring flight safety and structural reliability
- Verifying mission-performance capability
- Validating operational readiness for wildfire-response applications

Table 14: System Validation and Acceptance

Requirement	Source	Target/Threshold	Verification Method	Current Value	Status
Maximum takeoff weight (MTOW)	DBVF	≤ 25 lb	Scale measurement	18.02 lb	Pass
Propulsion battery voltage limit	DBVF	≤ 6 S	Battery spec sheet	6S	Pass
Separate propulsion and avionics power	DBVF	Required	Wiring inspection	6S + 3S	Pass
External propulsion battery disconnect	DBVF	Required	Inspection and functional test	Satisfied	Pass
Communications kill provision	DBVF	Required	Functional test	Satisfied in wiring	Pass
Minimum flight endurance	DBVF	≥ 10 min	Calculations and hover log	10.6 min	Pass/TBD
Hover thrust margin, T/W	Derived	≥ 2.0 at design weight	Calculated from thrust data	2.66	Pass
Hover throttle margin	Derived	Hover $\leq 60\%$ throttle	Flight log data	$\sim 47\%$	Pass
Structural factor of safety	Derived	≥ 2.0	FEA results	Table 11	Pass
Maximum arm deflection under load	Derived	≤ 1 mm	FEA displacement	0.2064 mm	Pass
Resonance separation at hover	Derived	$f_1 \geq 1.45 * 59.63 \text{ Hz}$	Modal analysis	87.891 Hz	Pass
Propeller clearance	Derived	≥ 3 in	CAD measurement	4.5 in	Pass
Safe ground and flight test plan	Derived	Documented	Test procedure	Complete	Pass
Payload secure mounting	Derived	No slip/release	Ground shake test	Payload stays secure	Pass

8.1 Testing Strategy

The testing strategy for Pegasus was intentionally designed around progressive risk reduction and subsystem validation prior to full-flight operations. Because the aircraft integrated multiple tightly coupled subsystems: including propulsion, avionics, controls, payload systems, structural components, and autonomous navigation, it was critical to isolate and verify individual subsystems before conducting integrated mission testing.

The overall testing methodology followed a bottom-up validation process beginning with analytical verification and static subsystem inspection before advancing toward powered ground tests, hover validation, and autonomous mission execution. This structured progression reduced the likelihood of catastrophic integration failures and allowed engineering issues to be corrected incrementally.

Testing procedures were developed around the following principles:

Incremental Complexity

Each stage of testing introduced only one additional layer of operational complexity at a time. For example:

- Structural integrity was validated before propulsion spin-up
- Propulsion behavior was validated before free-flight operation

- Hover stability was verified before translational maneuvering
- Manual flight capability was confirmed before autonomous mission testing

This incremental methodology ensured that subsystem-level issues could be isolated efficiently without introducing unnecessary operational risk.

Conservative Operating Margins

All tests were conducted using conservative throttle limits, reduced flight altitudes, and restricted operating envelopes during early validation phases. The aircraft was intentionally operated below maximum performance conditions during initial testing to protect hardware and maintain adequate recovery margins.

This conservative philosophy was especially important because wildfire-response aircraft prioritize reliability and stable operation rather than aggressive maneuverability or peak-speed performance.

Data-Driven Validation Testing relied heavily on telemetry logging, vibration monitoring, propulsion analysis, battery monitoring, and flight-data review. IMU data, throttle response, ESC temperatures, battery voltage sag, and vibration signatures were continuously monitored throughout testing to verify system health and identify abnormal behavior.

Analytical models and simulation predictions developed earlier in the project were also compared directly against measured test data to validate subsystem sizing assumptions and design calculations.

Safety-Centered Procedures Safety was treated as a primary requirement throughout all validation activities. Ground testing was performed with the aircraft restrained to fixed test stands, while flight tests utilized controlled open-field environments with designated safety perimeters and emergency shutdown capability. Pre-flight inspection procedures, communication protocols, and fail-safe verification routines were incorporated into all testing operations.

The resulting testing strategy balanced engineering rigor with operational practicality while ensuring that the aircraft could be validated systematically before progressing toward advanced mission operations.

8.2 Ground Testing

Ground testing represented the first major physical validation stage of the Pegasus aircraft and focused on verifying structural integrity, propulsion integration, electrical reliability, vibration behavior, and subsystem functionality before attempting free flight. Because multirotor aircraft can experience severe instability or catastrophic failure from improperly integrated propulsion systems or excessive vibration, thorough ground testing was considered essential prior to hover operations.

The aircraft was mounted securely to a rigid test stand during initial propulsion testing to prevent unintended lift-off while allowing safe motor operation across increasing throttle ranges.

Ground-testing procedures included:

- Structural inspections
- Fastener verification
- ESC calibration
- Motor direction verification
- Power-system validation
- Telemetry checks
- RC communication testing
- Fail-safe verification
- Vibration analysis

Mechanical inspections were first conducted to verify:

- Arm alignment
- Clamp security
- Structural attachment integrity
- Propeller clearance
- Payload retention
- Wiring management

Special attention was given to the arm-to-frame interfaces because these joints experienced significant vibratory loading during propulsion operation.

Following mechanical inspection, propulsion-system testing was conducted incrementally in approximately 10% throttle intervals while monitoring:

- Current draw
- ESC temperatures
- Motor temperatures
- Vibration levels

- Structural resonance behavior
- Telemetry stability

One of the most important objectives during ground testing was verifying that structural natural frequencies remained sufficiently separated from rotor excitation frequencies during hover operation. Modal analysis predicted a first structural natural frequency near 87.8 Hz, while hover rotor excitation occurred near 59.6 Hz, producing an acceptable resonance separation ratio of approximately 1.47.

Ground vibration testing confirmed that no dangerous oscillatory behavior occurred within the intended operating envelope. Additionally, propulsion-system validation demonstrated that:

- Hover throttle remained below 60%
- The thrust-to-weight ratio exceeded 2.0
- Motor and ESC temperatures remained within acceptable operating limits
- Structural deflection remained below allowable thresholds

The ground-testing phase also validated the aircraft’s electrical architecture, including separate propulsion and avionics power systems, external battery disconnect capability, and communication-based kill-switch functionality.

Overall, ground testing confirmed that the aircraft was structurally sound, electrically reliable, and mechanically stable enough to progress safely into hover testing.

8.3 Hover Testing

Hover testing represented the first stage of free-flight validation and focused on verifying aircraft stability, controllability, propulsion performance, vibration behavior, and thermal performance during sustained hover conditions.

Because hover operation is one of the most demanding flight regimes for multirotor aircraft, especially in wildfire-response missions where stationary observation and payload deployment are common, stable hover performance was treated as a primary mission requirement.

Initial hover testing began with short-duration “hop tests” performed at very low altitude to confirm:

- Correct control response
- Stable attitude behavior
- Accurate flight-controller orientation
- Basic propulsion balance

These tests were intentionally brief to minimize risk while validating the aircraft’s fundamental control behavior.

After successful hop testing, sustained hover tests were conducted at approximately 0.5–1.5 meters altitude while monitoring:

- Hover throttle percentage
- IMU vibration levels
- Motor temperatures
- ESC temperatures
- Battery current draw
- Flight stability
- Oscillation behavior

A major objective during hover testing was confirming that rotor-induced vibration did not excite structural resonance modes or degrade sensor performance. Excessive vibration in multirotor systems can corrupt IMU measurements, destabilize control loops, and negatively affect autonomous navigation accuracy.

Flight logs confirmed that the aircraft maintained stable hover behavior without significant oscillation or control instability. Hover throttle stabilized near approximately 47%, validating earlier propulsion sizing calculations and confirming that adequate thrust margin remained available for maneuvering and gust rejection.

Hover endurance testing also demonstrated mission durations exceeding ten minutes while maintaining acceptable battery reserve margins, satisfying the DBVF minimum endurance requirement.

The hover-testing phase additionally validated:

- Stable center-of-gravity placement
- Effective control tuning
- Reliable ESC synchronization
- Proper power-distribution behavior
- Consistent flight-controller operation

Overall, hover testing confirmed that Pegasus possessed the stable low-speed flight characteristics necessary for wildfire reconnaissance and payload deployment missions.

8.4 Basic Controls Testing

Basic controls testing focused on validating manual flight capability, attitude response, translational maneuvering, yaw authority, and pilot controllability before transitioning into autonomous mission operations.

Although Pegasus was designed primarily as an autonomous wildfire-response platform, safe manual override capability remained critical for:

- Flight recovery
- Emergency intervention
- Precision maneuvering
- Initial tuning
- Safety verification

Testing evaluated all primary flight-control axes:

- Roll
- Pitch
- Yaw
- Throttle response

Initial controls testing emphasized low-speed translational maneuvers and gentle attitude inputs to confirm:

- Stable flight-controller tuning
- Predictable aircraft response
- Proper ESC synchronization
- Balanced propulsion output
- Smooth throttle transitions

Special attention was given to yaw response because multirotor yaw authority can degrade under high thrust loading conditions. The hexacopter architecture provided improved rotational control authority compared to some quadcopter configurations due to increased distributed thrust capability.

Flight-control tuning was refined iteratively throughout testing using telemetry logs and pilot feedback. PID gain adjustments were performed conservatively to reduce oscillation tendency while maintaining sufficient responsiveness for stable maneuvering in turbulent environments.

Additional testing validated:

- Return-to-level behavior
- Hover hold stability
- Altitude response
- RC communication reliability
- Flight-mode transitions
- Failsafe activation behavior

The controls subsystem demonstrated stable and predictable handling characteristics throughout manual testing, reinforcing the design philosophy that prioritized reliability and operational stability over aggressive maneuverability.

Because wildfire-response operations emphasize persistent observation and payload precision rather than high-speed flight, the aircraft was intentionally tuned for smooth and stable response behavior instead of rapid acrobatic control authority.

8.5 Autonomous Mission Testing

Autonomous mission testing represented the final and most integrated validation phase of the Pegasus project. These tests evaluated the aircraft's ability to execute mission objectives using autonomous navigation, waypoint following, telemetry communication, and payload-system integration while maintaining stable flight behavior.

The autonomous testing phase validated several major mission functions:

- GPS waypoint navigation
- Autonomous takeoff and landing
- Position hold
- Autonomous hover
- Telemetry communication
- Payload-system operation
- Mission sequencing
- Return-to-launch functionality

Testing began with simple waypoint-navigation exercises involving short-distance autonomous translations between predefined GPS coordinates. These tests verified GPS integration accuracy, flight-controller mission logic, and autonomous control stability.

After basic navigation validation, more complex mission profiles were conducted involving:

- Autonomous takeoff
- Hover stabilization
- Translational waypoint flight
- Return-to-hover operations
- Controlled landing sequences

Payload-system functionality was also evaluated during integrated mission testing. The magnetic linear-actuator payload system was tested to verify:

- Payload retention security
- Actuator deployment reliability
- Repeatable payload release behavior
- Stable aircraft response during payload transfer

These tests demonstrated that the payload mechanism could operate successfully without destabilizing the aircraft or compromising flight safety.

Telemetry reliability and mission-state monitoring were continuously evaluated throughout autonomous testing to ensure that communication links remained stable during mission execution. Flight logs and telemetry data confirmed reliable waypoint tracking, stable hover behavior, and acceptable positional accuracy.

Autonomous mission testing also validated several critical safety functions including:

- Failsafe return-to-home behavior
- Communication-loss handling
- Battery failsafe response
- Controlled landing routines

Overall, the autonomous testing phase demonstrated that Pegasus could perform the core mission tasks required for wildfire-response operations while maintaining stable flight characteristics, reliable telemetry integration, and safe autonomous behavior.

The successful completion of autonomous mission testing validated the aircraft as a functional and operationally relevant wildfire-response eVTOL platform capable of supporting reconnaissance, environmental monitoring, and lightweight payload deployment missions under realistic operating conditions.

9 Professional Issues and Constraints

9.1 Science, Technology & Society

The primary beneficiaries of this system are wildfire response teams, emergency management agencies, and communities located in fire-prone regions. By providing rapid aerial situational awareness within the critical first 15-30 minutes of ignition, the system enhances decision-making and improves response efficiency, ultimately helping to protect lives and property. However, certain groups may be indirectly disadvantaged, including smaller or underfunded fire departments that may lack the resources to adopt such technology. Additionally, the increased use of autonomous aerial systems may shift reliance away from traditional reconnaissance methods, potentially reducing the role of manual observation and introducing a dependency on technological systems.

Unanticipated consequences may include increased airspace congestion during emergency response operations, particularly if multiple unmanned systems are deployed simultaneously. While the system is designed to operate within visual line-of-sight (VLOS), coordination with other aerial assets remains critical to avoid interference. The integration of this technology may also alter operational workflows by prioritizing real-time aerial data, which can improve safety but may reduce redundancy if alternative observation methods are underutilized. As a result, the system is best implemented as a complementary tool that enhances, rather than replaces, existing wildfire response strategies.

9.2 Civic Engagement

The operation of this system is governed primarily by regulations established by the Federal Aviation Administration (FAA), specifically under Part 107 rules for small unmanned aircraft systems. These regulations impose constraints such as maintaining visual line-of-sight (VLOS), altitude limits, and restrictions on operating near people or in controlled airspace without authorization. In addition, coordination with local and state emergency response agencies is required to ensure safe and effective deployment during wildfire incidents. Professional standards from organizations such as the American Society of Mechanical Engineers (ASME) and the Institute of Electrical and Electronics Engineers (IEEE) inform the design and testing of the system, particularly in areas related to structural integrity, electrical safety, and system reliability.

The current prototype addresses these requirements through the implementation of safety-focused design features, including a communications-based kill switch, independent propulsion and avionics power systems, and structured ground and flight testing procedures. By operating within VLOS constraints, the system aligns with current regulatory requirements without the need for additional waivers. However, future policy changes, such as expanded allowances for beyond-visual-line-of-sight (BVLOS) operations or stricter UAV restrictions, could significantly impact the system's scalability and operational flexibility. Continued engagement with regulatory developments will be essential for long-term deployment.

9.3 Health, Safety, and Risk

Health and safety considerations were central to the Pegasus design because the system combines high-speed propellers, lithium-polymer batteries, autonomous flight-control software, high-current electrical systems, and payload deployment capability. These features create potential hazards to operators, emergency personnel, bystanders, property, and other aircraft if the system is operated improperly or if a subsystem fails. To reduce these risks, the prototype incorporated safety-focused features including independent propulsion and avionics power systems, a communications-based kill switch, structured ground testing procedures, conservative throttle limits, and incremental flight validation. The current prototype was designed and tested as a senior-design and DBVF competition aircraft, not as a fully certified emergency-response aircraft; additional regulatory review, environmental testing, pilot training, and operational procedures would be required before real wildfire deployment.

9.4 Environmental Sustainability

Environmental considerations influenced several aspects of the system’s design. The use of electric propulsion eliminates direct emissions during operation, making the system more environmentally friendly than fuel-powered aerial alternatives. Additionally, the modular design approach allows for component replacement, repair, and future upgrades, extending the system’s functional lifespan and reducing overall material waste. These design choices support both operational sustainability and long-term usability.

Despite these advantages, environmental impacts are still present in the production and lifecycle of the system. Materials such as aluminum, carbon fiber, and lithium-polymer batteries require energy-intensive manufacturing processes and resource extraction. Furthermore, improper disposal of electronic components and batteries at the end of life could contribute to environmental harm. To mitigate these effects, the design emphasizes durability, maintainability, and the potential for responsible recycling of key components. Importantly, the system’s ability to support faster and more effective wildfire response may help reduce large-scale environmental damage, providing an indirect but significant environmental benefit.

10 Ethics

10.1 Ethical Justification for the Project

The ethical justification for Pegasus is based on its intended role as a tool for improving wildfire response while reducing unnecessary human exposure to danger. Wildfires can develop rapidly, and emergency personnel often face limited situational awareness because of smoke, wind, difficult terrain, road closures, and the constantly changing fire boundaries. Pegasus was developed to support rapid aerial reconnaissance, environmental monitoring, and lightweight payload delivery without immediately placing responders in hazardous areas.

From an engineering ethics perspective, this project is justified because its goal aligns with the responsibility of engineers to protect public safety, health, and welfare. Pegasus is not intended to replace firefighters, pilots, or any other first responders. Instead, it is designed as a support tool that can help trained responders make faster and better-informed decisions during time-sensitive wildfire events. Its ethical value comes from using autonomous flight to improve situational awareness while preserving human oversight over mission planning, flight operation, and payload deployment.

10.2 Engineering Characters

The Pegasus project shows that being a good engineer requires more than building a system that works. A good engineer must make careful decisions under uncertainty, communicate clearly with a team, recognize technical limits, and prioritize safety over performance when the two conflict. In this project, those responsibilities appeared through conservative propulsion sizing, structural validation, modal analysis, incremental testing, fail-safe design, and honest recognition that the prototype would need more testing before real wildfire deployment.

One important lesson from the project is that ethical engineering requires humility about models and simulations. The finite element analysis, modal analysis, and endurance estimates helped guide the design, but they relied on assumptions about loads, boundary conditions, vibration, battery behavior, and real flight conditions. Because of this uncertainty, the team had to treat simulation results as engineering evidence rather than absolute truth. This is why physical testing, conservative margins, and clear documentation were important parts of the design process.

The project also showed that good engineers must be willing to choose safety over maximum performance. For example, the modal analysis showed that the hexacopter arm's first natural frequency was close to the estimated full-throttle excitation frequency. Rather than ignoring this result, the team selected a conservative throttle limit during testing to reduce the chance of sustained operation near resonance. This decision reflects engineering restraint and responsibility because predictable, safe behavior was prioritized over pushing the aircraft to its maximum possible operating range.

10.3 Ethical Pitfalls, Safety, Risk, the Public, and Informed Consent

Despite its humanitarian purpose, Pegasus raises several ethical risks. The most direct concern is physical safety. The aircraft contains high-speed propellers, lithium-polymer batteries, high-current electrical systems, autonomous control software, and enough mass to cause injury or property damage if control is lost. For this reason, ethical operation requires preflight inspections, controlled test areas, trained operators, emergency shutdown capability, visual-line-of-sight operation, and compliance with applicable FAA regulations.

The public affected by Pegasus includes more than the design team. It includes firefighters, emergency responders, bystanders near a test or deployment area, nearby property owners, communities in wildfire-prone regions, and other aircraft operating in emergency airspace. Some of these groups may benefit from improved situational awareness, while others may be exposed to risk without directly choosing that exposure. This makes risk communication and responsible operating procedures essential.

Autonomy and payload delivery also create ethical concerns. Autonomous waypoint navigation can reduce operator workload, but it may also create overreliance on software if operators trust the system too much. Similarly, a payload-capable drone designed for sensors or emergency supplies could be misused if adapted for harmful materials or unauthorized surveillance. These risks support the need for human-in-the-loop oversight, benign payload limits, secure communication links, mission logging, and clear authorization before payload release.

The principle of informed consent applies differently in testing and real emergency use. In a university or competition test environment, participants can be informed of the risks, safety perimeter, operating limits, and emergency procedures before flight. In a real wildfire environment, however, residents, evacuees, or bystanders may not be able to personally consent to every risk. In that case, consent must be represented through public agencies, emergency-response authorities, regulatory approval, and transparent operating procedures. The ethical responsibility of the engineers and operators is to make risks understandable, avoid unnecessary exposure, and ensure that deployment decisions are made by accountable authorities.

Overall, Pegasus is ethically justified only if it is treated as a carefully tested emergency-response support tool, not as a fully deployable wildfire aircraft in its current prototype form. Future real-world use would require additional environmental testing, regulatory review, operator training, and safety validation. This protects the public while still allowing the project to demonstrate the potential value of autonomous eVTOL systems for wildfire response.

11 Conclusions

The Pegasus wildfire-response eVTOL system successfully demonstrated the feasibility of a modular autonomous multicopter platform designed around the operational demands of early-stage wildfire response. Throughout the project, the aircraft was developed using a systems-engineering approach that translated real-world operational challenges, stakeholder needs, and DBVF competition requirements into an integrated aerospace platform emphasizing stability, reliability, modularity, and autonomous mission capability.

The final aircraft combined a hexacopter propulsion architecture, structurally optimized aluminum airframe, autonomous flight-control system, modular payload integration, adaptive landing gear, and distributed avionics systems into a cohesive wildfire-response platform. Rather than prioritizing aggressive flight performance or maximum aerodynamic efficiency, the design philosophy consistently focused on dependable hover stability, operational robustness, rapid deployment capability, and adaptability to uncertain environments. This philosophy guided major decisions across propulsion sizing, structural design, controls integration, payload architecture, and manufacturing methods.

Analytical modeling and validation demonstrated that Pegasus met or exceeded the majority of the project’s derived engineering requirements. Propulsion analysis confirmed sufficient thrust margin and hover efficiency for stable payload operations, while finite element analysis and modal validation verified structural stiffness, lightweight performance, and acceptable vibration behavior. Testing further demonstrated stable hover capability, reliable subsystem integration, autonomous waypoint navigation, and successful mission execution under realistic operating conditions. Collectively, these results validated the aircraft as a capable proof-of-concept platform for wildfire reconnaissance and lightweight emergency-response operations.

The project also highlighted the importance of multidisciplinary integration in modern aerospace systems. Successful aircraft development required balancing tradeoffs between endurance, structural weight, controllability, manufacturability, maintainability, and operational reliability. In many cases, the most effective engineering decisions were not those that maximized theoretical performance, but rather those that improved system robustness, field-serviceability, and mission consistency. This became especially important given the unpredictable nature of wildfire environments and the operational demands placed on emergency-response systems.

Beyond technical performance, Pegasus demonstrated the broader potential of autonomous eVTOL systems within emergency-response operations. Autonomous aerial systems offer the ability to rapidly provide situational awareness, environmental monitoring, payload deployment, and remote reconnaissance while reducing risk to personnel and minimizing deployment time. As wildfire frequency and severity continue to increase, systems such as Pegasus may contribute to future “Drone-as-First-Responder” operations by supplementing traditional firefighting resources with rapidly deployable autonomous support platforms.

Overall, the Pegasus project successfully achieved its primary objectives as both a DBVF

competition aircraft and a practical engineering exploration into autonomous wildfire-response technology. The final system represents a functional integration of propulsion, structures, controls, manufacturing, and autonomous systems engineering into a unified platform capable of supporting realistic mission-oriented operations. While additional refinement and testing would be required before operational deployment in real wildfire conditions, the project established a strong technical foundation for future development of autonomous emergency-response eVTOL systems.

11.1 Future Work

Although Pegasus successfully demonstrated the feasibility of an autonomous wildfire-response eVTOL platform, several opportunities for future improvement were identified throughout the design, manufacturing, and testing process. Future development efforts would primarily focus on reducing overall system weight, improving payload integration reliability, expanding autonomous capability, and conducting more extensive real-world validation testing.

One of the largest areas for improvement involves structural weight reduction through the use of advanced composite materials. Due to project budget constraints, the aircraft primarily utilized aluminum structural components because of their affordability, manufacturability, and reliability during rapid prototyping. However, future iterations of Pegasus could significantly benefit from the integration of carbon-fiber structural members, composite sandwich panels, and lightweight additive-manufactured components. Carbon-fiber propulsion arms and central plates could substantially reduce structural mass while maintaining or improving stiffness characteristics. Reducing structural weight would directly improve endurance, payload capacity, thrust margin, and maneuverability while also lowering propulsion power requirements during hover operation. Additional topology optimization and composite layup analysis could further improve stiffness-to-weight performance beyond what was achievable within the scope of the current project.

Future work would also focus on improving the payload pickup and deployment mechanism. While the magnetic linear-actuator system demonstrated reliable operation during testing, several alternative payload-interface concepts were identified that could improve attachment consistency, autonomous alignment, and mission flexibility. One possible improvement involves integrating QR-code or visual fiducial marker systems that would allow onboard cameras and autonomous software to more accurately align the aircraft during payload pickup operations. Additional concepts include permanent-electromagnet hybrid systems that would reduce actuator complexity and improve holding reliability while minimizing power consumption during sustained transport operations. Conical or funnel-shaped payload docking geometries could also simplify payload engagement by passively guiding payloads into alignment during descent, reducing precision requirements during autonomous pickup maneuvers. These improvements could significantly enhance autonomous mission reliability in real wildfire environments where visibility, terrain, and environmental disturbances complicate precision operations.

Another important area for future development involves expanded vibration testing and ex-

perimental structural validation. Much of the vibration and modal analysis conducted during this project relied on computational finite element analysis and analytical modeling. While these simulations provided valuable insight into structural behavior and resonance separation, future work should include extensive physical vibration testing of the completed aircraft. Experimental modal analysis using accelerometers, frequency sweeps, and operational vibration measurements during hover could provide more accurate characterization of real-world structural dynamics and rotor-induced excitation behavior. Additional physical testing would improve confidence in long-term structural durability, avionics isolation effectiveness, and flight-control stability under sustained operating conditions.

Autonomous flight capability also presents a major opportunity for future expansion. The current Pegasus platform demonstrated waypoint navigation and stable autonomous operation; however, future versions could incorporate advanced obstacle avoidance, terrain-following algorithms, autonomous target tracking, and AI-assisted wildfire monitoring systems. Integration of thermal imaging cameras, LiDAR systems, and onboard computer-vision processing could allow the aircraft to autonomously detect hotspots, map fire perimeters, identify safe access routes for responders, and improve situational awareness during low-visibility conditions. Machine-learning-based path planning and adaptive mission execution could further improve the aircraft's ability to operate in rapidly changing wildfire environments.

Additional future work could focus on improving environmental survivability for deployment in extreme wildfire conditions. Future aircraft iterations may incorporate thermal shielding, insulated avionics compartments, conformal protective coatings, and active cooling strategies to improve resistance to elevated temperatures, smoke exposure, ash contamination, and harsh environmental conditions. Research into thermally resistant composite materials and protected sensor integration could further expand operational capability near active fire zones.

Endurance improvements also remain an important long-term development goal. While Pegasus successfully met its mission endurance targets, future iterations could explore higher energy-density battery technologies, optimized propeller geometries, improved aerodynamic efficiency, or hybrid power architectures to extend operational flight time. Longer endurance would significantly improve mission usefulness for persistent monitoring, communication relay operations, and extended wildfire reconnaissance missions.

Finally, future development should include expanded field testing in more realistic operational environments. While the current project successfully validated subsystem integration and autonomous mission execution, additional outdoor testing under stronger wind conditions, uneven terrain, elevated temperatures, and degraded visibility environments would provide valuable operational data. Collaboration with firefighters, emergency-response agencies, and public safety drone operators could further refine mission requirements and help transition the platform toward practical real-world deployment scenarios.

Overall, Pegasus established a strong engineering foundation for future wildfire-response UAV development. The lessons learned throughout the project identified numerous opportunities for refinement and demonstrated the significant potential of autonomous eVTOL systems

to support emergency-response operations through rapid deployment, improved situational awareness, and adaptable mission capability.

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A MATLAB Code for the Monte Carlo Analysis

```

1 %% 2026 VFS-DBVF Wildfire Drone Competition - Sensitivity Analysis
2 % Calibrated to a competition-like DBVF mission profile
3 % AUW envelope: 18-22 lb, mode at 20 lb
4 % Success requirement: minimum 10 min flight time
5
6 clear; clc; close all;
7
8 %% Mission Requirements
9 mission.min_flight_time = 10.0;      % minutes (success requirement)
10 mission.max_weight      = 25.0;      % lb (competition limit)
11 mission.host_payload    = 0.5;       % lb
12 mission.sandbag_weight  = 0.5;       % lb each
13 mission.num_sandbags    = 5;         % bags considered in the mission
    payload set
14 mission.total_payload   = mission.host_payload + mission.num_sandbags *
    mission.sandbag_weight;
15
16 %% Baseline Design Data (mass sanity check only)
17 baseline.num_motors = 6;              % hexacopter
18
19 % Motors: T-Motor MN4110 KV340
20 baseline.motor_thrust = 3634;         % g per motor at max throttle (
    estimate)
21 baseline.motor_weight = 152;         % g each
22 baseline.motor_kv      = 340;
23
24 % Manufacturer static data: MN4110 KV340 + 18x6.1 prop on ~6S
25 % Used to map required hover thrust per motor (g) -> electrical power per
    motor (W)
26 mfg_thrust_g = [1079 1162 1232 1314 1390 1477 1553 1622 1706 1786 1862
    1958 2035 2123 2216 2300 2394 2483 2564 2659 2745 3187 3634];
27 mfg_power_W  = [ 97  108  117  129  141  155  168  178  194  208  222
    240  257  274  291  309  330  350  371  393  415  537  708];
28
29 % Low-thrust anchor so interpolation behaves reasonably below the first
    data point
30 P_idle_W = 25;    % W per motor near idle / very low thrust
31 mfg_thrust_g_ext = [0, mfg_thrust_g];
32 mfg_power_W_ext  = [P_idle_W, mfg_power_W];
33
34 % Calibrated mission-average power factor
35 % This is intentionally below 1.0 because the actual DBVF mission is not
36 % steady hover the entire time; it includes landing, low-power segments,
37 % short repositioning, and ground-contact portions.
38 mission_avg_power_factor = 0.90;
39
40 % Main battery: Zeee 6S 10,000 mAh
41 baseline.battery_capacity = 10000;    % mAh
42 baseline.battery_voltage  = 22.2;     % V
43 baseline.battery_weight   = 1150;     % g
44 baseline.battery_c_rating = 120;

```

```

45
46 % Avionics battery: 3S 3300 mAh
47 baseline.avionics_battery_weight = 279; % g
48
49 % Structure / electronics / payload system masses
50 baseline.frame_weight = 800; % g
51 baseline.arms_hardware = 200; % g
52 baseline.pdb = 81.93; % g
53 baseline.step_downs = 150; % g
54 baseline.telemetry = 14; % g
55 baseline.gps = 60; % g
56 baseline.power_module = 14; % g
57 baseline.flight_controller = 53; % g
58 baseline.receiver = 4.6; % g
59 baseline.lidar = 38; % g
60 baseline.flight_computer = 100; % g
61 baseline.gps_mast = 16.6; % g
62 baseline.remote_id = 14; % g
63 baseline.kill_plug = 15; % g
64 baseline.comms_switch = 2; % g
65 baseline.prop_weight = 31.15; % g each
66 baseline.esc_weight = 5.8; % g each
67 baseline.payload_mechanism = 400; % g
68 baseline.wiring_misc = 200; % g
69
70 % Baseline empty-weight calculation
71 weight_structure = baseline.frame_weight + baseline.arms_hardware;
72 weight_motors = baseline.num_motors * baseline.motor_weight;
73 weight_props = baseline.num_motors * baseline.prop_weight;
74 weight_escs = baseline.num_motors * baseline.esc_weight;
75 weight_electronics = baseline.pdb + baseline.step_downs + baseline.
    telemetry + ...
76 baseline.gps + baseline.power_module + baseline.
    flight_controller + ...
77 baseline.receiver + baseline.lidar + baseline.
    flight_computer + ...
78 baseline.gps_mast + baseline.remote_id + baseline.
    kill_plug + ...
79 baseline.comms_switch;
80 weight_batteries = baseline.battery_weight + baseline.
    avionics_battery_weight;
81
82 baseline.empty_weight_g = weight_structure + weight_motors + weight_props
    + ...
83 weight_escs + weight_electronics +
    weight_batteries + ...
84 baseline.payload_mechanism + baseline.
    wiring_misc;
85 baseline.empty_weight_lb = baseline.empty_weight_g / 453.6;
86
87 nominal_loaded_weight_lb = baseline.empty_weight_lb + mission.
    total_payload;
88
89 fprintf('=== BASELINE MASS SANITY CHECK ===\n');

```

```

90 fprintf('Estimated empty weight:                %.2f lb\n', baseline.
    empty_weight_lb);
91 fprintf('Estimated loaded weight (+3.0 lb payload): %.2f lb\n',
    nominal_loaded_weight_lb);
92 fprintf('Modeled AUW envelope used in Monte Carlo: %.1f to %.1f lb (mode
    %.1f lb)\n\n', 18.0, 22.0, 20.0);
93
94 if nominal_loaded_weight_lb < 18.0
95     fprintf('[NOTE] Component build-up estimate is below the modeled AUW
        envelope.\n');
96     fprintf('        This is acceptable if the 18-22 lb envelope is being
        used as the final measured AUW range.\n\n');
97 elseif nominal_loaded_weight_lb > 22.0
98     fprintf('[NOTE] Component build-up estimate is above the modeled AUW
        envelope.\n');
99     fprintf('        Reconcile the mass budget if you intend to claim the
        18-22 lb AUW range.\n\n');
100 else
101     fprintf('[OK] Baseline loaded-weight estimate is consistent with the
        modeled 18-22 lb AUW envelope.\n\n');
102 end
103
104 %% Active Uncertainty Ranges
105 % Only parameters that are actually used in this calibrated model are
    varied.
106 % Manufacturer thrust-power data already embeds propulsion electrical
    losses,
107 % so separate motor/ESC efficiency sweeps are not included here.
108
109 battery_capacity_factor = [0.90, 0.98];    % healthy packs, near-
    competition condition
110 wind_speed_range       = [0, 12];          % mph
111 temperature_factor     = [0.95, 1.00];    % mild weather / pre-warmed pack
112 usable_battery_fraction = 0.85;           % landed with reserve, less
    pessimistic than old model
113
114 %% Monte Carlo Simulation
115 num_simulations = 3000;
116 fprintf('Running %d Monte Carlo simulations...\n\n', num_simulations);
117
118 % Preallocate results
119 results.total_weight    = zeros(num_simulations, 1);
120 results.weight_margin   = zeros(num_simulations, 1);
121 results.hover_throttle = zeros(num_simulations, 1);
122 results.flight_time     = zeros(num_simulations, 1);
123 results.max_payloads    = zeros(num_simulations, 1);
124 results.points_possible = zeros(num_simulations, 1);
125 results.mission_success = false(num_simulations, 1);
126
127 % Store sampled parameters for sensitivity analysis
128 params.auw              = zeros(num_simulations, 1);
129 params.batt_factor      = zeros(num_simulations, 1);
130 params.wind             = zeros(num_simulations, 1);
131 params.temp_factor      = zeros(num_simulations, 1);

```

```

132
133 for i = 1:num_simulations
134     % Random samples from active uncertainty ranges
135     batt_factor = battery_capacity_factor(1) + rand() * diff(
        battery_capacity_factor);
136     wind        = wind_speed_range(1)          + rand() * diff(
        wind_speed_range);
137     temp_factor = temperature_factor(1)         + rand() * diff(
        temperature_factor);
138
139     % Total AUW sampled with a triangular distribution: 18-22 lb, mode 20
        lb
140     a_AUW = 18; c_AUW = 20; b_AUW = 22;
141     u_AUW = rand();
142     if u_AUW < (c_AUW - a_AUW) / (b_AUW - a_AUW)
143         total_weight_lb = a_AUW + sqrt(u_AUW * (b_AUW - a_AUW) * (c_AUW -
            a_AUW));
144     else
145         total_weight_lb = b_AUW - sqrt((1 - u_AUW) * (b_AUW - a_AUW) * (
            b_AUW - c_AUW));
146     end
147     total_weight_g = total_weight_lb * 453.6;
148
149     % Store sampled parameters
150     params.auw(i)          = total_weight_lb;
151     params.batt_factor(i) = batt_factor;
152     params.wind(i)         = wind;
153     params.temp_factor(i) = temp_factor;
154
155     results.total_weight(i) = total_weight_lb;
156     results.weight_margin(i) = mission.max_weight - total_weight_lb;
157
158     % Weight-limit check
159     if total_weight_lb > mission.max_weight
160         results.max_payloads(i)    = 0;
161         results.hover_throttle(i)  = NaN;
162         results.flight_time(i)     = NaN;
163         results.points_possible(i) = 0;
164         results.mission_success(i) = false;
165         continue;
166     else
167         results.max_payloads(i) = mission.num_sandbags;
168     end
169
170     % Hover throttle estimate
171     max_thrust_g = baseline.num_motors * baseline.motor_thrust;
172     hover_throttle_pct = (total_weight_g / max_thrust_g) * 100;
173     results.hover_throttle(i) = hover_throttle_pct;
174
175     % Power estimate from manufacturer thrust-power map
176     T_per_motor_g = total_weight_g / baseline.num_motors;
177     T_query = min(T_per_motor_g, max(mfg_thrust_g_ext));
178     P_hover_per_motor_W = interp1(mfg_thrust_g_ext, mfg_power_W_ext,
        T_query, 'pchip', 'extrap');

```

```

179 P_hover_total_W = baseline.num_motors * P_hover_per_motor_W;
180
181 % Modest wind penalty so we do not double-count conservatism
182 wind_factor = 1 + (wind / 12) * 0.12; % up to +12% power at 12 mph
183
184 % Mission-average electrical power
185 power_total_W = P_hover_total_W * mission_avg_power_factor *
    wind_factor;
186
187 % Available energy and flight time
188 battery_energy_Wh = (baseline.battery_capacity * batt_factor *
    temp_factor * baseline.battery_voltage) / 1000;
189 usable_energy_Wh = battery_energy_Wh * usable_battery_fraction;
190 flight_time_min = (usable_energy_Wh / power_total_W) * 60;
191 results.flight_time(i) = flight_time_min;
192
193 % Mission success criteria
194 results.mission_success(i) = (flight_time_min >= mission.
    min_flight_time) && (total_weight_lb <= mission.max_weight);
195
196 if results.mission_success(i)
197     % Simplified scoring placeholder
198     points_FM1 = 20;
199     points_FM2 = mission.num_sandbags * 5;
200     points_FM3 = mission.num_sandbags * 5;
201     results.points_possible(i) = points_FM1 + points_FM2 + points_FM3;
202 else
203     results.points_possible(i) = 0;
204 end
205 end
206
207 %% Results Summary
208 valid_hover = ~isnan(results.hover_throttle);
209 valid_flight = ~isnan(results.flight_time);
210
211 success_rate = mean(results.mission_success) * 100;
212 timeout_rate = mean(results.flight_time(valid_flight) < mission.
    min_flight_time) * 100;
213 overweight_rate = mean(results.total_weight > mission.max_weight) * 100;
214
215 fprintf('=== SENSITIVITY ANALYSIS RESULTS ===\n\n');
216 fprintf('Simulated %d scenarios with varying conditions\n\n',
    num_simulations);
217
218 fprintf('WEIGHT ANALYSIS:\n');
219 fprintf('  Mean total weight:          %.2f lb\n', mean(results.total_weight
    ));
220 fprintf('  Weight range:              %.2f to %.2f lb\n', min(results.
    total_weight), max(results.total_weight));
221 fprintf('  Mean weight margin:         %.2f lb\n', mean(results.
    weight_margin));
222 fprintf('  Probability overweight:      %.1f%%\n\n', overweight_rate);
223
224 fprintf('PERFORMANCE ANALYSIS:\n');

```

```

225 fprintf(' Mean hover throttle:      %.1f%%\n', mean(results.
    hover_throttle(valid_hover)));
226 fprintf(' Hover throttle range:      %.1f%% to %.1f%%\n', min(results.
    hover_throttle(valid_hover)), max(results.hover_throttle(valid_hover)))
    ;
227 fprintf(' Mean flight time:          %.1f min\n', mean(results.flight_time
    (valid_flight)));
228 fprintf(' Flight time range:         %.1f to %.1f min\n', min(results.
    flight_time(valid_flight)), max(results.flight_time(valid_flight)));
229 fprintf(' Probability below 10 min: %.1f%%\n\n', timeout_rate);
230
231 fprintf('MISSION SUCCESS:\n');
232 fprintf(' Overall success rate:      %.1f%%\n', success_rate);
233 if any(results.mission_success)
234     fprintf(' Mean points on success:  %.0f points\n\n', mean(results.
        points_possible(results.mission_success)));
235 else
236     fprintf(' Mean points on success:  N/A (no successful cases)\n\n');
237 end
238
239 %% Sensitivity Analysis
240 fprintf('=== SENSITIVITY ANALYSIS: ACTIVE FACTORS ===\n\n');
241
242 corr_auw      = safeCorr(params.auw, results.flight_time);
243 corr_battery  = safeCorr(params.batt_factor, results.flight_time);
244 corr_temp     = safeCorr(params.temp_factor, results.flight_time);
245 corr_wind     = safeCorr(params.wind, results.flight_time);
246
247 fprintf('Impact on FLIGHT TIME (absolute Pearson correlation):\n');
248 fprintf(' AUW:                        %.3f %s\n', abs(corr_auw), getRating(abs
    (corr_auw)));
249 fprintf(' Battery capacity:           %.3f %s\n', abs(corr_battery), getRating
    (abs(corr_battery)));
250 fprintf(' Temperature:                 %.3f %s\n', abs(corr_temp), getRating(
    abs(corr_temp)));
251 fprintf(' Wind speed:                   %.3f %s\n\n', abs(corr_wind), getRating(
    abs(corr_wind)));
252
253 fprintf('Interpretation:\n');
254 fprintf(' 0.0-0.3 = Weak effect\n');
255 fprintf(' 0.3-0.7 = Moderate effect\n');
256 fprintf(' 0.7-1.0 = Strong effect\n\n');
257
258 fprintf('Inactive factors in this cleaned model:\n');
259 fprintf(' - Pilot factor: intentionally removed (zero effect)\n');
260 fprintf(' - Payload weight variation: absorbed into the 18-22 lb AUW
    envelope\n');
261 fprintf(' - Separate motor/ESC efficiency sweeps: omitted to avoid double
    -counting losses already present in manufacturer power data\n\n');
262
263 %% Visualization
264 figure('Position', [50, 50, 1400, 900], 'Name', 'Sensitivity Analysis
    Results', 'Color', 'w');
265

```

```

266 % 1) Weight distribution
267 subplot(3,3,1)
268 histogram(results.total_weight, 40, 'FaceColor', [0.3 0.6 0.9], 'EdgeColor',
    'k', 'DisplayName', 'Simulation Trials')
269 hold on;
270 xline(mission.max_weight, 'r--', 'LineWidth', 2.5, 'Label', '25 lb limit',
    'DisplayName', '25 lb limit');
271 xline(mean(results.total_weight), 'g--', 'LineWidth', 2, 'Label', 'Mean',
    'DisplayName', 'Mean');
272 xlabel('Total Weight (lb)', 'FontWeight', 'bold');
273 ylabel('Frequency');
274 title('Weight Distribution');
275 grid on;
276 legend('Location', 'best');
277
278 % 2) Flight time distribution
279 subplot(3,3,2)
280 histogram(results.flight_time(valid_flight), 40, 'FaceColor', [0.4 0.8
    0.4], 'EdgeColor', 'k', 'DisplayName', 'Simulation Trials')
281 hold on;
282 xline(mission.min_flight_time, 'r--', 'LineWidth', 2.5, 'Label', '10 min
    target', 'DisplayName', '10 min target');
283 xline(mean(results.flight_time(valid_flight)), 'g--', 'LineWidth', 2, '
    Label', 'Mean', 'DisplayName', 'Mean');
284 xlabel('Flight Time (min)', 'FontWeight', 'bold');
285 ylabel('Frequency');
286 title('Flight Time Distribution');
287 grid on;
288 legend('Location', 'best');
289
290 % 3) Hover throttle distribution
291 subplot(3,3,3)
292 histogram(results.hover_throttle(valid_hover), 40, 'FaceColor', [0.9 0.5
    0.3], 'EdgeColor', 'k', 'DisplayName', 'Simulation Trials')
293 hold on;
294 xline(50, 'r--', 'LineWidth', 2, 'Label', '50% throttle', 'DisplayName', '
    50% reference');
295 xline(mean(results.hover_throttle(valid_hover)), 'g--', 'LineWidth', 2, '
    Label', 'Mean', 'DisplayName', 'Mean');
296 xlabel('Hover Throttle (%)', 'FontWeight', 'bold');
297 ylabel('Frequency');
298 title('Hover Throttle Distribution');
299 grid on;
300 legend('Location', 'best');
301
302 % 4) Wind vs flight time
303 subplot(3,3,4)
304 scatter(params.wind, results.flight_time, 30, 'filled', 'MarkerFaceAlpha',
    0.5)
305 hold on;
306 p = polyfit(params.wind, results.flight_time, 1);
307 x_trend = linspace(min(params.wind), max(params.wind), 100);
308 y_trend = polyval(p, x_trend);
309 plot(x_trend, y_trend, 'r-', 'LineWidth', 2);

```

```

310 xlabel('Wind Speed (mph)', 'FontWeight', 'bold');
311 ylabel('Flight Time (min)');
312 title(sprintf('Wind Impact (r = %.3f)', corr_wind));
313 grid on;
314
315 % 5) Battery factor vs flight time
316 subplot(3,3,5)
317 scatter(params.batt_factor, results.flight_time, 30, 'filled', '
    MarkerFaceAlpha', 0.5)
318 hold on;
319 p = polyfit(params.batt_factor, results.flight_time, 1);
320 x_trend = linspace(min(params.batt_factor), max(params.batt_factor), 100);
321 y_trend = polyval(p, x_trend);
322 plot(x_trend, y_trend, 'r-', 'LineWidth', 2);
323 xlabel('Battery Capacity Factor', 'FontWeight', 'bold');
324 ylabel('Flight Time (min)');
325 title(sprintf('Battery Impact (r = %.3f)', corr_battery));
326 grid on;
327
328 % 6) Temperature factor vs flight time
329 subplot(3,3,6)
330 scatter(params.temp_factor, results.flight_time, 30, 'filled', '
    MarkerFaceAlpha', 0.5)
331 hold on;
332 p = polyfit(params.temp_factor, results.flight_time, 1);
333 x_trend = linspace(min(params.temp_factor), max(params.temp_factor), 100);
334 y_trend = polyval(p, x_trend);
335 plot(x_trend, y_trend, 'r-', 'LineWidth', 2);
336 xlabel('Temperature Factor', 'FontWeight', 'bold');
337 ylabel('Flight Time (min)');
338 title(sprintf('Temperature Impact (r = %.3f)', corr_temp));
339 grid on;
340
341 % 7) Weight vs flight time
342 subplot(3,3,7)
343 scatter(results.total_weight, results.flight_time, 30, params.wind, '
    filled')
344 xlabel('Total Weight (lb)', 'FontWeight', 'bold');
345 ylabel('Flight Time (min)');
346 title(sprintf('AUW vs Flight Time (r = %.3f)', corr_auw));
347 colorbar;
348 grid on;
349
350 % 8) Success rate pie chart
351 subplot(3,3,8)
352 success_count = sum(results.mission_success);
353 fail_count = num_simulations - success_count;
354 pie([success_count, fail_count], {'Success', 'Failure'});
355 title(sprintf('Mission Success Rate: %.1f%%', success_rate));
356 colormap([0.4 0.8 0.4; 0.9 0.3 0.3]);
357
358 % 9) Active sensitivity bar chart
359 subplot(3,3,9)

```

```

360 sensitivities = [abs(corr_auw), abs(corr_battery), abs(corr_temp), abs(
    corr_wind)];
361 labels = {'AUW', 'Battery', 'Temp', 'Wind'};
362 bar(categorical(labels), sensitivities, 'FaceColor', [0.5 0.5 0.9]);
363 ylabel('Correlation Coefficient', 'FontWeight', 'bold');
364 title('Active Parameter Sensitivity');
365 ylim([0 1]);
366 grid on;
367 yline(0.3, 'r--', 'Moderate');
368 yline(0.7, 'r--', 'Strong');
369
370 sgtitle('VFS DBVF Wildfire Drone - Monte Carlo Sensitivity Analysis', ...
371         'FontSize', 16, 'FontWeight', 'bold');
372
373
374 % Presentation-friendly white background styling
375 % This overrides MATLAB dark-theme figure defaults and makes the figure
376 easier
377 % to paste into Google Slides / PowerPoint.
378 set(gcf, 'Color', 'w', 'InvertHardcopy', 'off');
379
380 all_axes = findall(gcf, 'Type', 'axes');
381 for ax_idx = 1:numel(all_axes)
382     set(all_axes(ax_idx), ...
383         'Color', 'w', ...
384         'XColor', 'k', ...
385         'YColor', 'k', ...
386         'GridColor', [0.75 0.75 0.75], ...
387         'MinorGridColor', [0.85 0.85 0.85], ...
388         'FontWeight', 'bold', ...
389         'LineWidth', 1.0);
390 end
391
392 all_text = findall(gcf, 'Type', 'text');
393 for txt_idx = 1:numel(all_text)
394     set(all_text(txt_idx), 'Color', 'k');
395 end
396
397 all_legends = findall(gcf, 'Type', 'legend');
398 for lgd_idx = 1:numel(all_legends)
399     set(all_legends(lgd_idx), ...
400         'Color', 'w', ...
401         'TextColor', 'k', ...
402         'EdgeColor', [0.5 0.5 0.5]);
403 end
404
405 all_colorbars = findall(gcf, 'Type', 'colorbar');
406 for cb_idx = 1:numel(all_colorbars)
407     set(all_colorbars(cb_idx), 'Color', 'k');
408 end
409 % Optional: automatically save a presentation-ready PNG next to the script
410 .

```

```

410 exportgraphics(gcf, 'DBVF_Monte_Carlo_Sensitivity_Analysis_white.png', '
    Resolution', 300, 'BackgroundColor', 'white');
411
412 %% Design Recommendations
413 fprintf('=== DESIGN RECOMMENDATIONS ===\n\n');
414
415 if mean(results.weight_margin) < 2
416     fprintf('[CRITICAL] Weight margin is below 2 lb on average.\n');
417     fprintf('        Reduce mass or recover margin before claiming a
        robust design.\n\n');
418 elseif mean(results.weight_margin) < 4
419     fprintf('[WARN] Weight margin is somewhat tight (%.1f lb average).\n',
        mean(results.weight_margin));
420     fprintf('        Some mass reduction would improve robustness.\n\n');
421 else
422     fprintf('[OK] Weight margin is adequate (%.1f lb average).\n\n', mean(
        results.weight_margin));
423 end
424
425 if mean(results.flight_time(valid_flight)) < (mission.min_flight_time +
    1.0)
426     fprintf('[CRITICAL] Mean flight time is too close to the 10 min
        requirement.\n');
427     fprintf('        More battery margin, less AUW, or lower mission-
        average power is recommended.\n\n');
428 elseif mean(results.flight_time(valid_flight)) < (mission.min_flight_time
    + 2.5)
429     fprintf('[WARN] Flight-time margin is modest (%.1f min average).\n',
        mean(results.flight_time(valid_flight)));
430     fprintf('        The design should work, but there is limited reserve.\n
        \n');
431 else
432     fprintf('[OK] Flight time has good margin (%.1f min average).\n\n',
        mean(results.flight_time(valid_flight)));
433 end
434
435 if mean(results.hover_throttle(valid_hover)) > 50
436     fprintf('[CRITICAL] Hover throttle is high (%.1f%% average).\n', mean(
        results.hover_throttle(valid_hover)));
437     fprintf('        Increase thrust capability or reduce weight.\n\n');
438 elseif mean(results.hover_throttle(valid_hover)) > 40
439     fprintf('[WARN] Hover throttle is above 40%% (%.1f%% average).\n',
        mean(results.hover_throttle(valid_hover)));
440     fprintf('        Control margin in wind/manoeuvring is reduced.\n\n');
441 else
442     fprintf('[OK] Hover throttle is in a comfortable range (%.1f%% average
        )\n\n', mean(results.hover_throttle(valid_hover)));
443 end
444
445 fprintf('KEY FOCUS AREAS (from the active sensitivity results):\n');
446 active_labels = {'AUW', 'Battery capacity/health', 'Temperature', 'Wind'};
447 active_values = [abs(corr_auw), abs(corr_battery), abs(corr_temp), abs(
    corr_wind)];

```

```

448 [sorted_values, sort_idx] = sort(active_values, 'descend');
449 for k = 1:length(sorted_values)
450     fprintf(' %d. %-24s : %.3f %s\n', k, active_labels{sort_idx(k)},
451             sorted_values(k), getRating(sorted_values(k)));
452 end
453 fprintf('\n');
454 fprintf('OVERALL ASSESSMENT:\n');
455 if success_rate >= 95
456     fprintf('[EXCELLENT] Design is robust (%.1f%% success).\n',
457             success_rate);
458     fprintf('          The calibrated model suggests the drone should
459             reliably complete the mission.\n');
460 elseif success_rate >= 85
461     fprintf('[GOOD] Design is solid (%.1f%% success).\n', success_rate);
462     fprintf('          Minor improvements could further increase reliability
463             .\n');
464 elseif success_rate >= 70
465     fprintf('[MARGINAL] Design is somewhat risky (%.1f%% success).\n',
466             success_rate);
467     fprintf('          Additional margin is recommended before relying on
468             this configuration.\n');
469 else
470     fprintf('[INADEQUATE] Design needs revision (%.1f%% success).\n',
471             success_rate);
472     fprintf('          The current configuration does not provide
473             enough mission margin.\n');
474 end
475 fprintf('\n=== END OF ANALYSIS ===\n');
476
477 %% Helper Functions
478 function rating = getRating(correlation)
479     if isnan(correlation)
480         rating = '(not applicable)';
481     elseif correlation < 0.3
482         rating = '(weak)';
483     elseif correlation < 0.7
484         rating = '(moderate)';
485     else
486         rating = '(strong)';
487     end
488 end
489
490 function r = safeCorr(x, y)
491     valid = ~(isnan(x) | isnan(y));
492     x = x(valid);
493     y = y(valid);
494
495     if numel(x) < 2 || std(x) == 0 || std(y) == 0
496         r = NaN;
497     else
498         r = corr(x, y);
499     end

```

494 `end`

Listing 1: MATLAB code for Monte Carlo sensitivity analysis